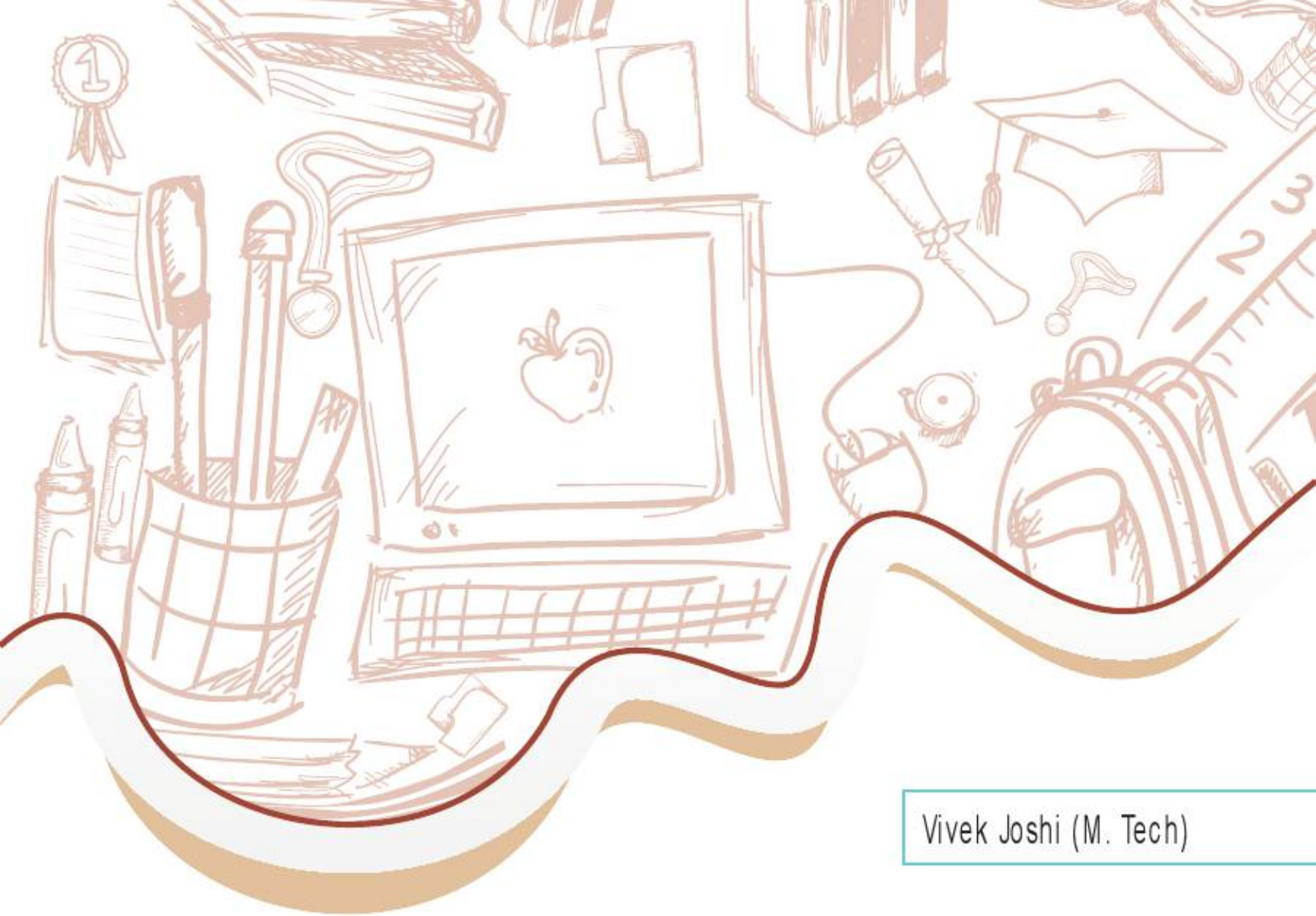
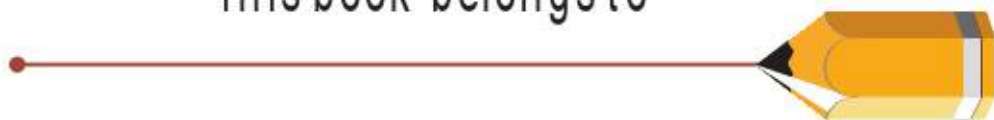




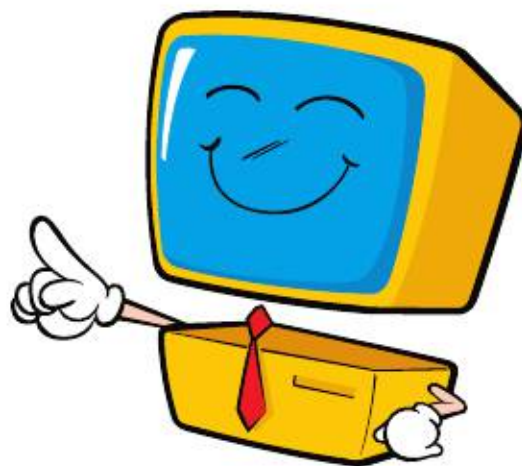
Vivek Joshi (M. Tech)

This book belongs to



Computer Interface

7



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PREFACE

In this modern age, computer plays an important role in our daily life. Computers are widely used everywhere for advancement and progress in our life. Mostly every field of our surrounding is depended on computers. So, all the schools and colleges are teaching computer as one of the main subject and preparing students to meet the challenges of the coming time.

This series **Computer Interface** of 8 book (I to VIII) written for the students studying in middle classes following **National Education Policy 2020** and covers every aspect which is related to computer according to the recent development in computer technology. This series is strictly written according to the latest syllabus based on the guidelines given by CBSE. Moreover the language used in this series is very easy to understand by the students, teachers and parents.

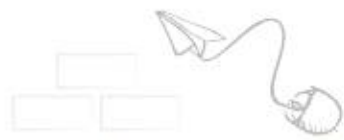
This series is designed to provide a complete knowledge with the latest information with lab exercises and more over.

We hope that students will enjoy learning computer with this series. Any suggestions for the improvement of these books will be highly appreciated.



Contents

 1.	Computer Memory	 5
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Computer Memory



Lesson at a Glance

Introduction

Memory Storage

Units of Memory

Types of Memory

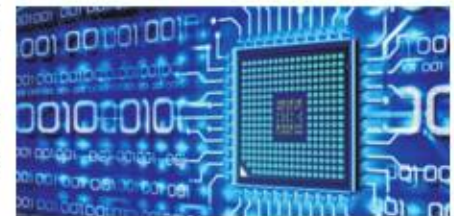


Introduction

One of the major advantages of computer is its storage capacity when huge amount of information can be stored. Infact inside the computer, there are different storage areas where it keeps data or information permanently or temporarily while working. This storage area is known as memory.

Memory Storage

Memory is that unit of the computer where data and instructions are stored. The memory unit of the computer is made up of one or more chips placed on the motherboard. All the data is stored on it in the form of 0 and 1 which is called binary digits or bits.



Units of Memory

Bit is the smallest unit of memory that a computer can process. In the computer, binary digit 0 and 1 are used to store data the instructions. These number are called bits.

A group of 4 bits is called a nibble.

A group of 8 bits is called a byte.

1 KB (Kilobyte)	= 2^{10} bytes	= 1024 bytes
1 MB (Megabyte)	= 2^{10} KB	= 1024 KB
1 GB (Gigabyte)	= 2^{10} MB	= 1024 MB
1 TB (Terabyte)	= 2^{10} GB	= 1024 GB
1 PB (Petabyte)	= 2^{10} TB	= 1024 TB

Must Know!



The computer only understands the language of 0's and 1's.

Types of Memory

A computer memory is classified into Primary memory and secondary memory.

Primary Memory

It is also known as main memory.

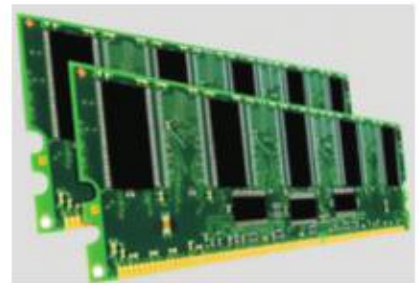
Main memory is directly or indirectly connected to the central processing unit via a memory bus.

The CPU continuously reads instructions stored there and executes them as required.

Example: RAM and ROM

RAM (Random Access Memory)

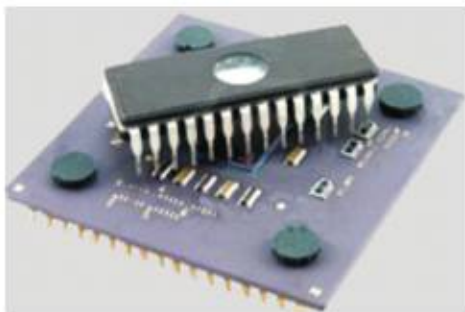
It is called Random Access Memory because any of the data in RAM can be accessed just as fast as any of the other data. It is used to write and read data from it. It is a volatile memory. When computer is off all the information that has been stored on it is lost.



There are two types of RAM:

DRAM (Dynamic Random Access Memory)

SRAM (Static Random Access Memory)



ROM (Read Only Memory)

This memory is used as the computer begins to boot up. It is a non-volatile memory. When computer is off all the information that has been stored on is not lost. ROM memory cannot be easily or quickly overwritten or modified.

Must Know!



RAM is a temporary memory and ROM is permanent memory.

There are three types of ROM:

PROM (Programmable Read Only Memory)

All the program are store on this chip. Once it is used, we cannot wipe it clean or store something else. It can be programmed only under special conditions.



EPROM (Erasable Programmable Read Only Memory).

We can erase its stored information by exposing it to ultraviolet light.

EEPROM (Electrically Erasable Programmable Read Only Memory).

We can erase its stored information using electrical signals.



Secondary Memory

Data and information are in different forms such as documents, photos, videos, animations, voices and sounds. That data and information can be stored in computer storage devices. Similarly, computer programmes which give instructions to the computer can be stored too.

Computer usually uses its input/output channels to access secondary storage and transfers the desired data using intermediate area in primary storage.

It is not directly accessible by the CPU.



Must Know!



Motherboard is an electronic board fixed inside the CPU box. The CPU and other internal parts of a computer are attached to it.

Hard Disk



The hard disk drive is the main, and usually largest, data storage device in a computer. The hard disk consists of a magnetised platter and a moving metal head. It can store anywhere from 160 gigabytes to 2 tera bytes.

Hard disk speed is the speed at which content can be read and written on a hard disk. A hard disk unit comes with a set rotation speed varying from 4500 to 7200 rpm.

Disk access time is measured in milliseconds.

Must Know!



Now days external hard disks are also available, which is not fixed inside the computer.

Magnetic Tapes

A magnetic tape is a thin plastic tape covered with a magnetic substance. It is used for recording sounds, images or computer data etc. The magnetic tape which was used often in the past has now become obsolete.



Floppy Disk

A floppy disk is a magnetic media device that can be used to store small files. It contains a capacity of 1.44 MB. It is used to carry data because it is an external storage device. But is becoming obsolete.



At present, there are disks which are produced by using the same technology used to produce the floppy disks such as zip disks and jazz disks. The capacity of these disks is greater than that of the floppy disks. But they are not used often.

CD-ROM

Compact Disc (CD) is the commonly used storage device. It can store 600 to 750 MB data on it. Data can be in the form of text, image, sound, video, movie etc. A laser beam is used to read CDs data.

Now a day's we got two types of CDs :

CD-R (Compact Disc-Recordable) in which data written cannot be changed or deleted.



CD-RW (Compact Disc-ReWritable) in which data can be rewritten or deleted.

DVD (Digital Versatile Disc)

It is similar to a CD but can store much more data. It can be recorded on one or both sides. They are of two types; DVD-R (data can be written once) and DVD - RWC (rewritable). It ranges from 4.7 GB to 8.5 GB.



Must Know!



To read the data stored on a CD you need to put it into a CD Drive.

Blue-ray Disc

It is similar to a CD or a DVD and hold the data upto 25 GB.

A Blue ray reader is required to use a Blu-ray disc.



USB Flash Drive



A small, portable flash memory card that plugs into a computer's USB port and functions as a portable hard drive.

Flash drives are available in sizes such as 1GB, 2 GB, 32 GB and more.

It is an easy way to transfer and store information.

Memory Card

An electronic flash memory storage disk commonly used in consumer electronic devices such as digital cameras, MP3 players, mobile phones, and other small portable devices.

Memory cards are usually read by connecting the device containing the card to your computer, or by using a USB card reader.



Must Know!



Now days instead of storing data on a local drive, many people are opting for cloud storage.



Work Station



A Tick (3) the correct option.

(NEP) Analytical Skills

1. It is the smallest unit of information which a computer can process and store.

(a) RAM

☐

(b) Bit

☐

(c) Nibble

☐

(d) ROM

☐

2. It is directly or indirectly connected to the central processing unit via a memory bus.

(a) Motherboard	☐	(b) Secondary memory	☐
(c) Main memory	☐	(d) Non-volatile	☐
3. When computer is off all the information that has been stored on is not lost.

(a) RAM	☐	(b) Bit	☐	(c) ROM	☐	(d) Byte	☐
---------	--------------------------	---------	--------------------------	---------	--------------------------	----------	--------------------------
4. The hard disk drive is the main, and usually largest, data storage device in a computer.

(a) DVD Drive	☐	(b) Hard disk	☐	(c) Pen drive	☐	(d) CD-Drive	☐
---------------	--------------------------	---------------	--------------------------	---------------	--------------------------	--------------	--------------------------
5. They are usually read by connecting the device containing the card to your computer.

(a) Memory cards	☐	(b) Volatile	☐
(c) Static memory	☐	(d) External hard disk	☐

B Fill in the blanks.

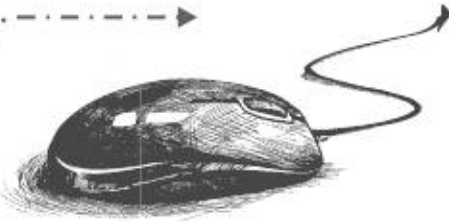
1. _____ memory is also known as the main memory of a computer.
2. A group of 4 bits is called a _____.
3. Pen drive is also known as _____ drive.
4. A computer _____ is stored in the form of bits and bytes.
5. A _____ is made up of metal plates coated to store data magnetically.
6. _____ refers to read and write memory.

C Write T for True or F for False.

1. A computer process the input data and store the information in form of 0 and 1. ☐
2. The bits are carried by electric circuits and stored on chip. ☐
3. The information in computer is stored in the form of bits and bytes. ☐
4. CD drive Disk can store anywhere from 160 gigabytes to 2 tera bytes. ☐
5. RAM memory cannot be easily or quickly overwritten or modified. ☐

D Answer the following questions.

1. What do you mean by computer memory?
2. Differentiate between RAM and ROM.
3. Define Hard Disk.
4. Write the features of DVD.
5. Differentiate between Primary and Secondary memory.
6. Write the uses of Blu-ray disc.

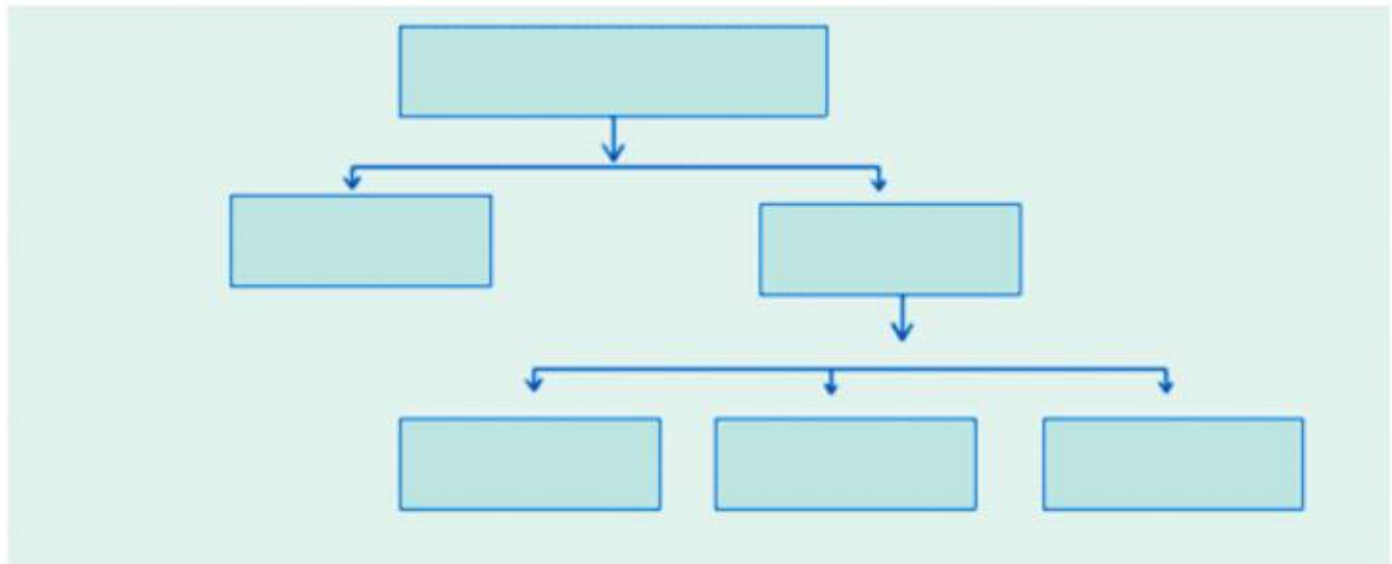


Work sheet



A Write the words to complete the flow chart.

(NEP) Creative Learning



RAM

PROM

ROM

EPROM

Primary Memory

EEPROM

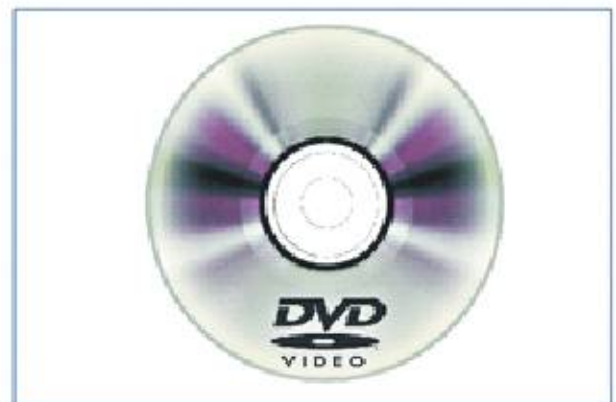
B Click the correct storage capacity of the given devices.



16 GB

8 GB

4.7 GB



1 GB

4.7 GB

700MB

About Operating System

Lesson at a Glance

Introduction
Different types of Operating System

Loading and Operating System
About Files and Folder

Introduction

An operating system is the interface between the computer hardware and the end user. Processing of data, running application, file management and handling the memory is all managed by the computer operating system. Windows, MAC, Android etc. are examples of operating system which are generally used nowadays.

Features Operating System

- It enables the hardware to communicate with the computer and used in an efficient manner.
- It works as an interface between the user and a computer.
- It provides an environment in which a user can execute programs.
- It works as a resource manager as it manages all hardware and software and allows it to communicate with various computer parts.
- It controls the functions of the Central Processing Unit (CPU).
- It helps to protect the computer through usernames and passwords.
- It manages the hardware memory of a computer very efficiently.

Loading an Operating System

The process of loading an operating system into main memory is called booting the system. Once the operating system is loaded into main memory, it usually remains in memory until the computer is turned off. The operating system controls the loading and manages the execution of each application program that is requested by the user.

Different Types of Operating Systems

Some examples for operating systems used in computers.

Microsoft DOS (MS DOS)

The Microsoft Disk Operating System, or MS-DOS, was released by Microsoft Corporation in 1981. MS-DOS was originally developed for IBM for their first personal computer system. Other personal computer operating systems exist, but by far the majority of personal computer software is written of MS-DOS. This single-user operating system in so widely used that it is often referred to simply as DOS.



Microsoft Windows

The windows software is the most popular operating system. It was developed to operate computer programs through mouse. In the beginning 3.1 version was introduced. There are rapid changes in computer software and in this scenario. Microsoft corporation, which developed windows, has introduced another version named window 95.



Apple Macintosh or Mac OS

The Apple Macintosh multiprogramming operating system provides a graphic interface that uses icons and windows.

The Macintosh operating system has set the standard for operating system ease of use and has been the model for most of the new graphic user interfaces developed for non-Macintosh system.



Linux

Linux is similar to Unix in operation but it is free. It also should not be used by anyone who does not understand the system and can be difficult to learn.



UNIX

The UNIX operating system was developed in the early 1970s by scientists at Bell Laboratories. It was specially designed to provide a way to manage a variety of scientific and specialized computer applications.

Today, most major computer manufacturers offer a multi user version of the UNIX operating system to run on their computers.
Some examples for operating systems used in mobile phones.

Android

Used in smart mobile phones.



iOS or iPhone OS

Used in Apple mobile phones such as iPhone, iPad, iPod.

Blackberry OS

Used in Blackberry mobile phones.



Windows Mobile OS

Used in mobile phones like Lumia, HTC.



Must Know!



The operating system provides a common method for opening/closing a program and managing files & folders, etc.

About File and Folder

What is a File?

A file is anything that is saved in the computer such as a letter, a song or a photo.
A file has two parts.

File name : Used to identify the file.

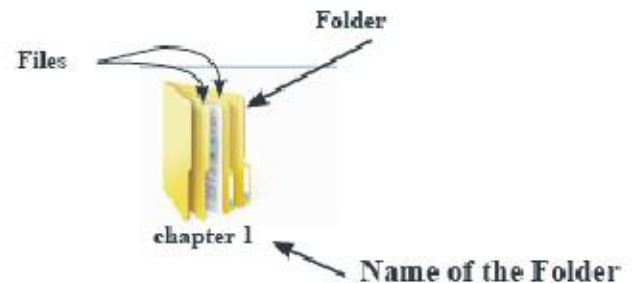
File Extension : Used to indicate the application software of the file.

sportmeet.docx
File name File Extension

This is a file named 'sportmeet', created by a word processing software.

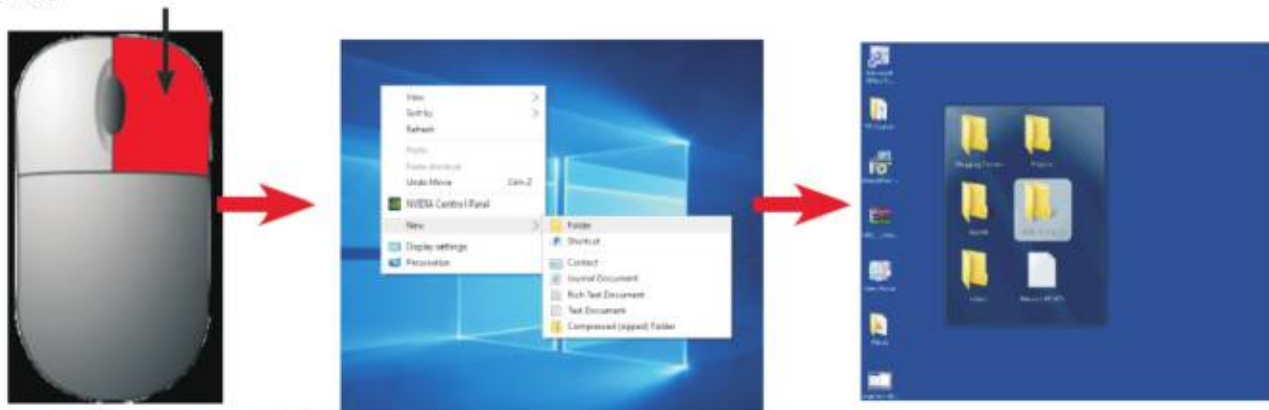
What is a Folder?

The place where files are stored is a folder. They are used to store files that are related to each other. It helps to keep files in order and to retrieve them easily and fast.



Creating a Folder

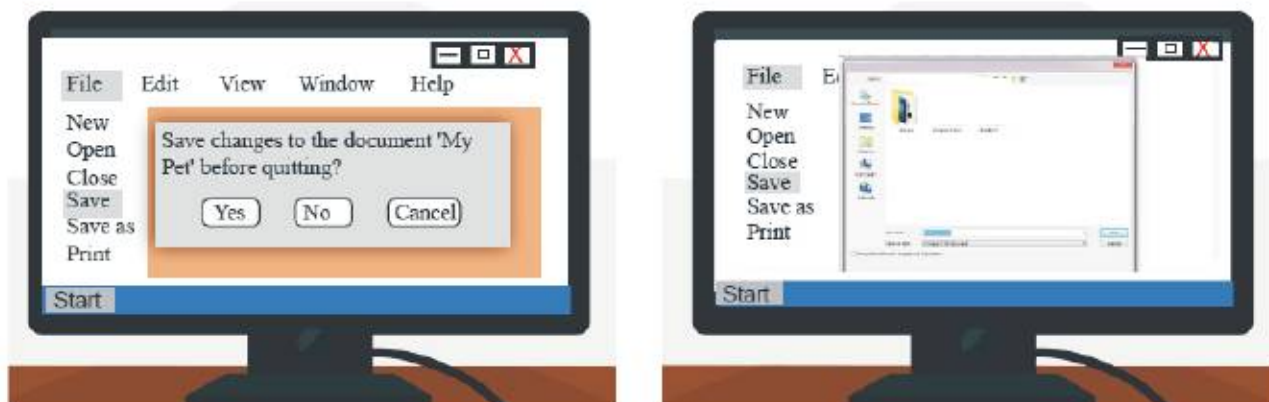
The methods used to create a folder can be different according to the operating system.



Saving Files in a Folder

Files created should be saved to reuse them later. To save a file, save or Save As commands are used.

When saving a file for the first time, despite the window selected, Save As command window will open.



After selecting the folder in which the file is to be stored, by giving the save command, the file is stored in the folder.

Opening a created Folder

The folder can be opened by taking the cursor on to the folder and double clicking on it.

Coping and Moving a Folder

A folder can be copied or moved to another location. It can be internal location or an external storage device.

First, take the cursor to the location of the folder and select it.

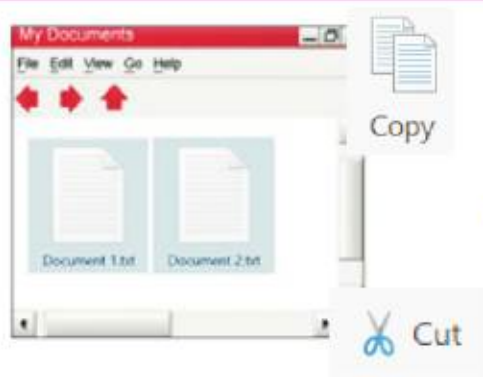
To Copy

Step 1

Firs, select the Copy command in the tool bar or the Copy command shown when the right button of the mouse is clicked.

Step 2

Then, go to the location where it should be pasted and select Paste command.



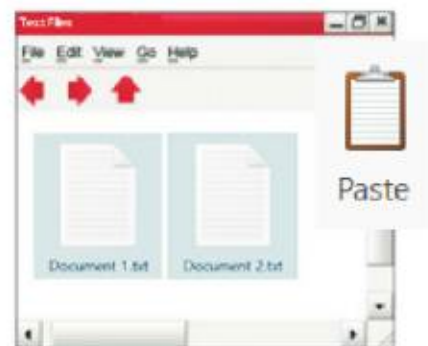
To Move

Step 1

First, select the Cut command in the tool bar or the Cut command shown when the right button of the mouse is clicked.

Step 2

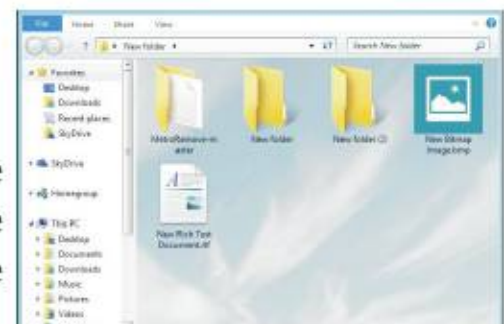
Then, go to the relevant location and select Paste command.



Changing the Name of a Folder

Step 1

To change the name of a folder, take the mouse pointer on to the folder, Then select Rename command after clicking the right button of the mouse.



Step 2

Type the required name.

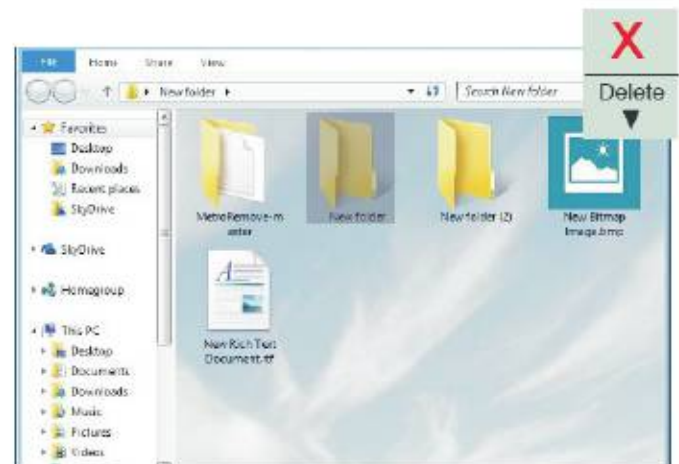
Deleting a Folder temporarily

Step 1

Select the folder that needs to be deleted.

Step 2

Press the Delete button on the keyboard.



Must Know!



If you delete a folder that is in an internal location such as in a hard disk, it is deleted temporarily. That means, it can be retrieved when needed.

If you delete a folder that is in an external storage device such as a pen drive, it is permanently deleted. It cannot be retrieved again.

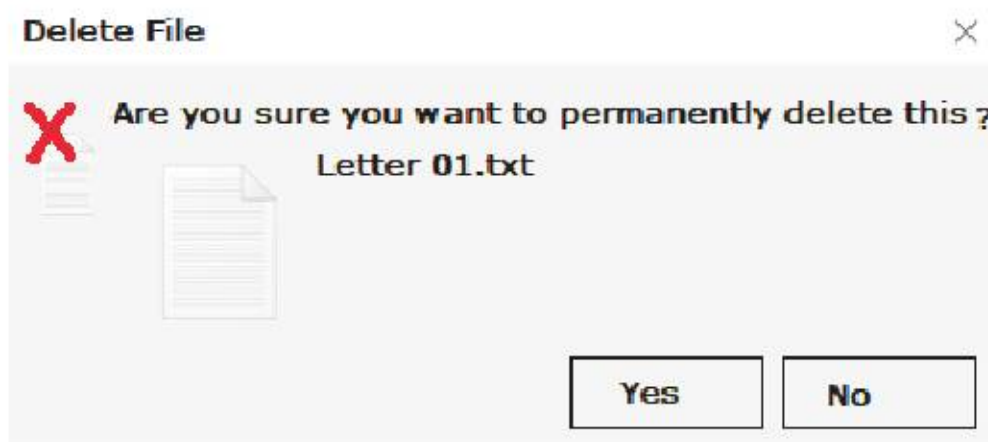
Delete a Folder Permanently

Step 1

To delete a file or a folder permanently, press Shift and Delete keys together.

Step 2

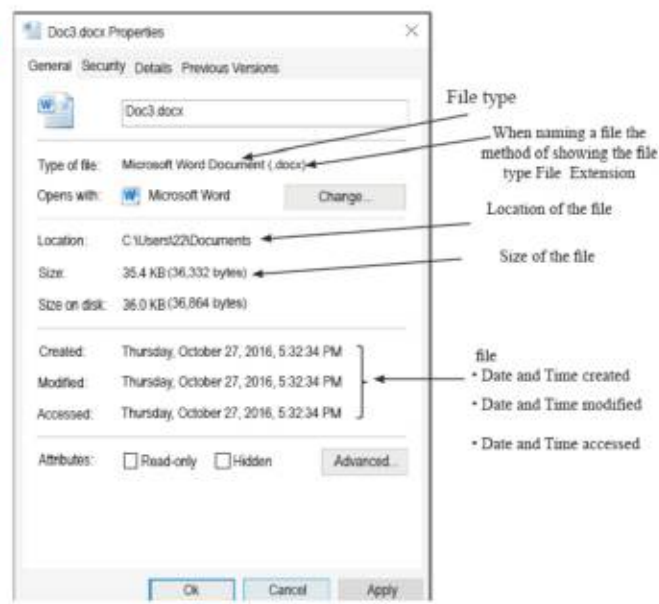
Select Yes command in the dialog box.



Identify the Characteristics of a File and a Folder

There are characteristics in a file or a folder such as size, type and date modified.

When the mouse pointer is taken on to the relevant file or folder and when right button is clicked a menu is displayed. Select the Properties command and you can see these characteristics.



Work Station



A Tick(3) the correct option.

(NEP) Analytical Skills

- It was originally developed for IBM for their first personal computer system.
 (a) MS-Windows ☐ (b) Linux ☐ (c) Unix ☐ (d) MS-DOS ☐
- It was developed to operate computer programs through mouse.
 (a) MS-DOS ☐ (b) Blackberry ☐
 (c) Microsoft Windows ☐ (d) Fedaro ☐
- This multiprogramming operating system provides a graphic interface that uses icons and windows.
 (a) Application ☐ (b) Ubuntu ☐
 (c) Apple Macintosh ☐ (d) Operating system ☐
- It was specially designed to provide a way to manage a variety of scientific and specialized computer applications.
 (a) MS-DOS ☐ (b) Unix ☐ (c) Apple ☐ (d) Google ☐

B Fill in the blanks.

- The _____, was released by Microsoft Corporation in 1981.
- In the beginning _____ version was the first windows operating system.
- _____ should not be used by anyone who does not understand the system and can be difficult to learn.
- Operating systems produced using Linux are: _____ and _____.

C Write T for True or F for False.

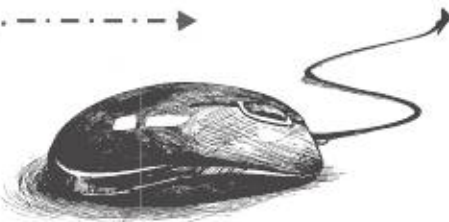
1. The operating system is necessary to fulfil the tasks using a computer. ☐
2. The windows software is the most popular operating system. ☐
3. Operating system helps to protect the computer through usernames and passwords. ☐
4. Operating system performs the tasks related to processing files and folders correctly. ☐

D Match the followings.

- | | |
|----------------------|---|
| 1. Linux | Used in smart mobile phones. |
| 2. Android | Used in Blackberry mobile phones. |
| 3. iOS or iPhone OS | Used in mobile phones like Lumia, HTC. |
| 4. Blackberry OS | Similar to Unix in operation but it is free. |
| 5. Windows Mobile OS | Used in mobile phones such as iPhone, iPad, iPod. |

E Answer the following questions.

1. List five tasks performed by the operating system.
2. What are computer storage devices?
3. What are the things that can be stored in computer storage device?
4. What is a file?
5. What is a folder?



Work sheet



■ **Observe the picture and write about them.**

(NEP) Creative Learning



More on Windows 10

Lesson at a Glance

Windows 10 Apps Management
Windows 10 Keyboard Shortcuts
Windows 10 Media Management

Windows 10 E-mail Management
Windows 10 System Tray

Windows 10 Apps Management

Windows comes with several useful applications already installed, which help you with many tasks, including browsing the Internet, managing your calendar, shopping etc.

You can access those applications in multiple ways. The basic way to open these applications is open the Search menu and click All Apps.

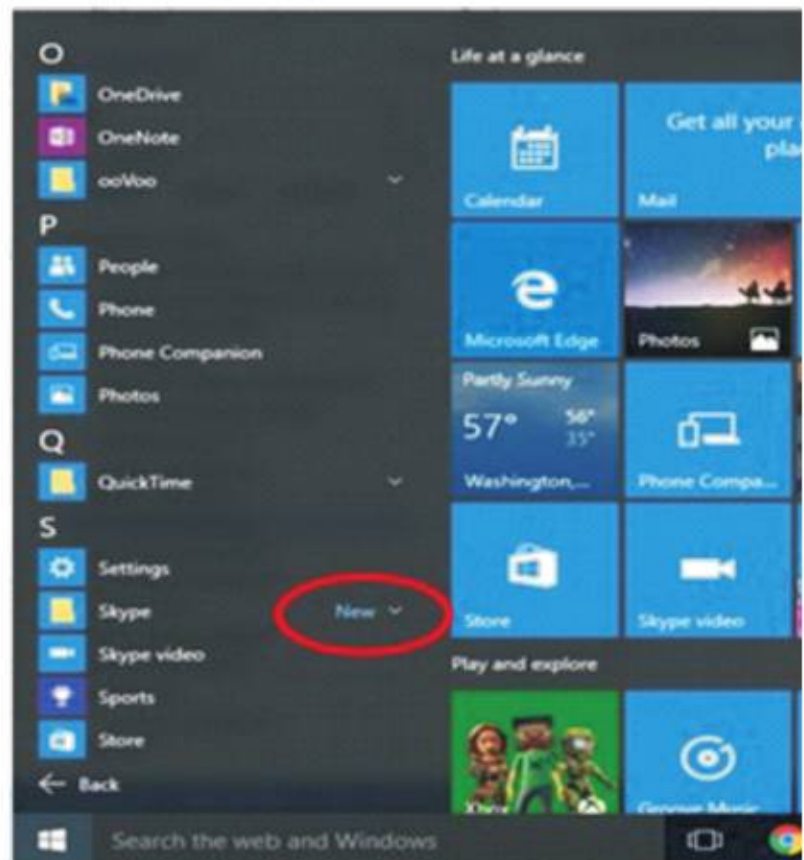
It will display all the applications installed in your computer. If there was a recently installed application, it will be labelled as “New” in the menu.

Uninstalling Applications

There are three ways in which you can remove an unwanted application:

- From the Start Menu
- From the Settings window
- From the Control Panel

Here are the steps for each way.



Must Know!



The Start menu is better with resizable tiles and other new capabilities to help you be more productive.

Uninstalling an Application from the Start Menu

Perhaps the easiest way to uninstall an application is from the Start Menu. To do this, follow these steps -

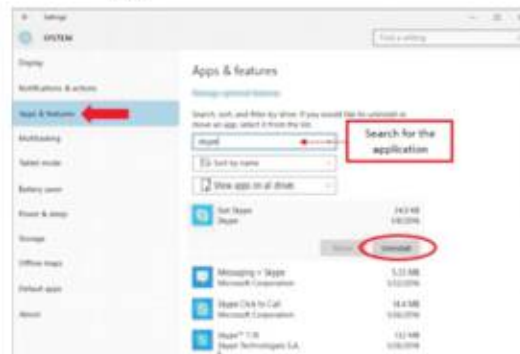
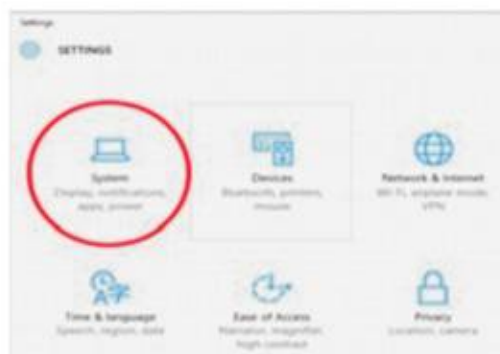
- Step 1** - Open the Start Menu and click on All Apps.
- Step 2** - Find the application you want to uninstall on the list, and right-click on it.
- Step 3** - On the menu, choose Uninstall. After this, follow the steps needed to complete the process. These steps will vary for each application.



Uninstalling an Application from the Settings window

A new and different way to uninstall an application is from the new Settings window. To do this, follow these steps -

- Step 1** - Open the Start Menu and click Settings.
- Step 2** - On the **SETTINGS** window, choose **System**.
- Step 3** - On the **SYSTEM** window, choose **Apps & features**.



- Step 4** - On the Apps & features window, search for the application you want to uninstall in the search box.
- Step 5** - From the results, click on the application and choose Uninstall.

Must Know!

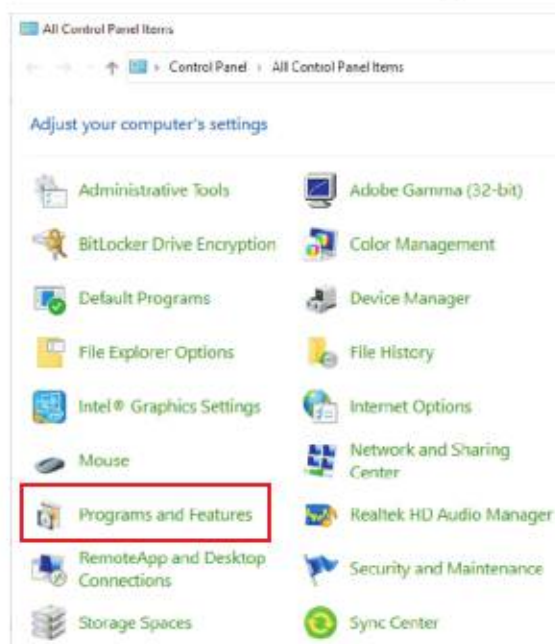
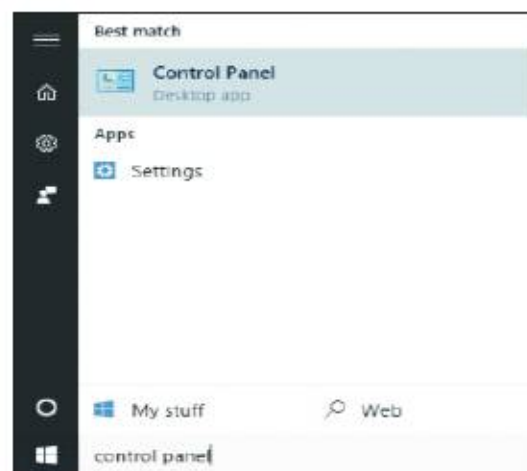


Select All apps at the bottom of the left pane to display a full list of programs and folders on your PC in alphabetical order.

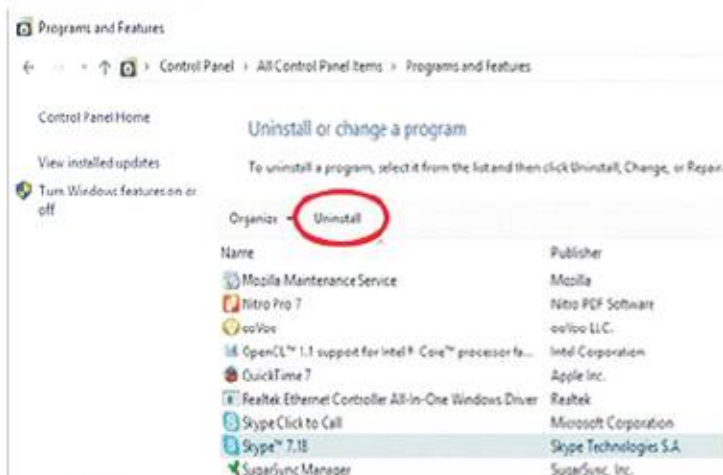
Uninstalling Applications from the Control Panel

The third way to uninstall an application is more traditional, because it uses the popular Control Panel window. To use this method, follow these steps -

- Step 1** - Open the Control Panel by searching for it in the Search bar.
- Step 2** - After the Control Panel is open, choose Programs and Features.



- Step 3** - The Programs and Features window will present a list of all the applications installed in your computer. To uninstall an application, just click on it and use the Uninstall button.



An alternate way is to right-click the application on the list.

This window also gives you other details about the applications you have installed like -

The date it was installed.

The size it occupies on the hard disk.

The specific version of this application.

Must Know!



Quick Actions are a set of tiles that give you access to frequently used settings and tasks (like Wi-Fi connection, or screen brightness).

Windows 10 Email Management

Windows 10 features the new Mail app to help you handle your e-mails. You can access the Mail app directly from the Start Menu.

If you are connected to Windows with a mail account, the Mail app will automatically configure itself to present you your e-mails. But, if you are using a local account, the Mail app will ask you to configure it.



Configuring the Mail App

The first time you access the Mail account, you will see this window.

To configure it, follow these steps -

Step 1 - Click on the Get started button.

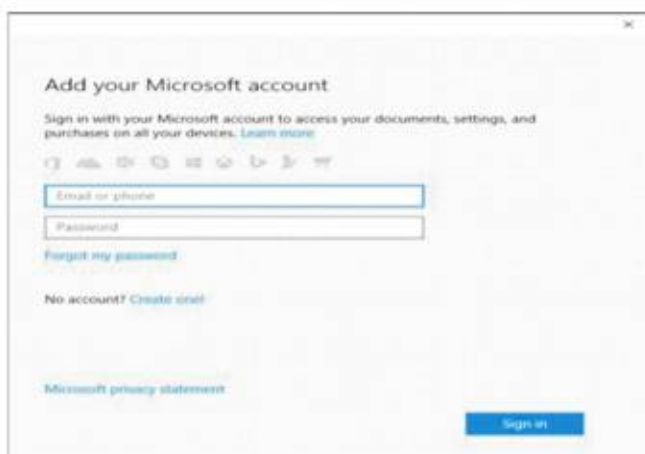
Step 2 - On the next window, add the account you want to use with Mail.



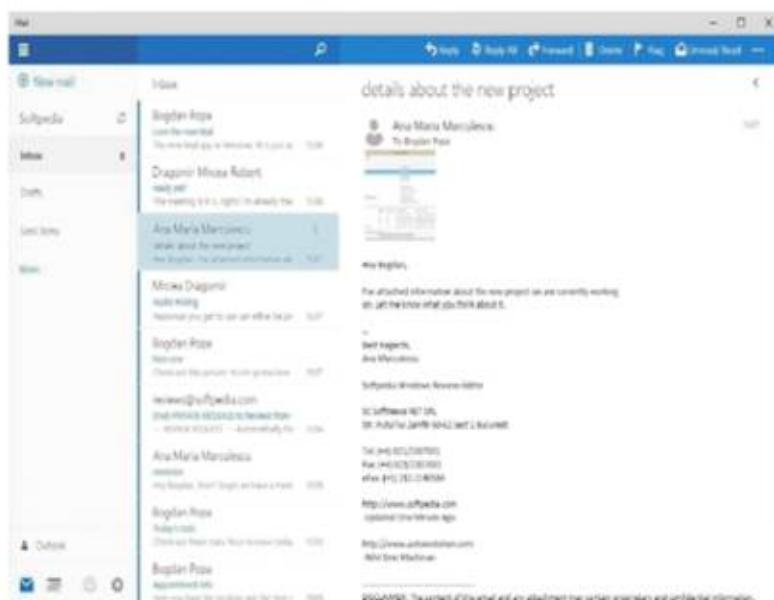
Step 3 - On the Choose an account window, select what type of account you want to configure. Mail is compatible with most of the current mail providers.



Step 4 - If you're using a Microsoft account, enter your username and password.



Step 5 - After a while, the Mail app will start loading your emails.





Some Windows apps require signing in to your Microsoft account before they are fully launched.

Windows 10 Keyboard Shortcuts

Like most Windows applications, there are several keyboard shortcuts you can use to make it easier or faster for some to perform certain tasks.

Most of the new Windows shortcuts use the Windows key () combined with other keys to perform several actions. The following are some of the most common or useful shortcuts used in Windows 10.

SHORTCUT	ACTION
Windows key 	Open the Start menu
Windows key  + S	Open the Windows Search
Windows key  + C	Open Cortana
Windows key  + Tab	Open Task View
Windows key  + D	Show or Hide the Desktop
Windows key  + L	Lock your account
Windows key  + A	Open the Action Center sidebar
Windows key  + I	Open the Settings window
Windows key  + E	Open the File Explorer window
Windows key  + PrintScreen	Takes a screenshot of your whole display and stores it in Pictures > Screenshots

Window Snapping

Other useful shortcuts introduced in Windows 10 are the ones used for Windows snapping. Snapping windows was introduced in Windows 7 and allows you to automatically arrange your windows making it easier to work with multiple applications at the same time.

Although you can do this with your mouse by dragging the windows to each side of the screen, Windows 10 introduced shortcuts to perform this task.

SHORTCUT	ACTION
Windows key  + Left arrow	Snap the active window on the left side of the screen
Windows key  + Right arrow	Snap the active window on the right side of the screen
Windows key  + Up arrow	Snap the active window on the top of the screen
Windows key  + Down arrow	Snap the active window on the bottom of the screen

Windows 10 System Tray

The System Tray is another name given to the Notification Area, which we can find in the right-side of the Windows Taskbar. The System Tray features different types of notifications and alerts from your computer like your Internet connection, or the volume level.

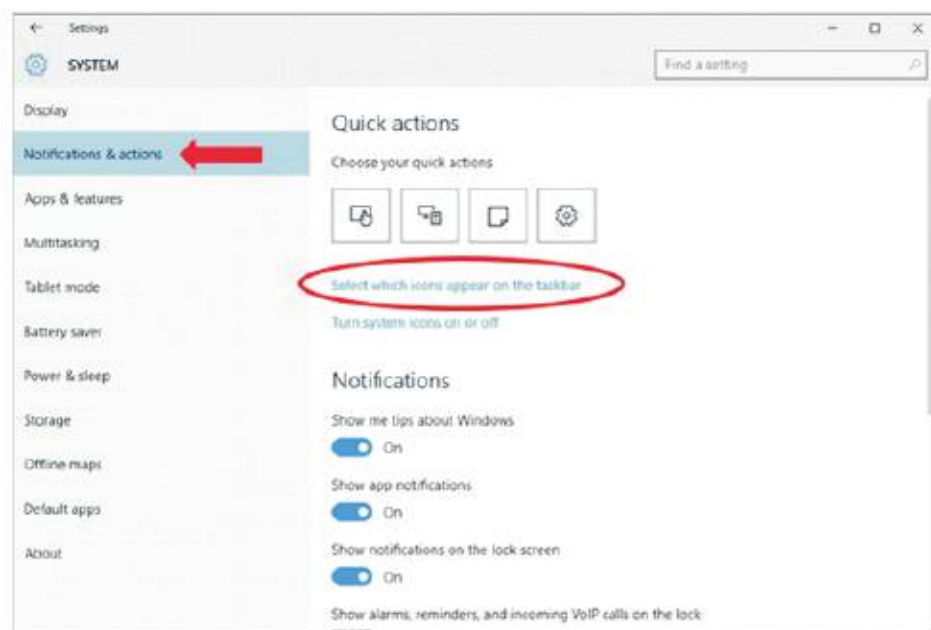


However, you can choose what icons and notifications appear here. To further customize the System Tray follow these steps -

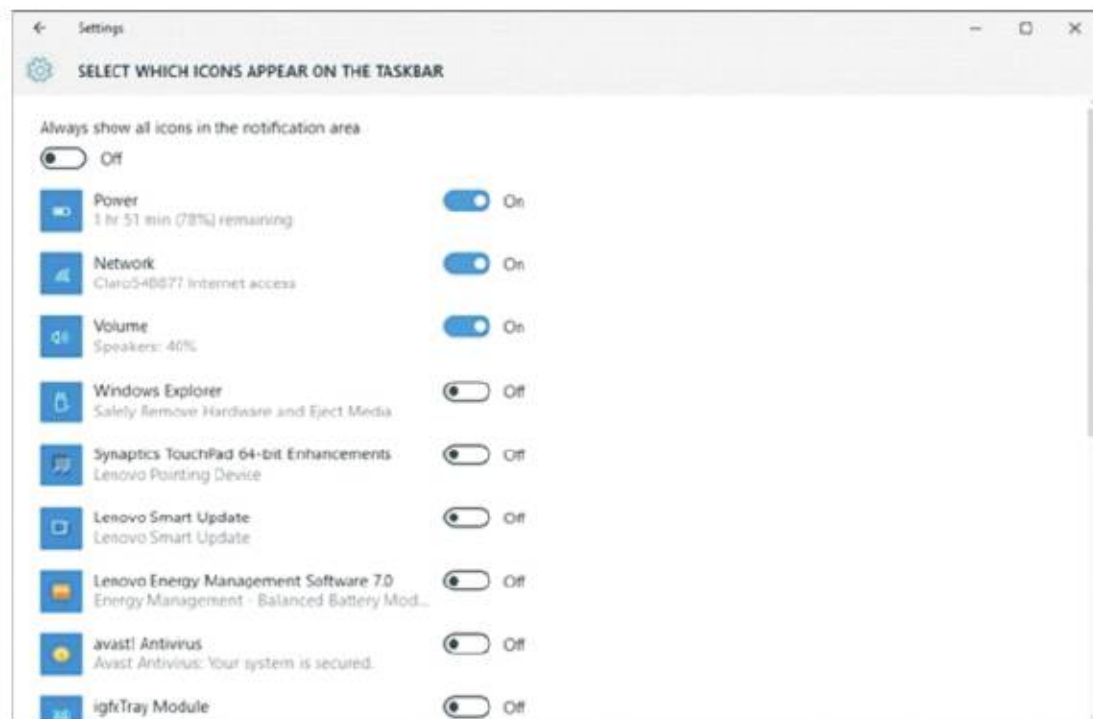
Step 1 - Go to the SETTINGS window and choose System.



Step 2 - In the SYSTEM window, select Notifications & actions. Here you can select the option that reads “Select which icons appear on the taskbar”.

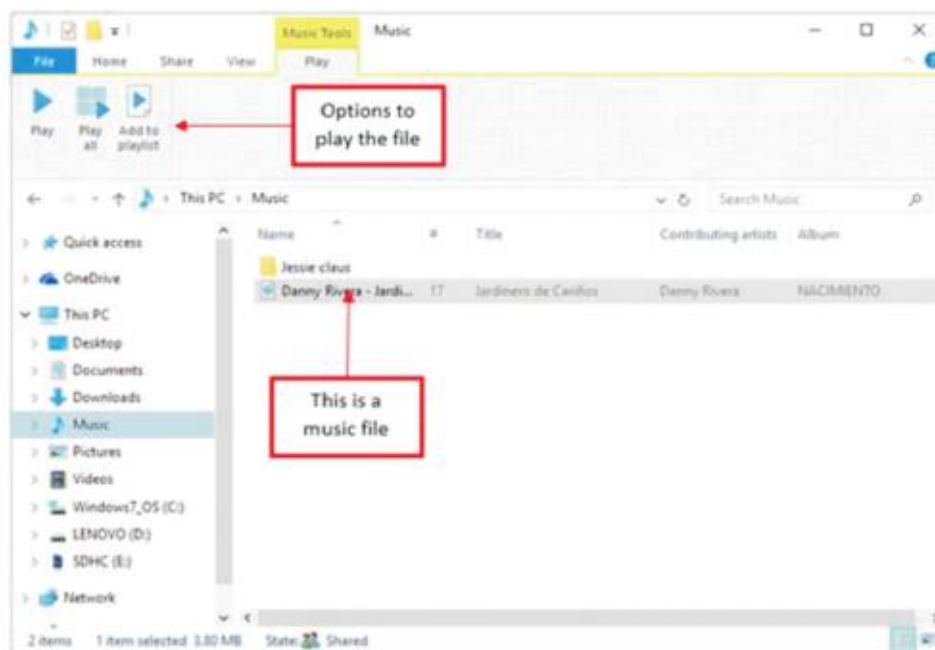


Step 3 - In the **SELECT WHICH ICONS APPEAR ON THE TASKBAR** window, you can turn on or off the icons in whichever way you prefer.



Windows 10 Media Management

Windows 10 offers multiple ways to work with your media files (music and videos). Some of the features needed to handle these files have been integrated into Windows to facilitate the way you access them. For example, if you select a music file, you will see the following ribbon on the top of your folder window.



You will see similar options when you select a video file.

Media Applications

Although you can have multiple media programs in your computer, Windows 10 includes several options for you to use.

For music files, Windows includes the traditional Windows Media Player. However, new Windows versions also include the Groove Music app (previously known as Xbox Music).



For video files, Windows also uses Windows Media Player, but now also includes the Movies & TV app.



Windows 10 also includes the Windows DVD Player app for playing DVD's in your computer.

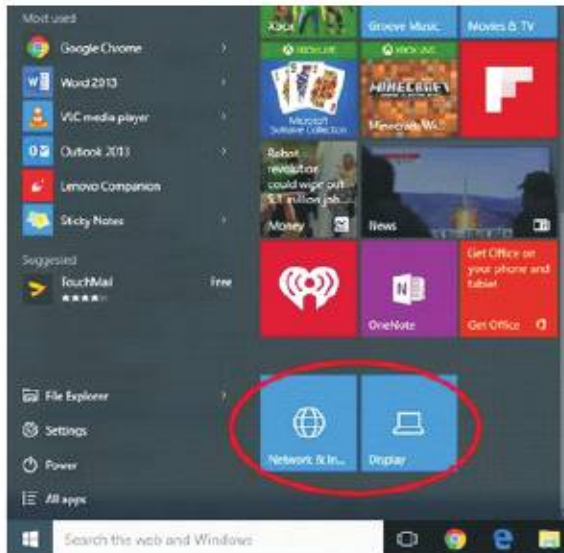
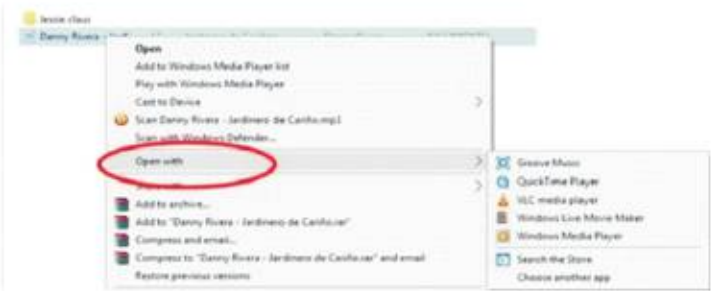


You can also choose which application you want to use to open your media file by following these steps -

Step 1 - Right-click any media file (music or video).

Step 2 - Select the Open with option.

Step 3 - Choose what application you want to use to open the media file.



Windows 10 - Favorite Settings

As we've seen already, Windows 10 uses the new Settings app for most of its configuration. And although the Settings app can be easily accessed through the Start menu, Windows 10 also allows you to “pin” your favorite setting apps to the Start menu for even easier access. A “pinned” setting app will appear directly in the tiles of your Start Menu.

Pin your Favorite Settings

To “pin” your favorite settings to the Start Menu, follow these steps -

Step 1 - Open the Settings window from the Start Menu.

Step 2 - In the Settings window, point to your favorite settings and right-click.



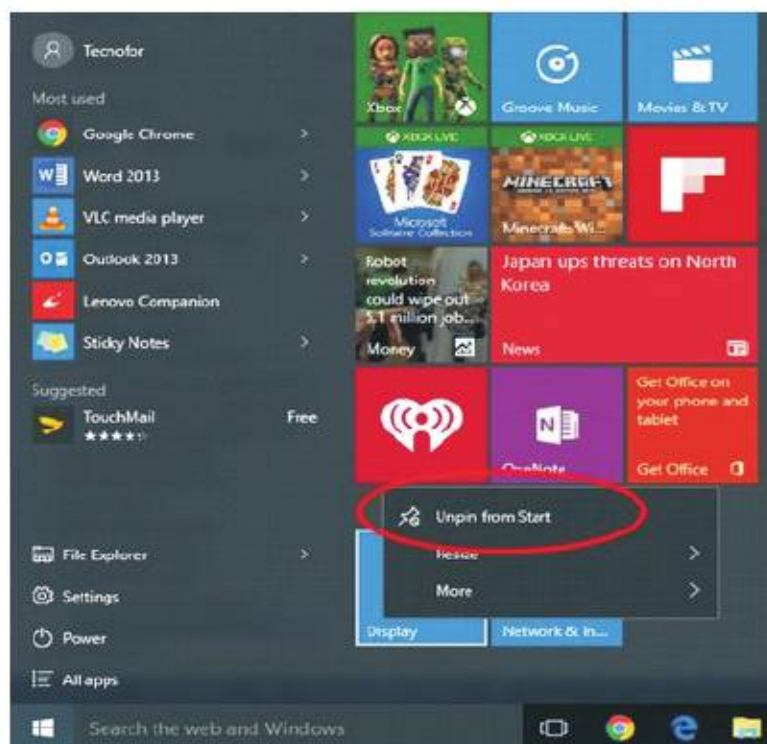
Step 3 - On the small menu that appears, select Pin to Start.

Step 4 - Click Yes on the screen to confirm the action. After this, you will see the Network & Internet app directly in the Start Menu.



Unpin your Favorite Settings

Alternatively, you can remove any pinned app from the Start Menu just by right-clicking on it and choosing Unpin from Start.



Must Know!



The System Tray is another name given to the Notification Area, which we can find in the right-side of the Windows Taskbar.

Windows 10 - Shortcuts

In Windows 10, a shortcut is simply an icon that directs you to an item (a document, application, or picture) that is located somewhere else in your computer. As the name implies, shortcuts make it easier for you to access that item

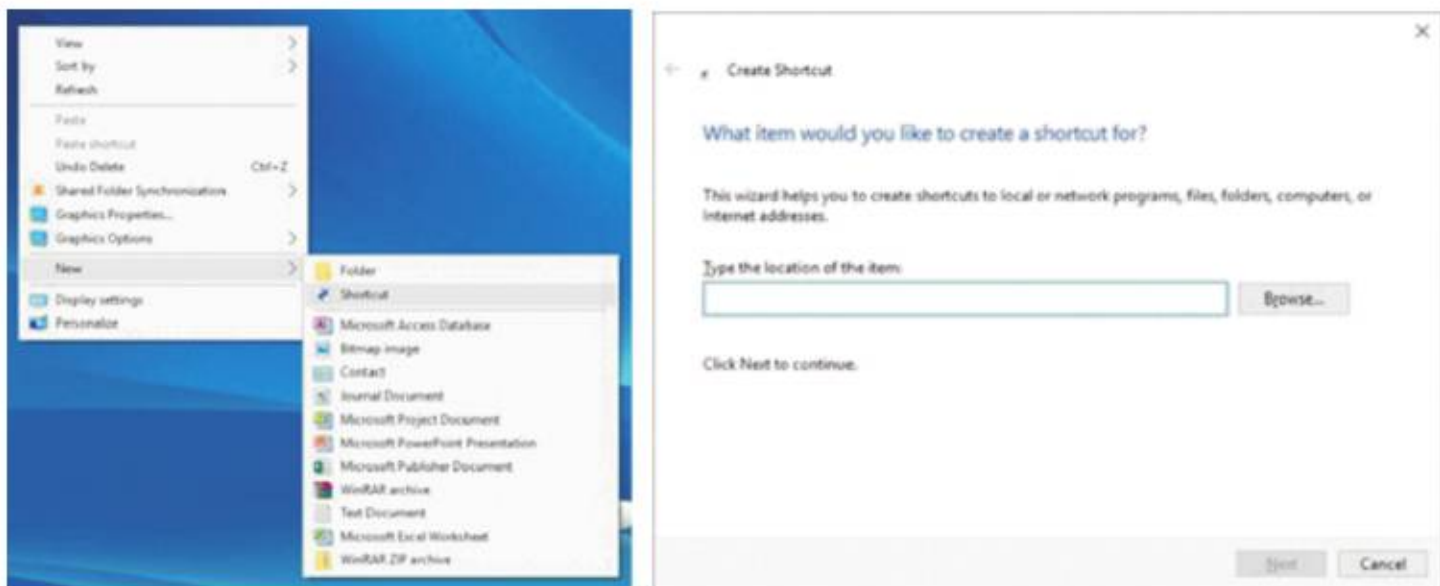


by being more accessible (usually on the Desktop). Shortcuts are identified by a small arrow in the corner of the icon.

Creating a Shortcut

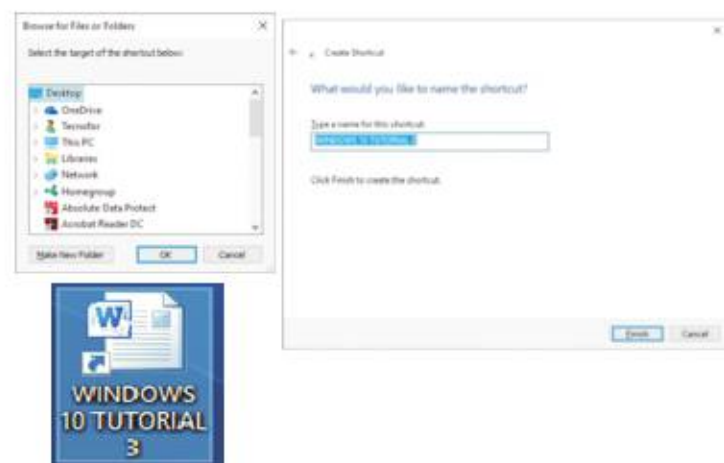
To create a shortcut in your Desktop, follow these steps -

- Step 1** - Position your mouse pointer over an empty space and right-click.
- Step 2** - In the menu, select New and then Shortcut.
- Step 3** - On the Create Shortcut window, you can either Browse or type the location of the item you want to access.



Choosing Browse will open another window, allowing you to search for the item in your computer.

- Step 4** - After choosing the item and clicking Next, you can assign or change the name with which the shortcut will be identified.
- Step 5** - After making any changes, click Finish. You will then see the shortcut in your Desktop.



Deleting the shortcut won't delete the file it accesses.



Work Station



A Tick(3) the correct option.

(NEP) Analytical Skills

- The easiest way to uninstall an application is from the:
(a) Program ☐ (c) Start Menu ☐ (b) Status bar ☐ (d) Control panel ☐
- A new and different way to uninstall an application is from the new:
(a) Control panel ☐ (c) Document window ☐
(b) Settings window ☐ (d) System tray ☐
- Takes a screenshot of your whole display and stores it in Pictures > Screenshots.
(a) Windowskey + PrintScreen ☐ (c) Windows key + Tab ☐
(b) Windowskey + D ☐ (d) Windows key + E ☐
- It allows you to automatically arrange your windows making it easier to work with multiple applications at the same time.
(a) Smapping windows ☐ (c) Snapping windows ☐
(b) Snaping windows ☐ (d) Snopping windows ☐
- It is another name given to the NotificationArea, which we can find in the right-side of the Windows Taskbar.
(a) Control Tray ☐ (c) Window Tray ☐
(b) Program Tray ☐ (d) System Tray ☐
- It is simply an icon that directs you to an item that is located somewhere else in your computer.
(a) Program ☐ (c) Shortcut ☐ (b) Control ☐ (d) Snapping ☐

B Fill in the blanks.

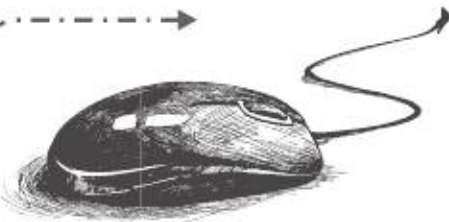
- Three ways in which you can remove an unwanted application: _____, _____ and _____.
- The third way to uninstall an application is more traditional, because it uses the popular _____.
- New Windows shortcuts use the _____ combined with other keys to perform several actions.
- Other useful shortcuts introduced in Windows 10 are the ones used for Windows _____.
- The _____ alerts from your computer like your Internet connection, or the volume level.
- New Windows versions also include the _____ previously known as Xbox Music.

C Write T for True or F for False.

1. An alternate way is to left-click the application on the list. ☐
2. You can access the Mail app directly from the Status Menu. ☐
3. Windows 10 offers multiple ways to work with your media files (music and videos). ☐
4. A “pinned” setting app will appear directly in the tiles of your Start Menu. ☐
5. Shortcuts are identified by a small arrow in the center of the icon. ☐
6. Windows 10 also allows you to “pin” your favorite setting apps to the Start menu for even easier access. ☐

D Answer the following questions.

1. What are the three ways to remove an unwanted application?
2. Write a note on Windows 10 - Email Management.
3. What is Window Snapping?
4. Define Windows 10 - System Tray.
5. Write a note on Windows 10 - Media Management.
6. Define Media Applications.
7. How to Pin and Unpin your Favorite Settings?
8. Define Windows 10 - Shortcuts.

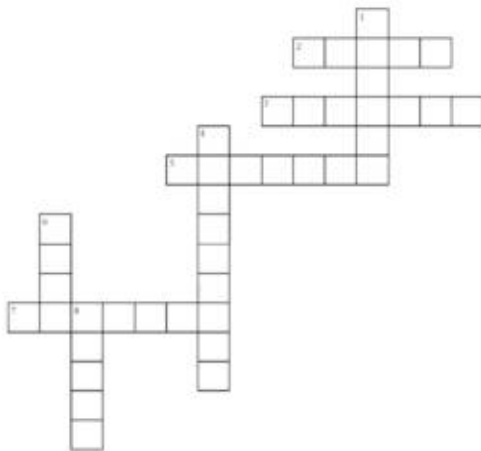


Work sheet



Use the hints given below to solve the crossword:

(NEP) Critical Thinking



Across	Down
2. These are small labelled pictures that appear on the desktop.	1. A place where different files are stored.
3. One of the most popular operating systems.	4. This is the picture in the background of the desktop.
5. The long bar that we see at the bottom of the desktop.	6. Whenever we do any work on a computer like typing a letter, drawing a picture, etc., we do it in the here.
7. The screen that appears when we switch on a computer.	8. This button is present at the bottom left of the screen on the taskbar.



MS-Excel (Formula and Functions)



Lesson at a Glance

Simple Formulas
Complex Formulas

Creating Formulas with Cell References
Working with cell references

Basic Functions



Simple Formulas

Formulas are the bread and butter of worksheet. Without formula, worksheet will be just simple tabular representation of data. Formulas use a variety of operators, functions to work with values and text. The values and text used in formulas can be located in other cells. This is called using a cell reference.

To perform basic mathematical operations, arithmetic operators are used. For example : (plus sign) for addition (+), minus sign for subtraction (-), asterisk for multiplication (*), forward slash for division (/), and caret (^) for exponents.

To create a simple formula in Excel :

A formula is an easy way to perform calculation. A formula always start with an equal sign (=). It must be written upto 1024 characters long. After writing the formula, Press Enter Key or Click the Enter Button (3) on the formula bar to display the result.

If the result of a formula is too large to be displayed in a cell, it may appear as pound signs (#####) instead of a value. This means the column is not wide enough to display the cell content. Simply increase the column width to show the cell content.

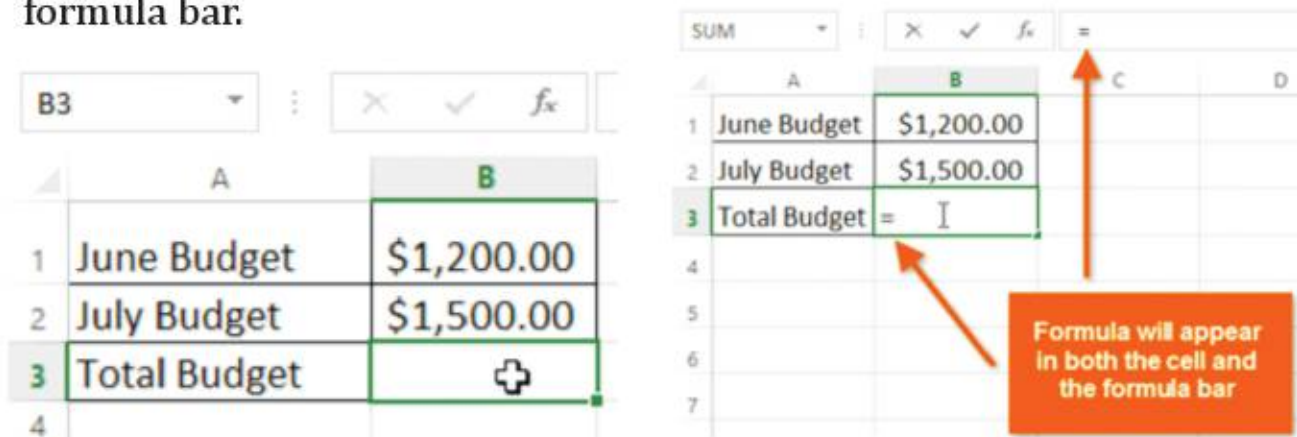
Creating Formulas with Cell References

When a formula contains a cell address, it is called a **cell reference**. Creating a formula with cell references is useful because you can update data in your worksheet without having to rewrite the values in the formula.

To create a formula :

Here below, we'll use a simple formula and cell references to calculate a budget.

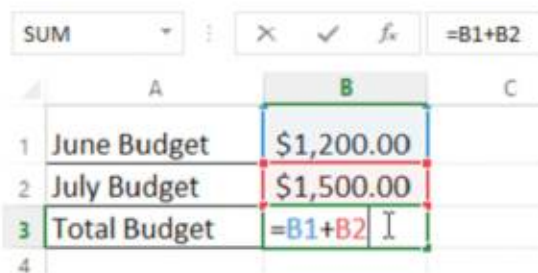
1. Select the cell that will contain the formula. Here, we'll select cell B3.
2. Type the equals sign (=). Notice how it appears in both the cell and the formula bar.



	A	B	C	D
1	June Budget	\$1,200.00		
2	July Budget	\$1,500.00		
3	Total Budget	=		
4				
5				
6				
7				

Formula will appear in both the cell and the formula bar

3. Type the cell address of the cell you want to reference first in the formula: cell B1 Here. A blue border will appear around the referenced cell.

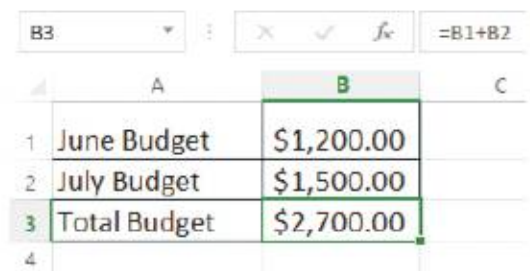


	A	B	C
1	June Budget	\$1,200.00	
2	July Budget	\$1,500.00	
3	Total Budget	=B1	
4			

4. Type the mathematical operator you want to use. Here, we'll type the addition sign (+).

5. Type the cell address of the cell you want to reference second in the formula: cell B2 Here. A red border will appear around the referenced cell.

6. Press Enter on your keyboard. The formula will be calculated, and the value will be displayed in the cell.

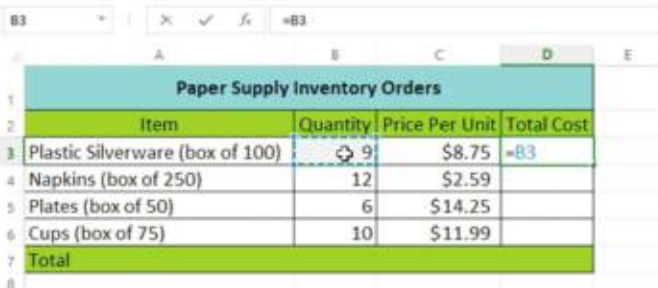


	A	B	C
1	June Budget	\$1,200.00	
2	July Budget	\$1,500.00	
3	Total Budget	\$2,700.00	
4			

To create a formula using the point-and-click method:

Rather than typing cell addresses manually, you can point and click on the cells you want to include in your formula. This method can save a lot of time and effort when creating formulas. Here, we'll create a formula to calculate the cost of ordering several boxes of plastic silverware.

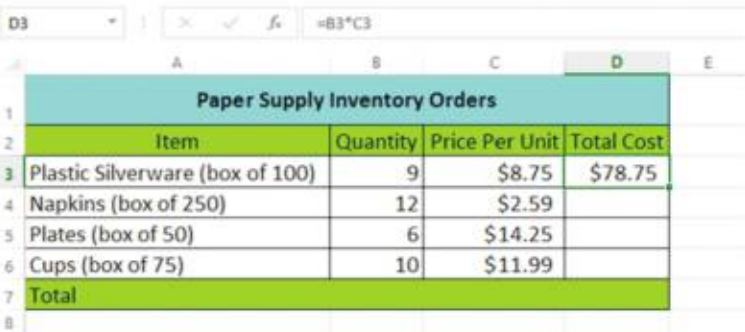
1. Select the cell that will contain the formula. Here, we'll select cell D3.



Paper Supply Inventory Orders			
Item	Quantity	Price Per Unit	Total Cost
Plastic Silverware (box of 100)	9	\$8.75	=B3
Napkins (box of 250)	12	\$2.59	
Plates (box of 50)	6	\$14.25	
Cups (box of 75)	10	\$11.99	
Total			

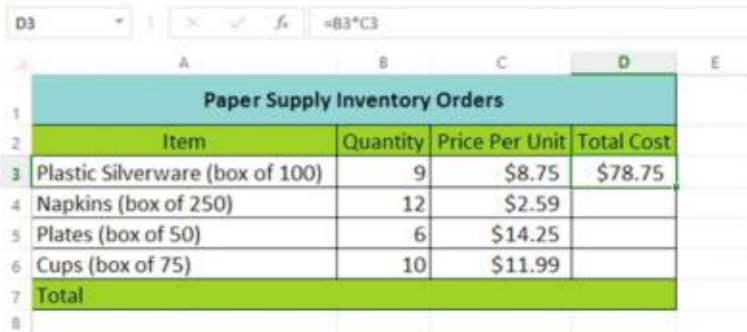
2. Type the equals sign (=).
3. Select the cell you want to reference first in the formula: cell B3 Here. The cell address will appear in the formula, and a dashed blue line will appear around the referenced cell.

4. Type the mathematical operator you want to use. Here, we'll type the multiplication sign (*).
5. Select the cell you want to reference second in the formula: cell C3 Here. The cell address will appear in the formula, and a dashed red line will appear around the referenced cell.



Paper Supply Inventory Orders			
Item	Quantity	Price Per Unit	Total Cost
Plastic Silverware (box of 100)	9	\$8.75	=B3*C3
Napkins (box of 250)	12	\$2.59	
Plates (box of 50)	6	\$14.25	
Cups (box of 75)	10	\$11.99	
Total			

6. Press Enter on your keyboard. The formula will be calculated, and the value will be displayed in the cell.



Paper Supply Inventory Orders			
Item	Quantity	Price Per Unit	Total Cost
Plastic Silverware (box of 100)	9	\$8.75	\$78.75
Napkins (box of 250)	12	\$2.59	
Plates (box of 50)	6	\$14.25	
Cups (box of 75)	10	\$11.99	
Total			

Complex Formulas

Complex formulas have more than one mathematical operation, such as $5+5-2$. To use Excel to calculate complex formulas, you'll need to understand the order of operations.

The order of operations

Excel calculates formulas based on the following order of operations:

1. Operations enclosed in parentheses
2. Exponential calculations (to the power of)
3. Multiplication and division, whichever comes first
4. Addition and subtraction, whichever comes first

To create a complex formula using the order of operations:

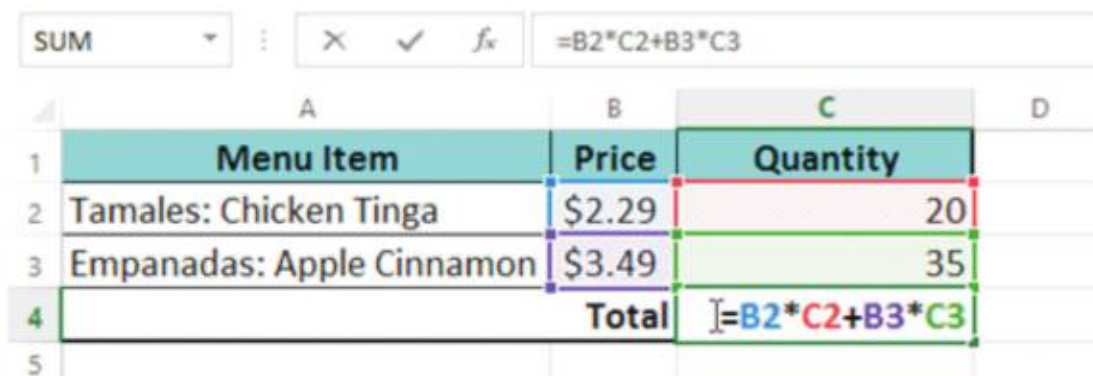
Here below, we will use cell references along with numerical values to create a complex formula that will calculate the total cost for a catering invoice. The formula will calculate the cost for each menu item and then add those values together.

1. Select the cell that will contain the formula. Here, we'll select cell C4.



	A	B	C	D
1	Menu Item	Price	Quantity	
2	Tamales: Chicken Tinga	\$2.29	20	
3	Empanadas: Apple Cinnamon	\$3.49	35	
4		Total		
5				

2. Enter your formula. Here, we'll type $=B2*C2+B3*C3$. This formula will follow the order of operations, first performing the multiplication: $2.29*20 = 45.80$ and $3.49*35 = 122.15$. It then will add those values together to calculate the total: $45.80+122.15$.



	A	B	C	D
1	Menu Item	Price	Quantity	
2	Tamales: Chicken Tinga	\$2.29	20	
3	Empanadas: Apple Cinnamon	\$3.49	35	
4		Total	$=B2*C2+B3*C3$	
5				

- Double-check your formula for accuracy, then press Enter on your keyboard. The formula will calculate and display the result. Here, the result shows that the total cost for the order is \$167.95.

C4 $\times$ $\checkmark$ f_x $=B2*C2+B3*C3$

	A	B	C	D
1	Menu Item	Price	Quantity	
2	Tamales: Chicken Tinga	\$2.29	20	
3	Empanadas: Apple Cinnamon	\$3.49	35	
4		Total	\$167.95	
5				

Working with Cell References

In order to maintain accurate formulas, it is necessary to understand how cell references respond when you copy or fill them to new cells in the worksheet.

Excel will interpret cell references as either relative or absolute.

Relative references

By default, all cell references are relative references. When copied across multiple cells, they change based on the relative position of rows and columns. For example, if you copy the formula $=A1+B1$ from row 1 to row 2, the formula will become $=A2+B2$. Relative references are especially convenient whenever you need to repeat the same calculation across multiple rows or columns.

To create and copy a formula using relative references:

- Select the cell that will contain the formula. Here, we'll select cell D2.

D2 $\times$ $\checkmark$ f_x

	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	\$2.99	15		
3	Empanadas: Chipotle Shrimp	\$3.99	10		
4	Empanadas: Black Bean & Plantain	\$2.49	20		
5	Tamales: Chicken Tinga	\$2.29	20		

C2 $\times$ $\checkmark$ f_x $=B2*C2$

	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	\$2.99	15	$=B2*C2$	
3	Empanadas: Chipotle Shrimp	\$3.99	10		
4	Empanadas: Black Bean & Plantain	\$2.49	20		
5	Tamales: Chicken Tinga	\$2.29	20		

- Enter the formula to calculate the desired value. Here, we'll type $=B2*C2$.

- Press Enter on your keyboard. The formula will be calculated, and the result will be displayed in the cell.

- Locate the fill handle in the bottom-right corner of the desired cell. Here, we'll locate the fill handle for cell D2.

D2 $\times$ $\checkmark$ f_x $=B2*C2$

	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	\$2.99	15	\$44.85	
3	Empanadas: Chipotle Shrimp	\$3.99	10		
4	Empanadas: Black Bean & Plantain	\$2.49	20		
5	Tamales: Chicken Tinga	\$2.29	20		
6	Tamales: Vegetable	\$2.29	30		
7	Arepas: Carnitas	\$2.89	10		
8	Arepas: Queso Blanco	\$2.49	20		

The fill handle

- Click, hold, and drag the fill handle over the cells you want to fill. Here, we'll select cells D3:D12.

	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	\$2.99	15	\$44.85	
3	Empanadas: Chipotle Shrimp	\$3.99	10	\$39.90	
4	Empanadas: Black Bean & Plantain	\$2.49	20	\$49.80	
5	Tamales: Chicken Tinga	\$2.29	20	\$45.80	
6	Tamales: Vegetable	\$2.29	30	\$68.70	
7	Arepas: Carnitas	\$2.89	10	\$28.90	
8	Arepas: Queso Blanco	\$2.49	20	\$49.80	
9	Empanadas: Apple Cinnamon	\$3.19	40	\$127.60	
10	Beverages: Horchata	\$1.89	25	\$47.25	
11	Beverages: Lemonade	\$1.89	35	\$66.15	
12	Beverages: Tamarindo	\$1.89	10	\$18.90	
13	Total				

	A	B	C	D	E
1	Menu Item	Price	Quantity	Total	
2	Empanadas: Beef Picadillo	\$2.99	15	\$44.85	
3	Empanadas: Chipotle Shrimp	\$3.99	10		
4	Empanadas: Black Bean & Plantain	\$2.49	20		
5	Tamales: Chicken Tinga	\$2.29	20		
6	Tamales: Vegetable	\$2.29	30		
7	Arepas: Carnitas	\$2.89	10		
8	Arepas: Queso Blanco	\$2.49	20		
9	Empanadas: Apple Cinnamon	\$3.19	40		
10	Beverages: Horchata	\$1.89	25		
11	Beverages: Lemonade	\$1.89	35		
12	Beverages: Tamarindo	\$1.89	10		
13	Total				

- Release the mouse. The formula will be copied to the selected cells with relative references, and the values will be calculated in each cell.

Absolute references

There may be times when you do not want a cell reference to change when copying or filling cells. An absolute reference is designated in the formula by the addition of a dollar sign (\$). It can precede the column reference, the row reference, or both.

\$A\$2:	The column and the row do not change when copied.
A\$2:	The row does not change when copied.
\$A2:	The column does not change when copied.

To create and copy a formula using absolute references:

	A	B	C	D	E
1	Sales Tax				7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	\$2.99	15		
4	Empanadas: Chipotle Shrimp	\$3.99	10		
5	Empanadas: Black Bean & Plantain	\$2.49	20		
6	Tamales: Chicken Tinga	\$2.29	20		

- Enter the formula to calculate the desired value. $= (B3 * C3) * \$E\1 .

- Select the cell that will contain the formula. Here, we'll select cell D3.

	A	B	C	D	E
1	Sales Tax				7.5%
2	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	\$2.99	15	$= (B3 * C3) * \$E\1	
4	Empanadas: Chipotle Shrimp	\$3.99	10		
5	Empanadas: Black Bean & Plantain	\$2.49	20		
6	Tamales: Chicken Tinga	\$2.29	20		

- Press Enter on your keyboard. The formula will calculate, and the result will display in the cell.
- Locate the fill handle in the bottom-right corner of the desired cell. Here, we'll locate the fill handle for cell D3.

Formula bar: `=B3*C3*$E$1`

	A	B	C	D	E
				Sales Tax	7.5%
	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	\$2.99	15	\$3.36	
4	Empanadas: Chipotle Shrimp	\$3.99	10		
5	Empanadas: Black Bean & Plantain	\$2.49	20		
6	Tamales: Chicken Tinga	\$2.29	20		
7	Tamales: Vegetable	\$2.29	30		
8	Arepas: Carnitas	\$2.89	10		
9	Arepas: Queso Blanco	\$2.49	20		

The fill handle is located in the bottom-right corner of cell D3.

Formula bar: `=B3*C3*$E$1`

Click, hold and drag the fill handle to copy the formula to adjacent cells

	A	B	C	D	E
				Sales Tax	Total
	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	\$2.99	15	\$3.36	
4	Empanadas: Chipotle Shrimp	\$3.99	10		
5	Empanadas: Black Bean & Plantain	\$2.49	20		
6	Tamales: Chicken Tinga	\$2.29	20		
7	Tamales: Vegetable	\$2.29	30		
8	Arepas: Carnitas	\$2.89	10		
9	Arepas: Queso Blanco	\$2.49	20		
10	Empanadas: Apple Cinnamon	\$3.19	40		
11	Beverages: Horchata	\$1.89	25		
12	Beverages: Lemonade	\$1.89	35		
13	Beverages: Tamarindo	\$1.89	10		
14					Total

- Click, hold, and drag the fill handle over the cells you want to fill: cells D4:D13 Here.

Formula bar: `=B3*C3*$E$1`

	A	B	C	D	E
				Sales Tax	7.5%
	Menu Item	Price	Quantity	Sales Tax	Total
3	Empanadas: Beef Picadillo	\$2.99	15	\$3.36	
4	Empanadas: Chipotle Shrimp	\$3.99	10	\$2.99	
5	Empanadas: Black Bean & Plantain	\$2.49	20	\$3.74	
6	Tamales: Chicken Tinga	\$2.29	20	\$3.44	
7	Tamales: Vegetable	\$2.29	30	\$5.15	
8	Arepas: Carnitas	\$2.89	10	\$2.17	
9	Arepas: Queso Blanco	\$2.49	20	\$3.74	
10	Empanadas: Apple Cinnamon	\$3.19	40	\$9.57	
11	Beverages: Horchata	\$1.89	25	\$3.54	
12	Beverages: Lemonade	\$1.89	35	\$4.96	
13	Beverages: Tamarindo	\$1.89	10	\$1.42	
14					Total

- Release the mouse. The formula will be copied to the selected cells with an absolute reference, and the values will be calculated in each cell.

Must Know!



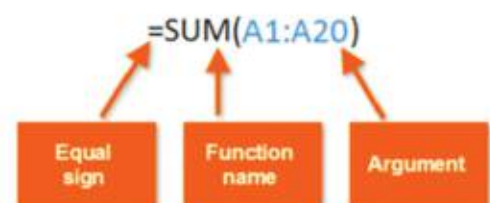
Cell references are the address of cells that contains some values.

Basic Functions

A function is a predefined formula that performs calculations using specific values in a particular order. One of the key benefits of functions is that they can save your time because you do not have to write the formula yourself. Excel has hundreds of functions to assist with your calculations.

The parts of a function

In order to work correctly, a function must be written a specific way, which is called the syntax. The basic syntax for a function is the equals sign



(=), the function name (SUM, for example), and one or more arguments. The function in the example below would add the values of the cell range A1:A20.

Working with arguments

Arguments can refer to both individual cells and cell ranges and must be enclosed within parentheses. You can include one argument or multiple arguments, depending on the syntax required for the function.

For example, the function `=AVERAGE(B1:B9)` would calculate the average of the values in the cell range B1:B9. This function contains only one argument.

Excel screenshot showing the formula bar with `=AVERAGE(B1:B9)` and a data table with values in column B.

	A	B
1		5
2		8
3		9
4		7
5		5
6		1
7		3
8		2
9		7
10		<code>=AVERAGE(B1:B9)</code>

Excel screenshot showing the formula bar with `=SUM(A1:A3,C1:C2,E1)` and a data table with values in columns A, C, and E.

	A	B	C	D	E
1	7		5		15
2	4		12		
3	23				
4					
5					

Multiple arguments must be separated by a comma. For example, the function `=SUM(A1:A3, C1:C2, E1)` will add the values of all the cells in the three arguments.

To create a basic function in Excel:

1. Select the cell that will contain the function. Here, we'll select cell C11.

Excel screenshot showing a table with an Autosum feature being applied to cell C11.

	A	B	C	D	E
1	Food Supply Inventory Orders (Non-Perishable Items)				
2	Item	Quantity	Price Per Unit	Total Cost	Date Ordered
3	Tomatoes (case of 12)	3	\$17.44	\$52.32	
4	Black Beans (case of 10)	5	\$20.14	\$100.70	
5	All Purpose Flour (50 lb.)	5	\$14.05	\$70.25	
6	Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45	
7	Brown Rice (25 lb.)	5	\$10.99	\$54.95	
8	Salsa, Medium (1 gallon)	12	\$8.47	\$101.64	
9	Olive Oil (2.5 gallon)	4	\$28.69	\$114.76	
10	Vegetable Wash (1 gallon)	2	\$8.99	\$17.98	
11	Average Price Per Unit				
12	Total Cost				
13					

Must Know!



Autosum feature quickly adds the data values stored in a row or column.

- Type the equals sign (=) and enter the desired function name. You can also select the desired function from the list of suggested functions that will appear below the cell as you type. Here, we'll type =AVERAGE.

COUNTA X ✓ f_x =AVERAGE(C3:C10)

Food Supply Inventory Orders (Non-Perishable Items)				
Item	Quantity	Price Per Unit	Total Cost	Date Ordered
Tomatoes (case of 12)	3	\$17.44	\$52.32	
Black Beans (case of 10)	5	\$20.14	\$100.70	
All Purpose Flour (50 lb.)	5	\$14.05	\$70.25	
Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45	
Brown Rice (25 lb.)	5	\$10.99	\$54.95	
Salsa, Medium (1 gallon)	12	\$8.47	\$101.64	
Olive Oil (2.5 gallon)	4	\$28.69	\$114.76	
Vegetable Wash (1 gallon)	2	\$8.99	\$17.98	
Average Price Per Unit		=AVERAGE(C3:C10)		
Total Cost				

- Press Enter on your keyboard. The function will be calculated, and the result will appear in the cell. Here, the average price per unit of items ordered was \$15.93.

To create a function using the Auto Sum command:

D12 X ✓ f_x

Food Supply Inventory Orders (Non-Perishable Items)				
Item	Quantity	Price Per Unit	Total Cost	Date Ordered
Tomatoes (case of 12)	3	\$17.44	\$52.32	
Black Beans (case of 10)	5	\$20.14	\$100.70	
All Purpose Flour (50 lb.)	5	\$14.05	\$70.25	
Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45	
Brown Rice (25 lb.)	5	\$10.99	\$54.95	
Salsa, Medium (1 gallon)	12	\$8.47	\$101.64	
Olive Oil (2.5 gallon)	4	\$28.69	\$114.76	
Vegetable Wash (1 gallon)	2	\$8.99	\$17.98	
Average Price Per Unit		\$15.93		
Total Cost				

COUNTA X ✓ f_x =AVERAGE

Food Supply Inventory Orders (Non-Perishable Items)				
Item	Quantity	Price Per Unit	Total Cost	Date Ordered
Tomatoes (case of 12)	3	\$17.44	\$52.32	
Black Beans (case of 10)	5	\$20.14	\$100.70	
All Purpose Flour (50 lb.)	5	\$14.05	\$70.25	
Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45	
Brown Rice (25 lb.)	5	\$10.99	\$54.95	
Salsa, Medium (1 gallon)	12	\$8.47	\$101.64	
Olive Oil (2.5 gallon)	4	\$28.69	\$114.76	
Vegetable Wash (1 gallon)	2	\$8.99	\$17.98	
Average Price Per Unit		=AVERAGE		

Type the function name or select it from the list of suggested functions

- AVERAGE Returns the average (arithmetic mean)
- AVERAGEA
- AVERAGEIF
- AVERAGEIFS

- Enter the cell range for the argument inside parentheses. Here, we'll type (C3:C10). This formula will add the values of cells C3:C10 and then divide that value by the total number of cells in the range to determine the average.

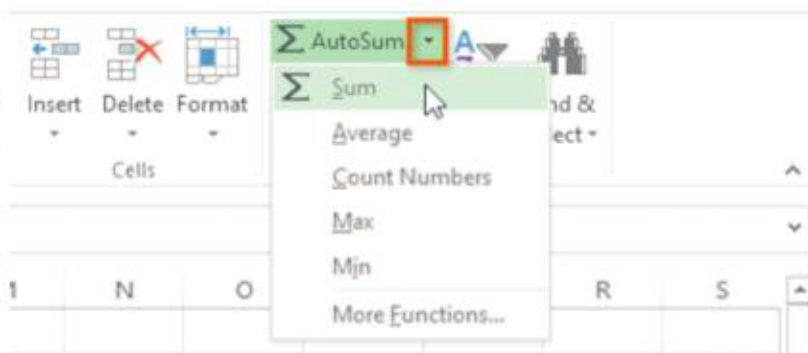
C11 X ✓ f_x =AVERAGE(C3:C10)

Food Supply Inventory Orders (Non-Perishable Items)				
Item	Quantity	Price Per Unit	Total Cost	Date Ordered
Tomatoes (case of 12)	3	\$17.44	\$52.32	
Black Beans (case of 10)	5	\$20.14	\$100.70	
All Purpose Flour (50 lb.)	5	\$14.05	\$70.25	
Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45	
Brown Rice (25 lb.)	5	\$10.99	\$54.95	
Salsa, Medium (1 gallon)	12	\$8.47	\$101.64	
Olive Oil (2.5 gallon)	4	\$28.69	\$114.76	
Vegetable Wash (1 gallon)	2	\$8.99	\$17.98	
Average Price Per Unit		\$15.93		
Total Cost				

The Auto Sum command allows you to automatically insert the most common functions into your formula, including SUM, AVERAGE, COUNT, MIN, and MAX.

- Select the cell that will contain the function. Here, we'll select cell D12.

- In the Editing group on the Home tab, locate and select the arrow next to the AutoSum command and then choose the desired function from the drop-down menu. Here, we'll select Sum.



- The selected function will appear in the cell. If logically placed, the AutoSum command will automatically select a cell range for the argument. Here, cells D3:D11 were selected automatically and their values will be added together to calculate the total cost. You can also manually enter the desired cell range into the argument.

COUNTA					=SUM(D3:D11)				
	A	B	C	D	E				
1	Food Supply Inventory Orders (Non-Perishable Items)								
2	Item	Quantity	Price Per Unit	Total Cost	Date Ordered				
3	Tomatoes (case of 12)	3	\$17.44	\$52.32					
4	Black Beans (case of 10)	5	\$20.14	\$100.70					
5	All Purpose Flour (50 lb.)	5	\$14.05	\$70.25					
6	Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45					
7	Brown Rice (25 lb.)	5	\$10.99	\$54.95					
8	Salsa, Medium (1 gallon)	12	\$8.47	\$101.64					
9	Olive Oil (2.5 gallon)	4	\$28.69	\$114.76					
10	Vegetable Wash (1 gallon)	2	\$8.99	\$17.98					
11	Average Price Per Unit		\$15.93						
12	Total Cost			=SUM(D3:D11)					
13						SUM(number1, [number2], ...)			
14									

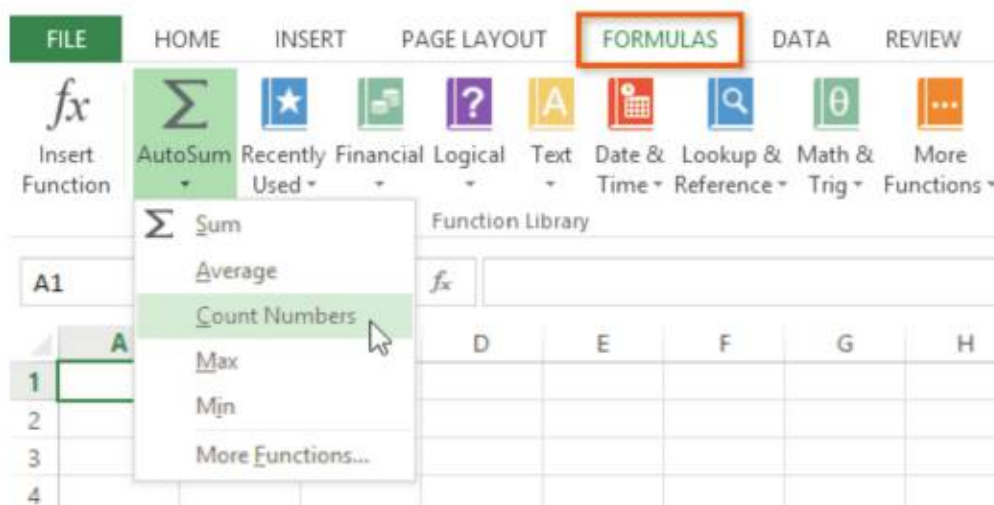
- Press Enter on your keyboard. The function will be calculated, and the result will appear in the cell. Here, the sum of D3:D11 is \$606.05.

D12					=SUM(D3:D11)				
	A	B	C	D	E				
1	Food Supply Inventory Orders (Non-Perishable Items)								
2	Item	Quantity	Price Per Unit	Total Cost	Date Ordered				
3	Tomatoes (case of 12)	3	\$17.44	\$52.32					
4	Black Beans (case of 10)	5	\$20.14	\$100.70					
5	All Purpose Flour (50 lb.)	5	\$14.05	\$70.25					
6	Corn Meal/Maza (25 lb.)	5	\$18.69	\$93.45					
7	Brown Rice (25 lb.)	5	\$10.99	\$54.95					
8	Salsa, Medium (1 gallon)	12	\$8.47	\$101.64					
9	Olive Oil (2.5 gallon)	4	\$28.69	\$114.76					
10	Vegetable Wash (1 gallon)	2	\$8.99	\$17.98					
11	Average Price Per Unit		\$15.93						
12	Total Cost			\$606.05					
13									

Must Know!



The AutoSum command can also be accessed from the Formulas tab on the Ribbon.



Work Station



A Tick (3) the correct option.

(NEP) Analytical Skills

- It is the bread and butter of worksheet.
 - Dollar sign ☐
 - Formula ☐
 - Equal sign ☐
 - Reference ☐
- The values and text used in formulas can be located in other cells.
 - Cellrange ☐
 - Function ☐
 - Parentheses ☐
 - Cell reference ☐
- When writing formulas for Excel is that all formulas must begin with an:
 - Plus sign ☐
 - Minus sign ☐
 - Equals sign ☐
 - Divide sign ☐
- You can use this reference to keep a row and/or column constant in the formula.
 - Relative ☐
 - Absolute ☐
 - Syntax ☐
 - Mixed ☐
- An absolute reference is designated in the formula by the addition of a:
 - Hash sign ☐
 - Star sign ☐
 - Pound sign ☐
 - Dollar sign ☐
- It is a predefined formula that performs calculations using specific values in a particular order.
 - Reference ☐
 - Syntax ☐
 - AutoSum ☐
 - Function ☐

B Fill in the blanks.

1. A _____ is an equation that performs a calculation.
2. If the result of a formula is too large to be displayed in a cell, it may appear as _____ instead of a value.
3. When a formula contains a cell address, it is called a _____.
4. Excel will interpret cell references as either _____ or _____.
5. Each function has a specific order called _____.
6. The basic syntax to create a formula with a function is to insert an _____, _____ and _____.

C Write T for True or F for False.

1. Excel can be used to calculate numerical information. ☐
2. Excel calculates formulas based on the following order of operations. ☐
3. If you copy a formula (=A1+B1) into row 2, the formula will change to become (=A1+B2). ☐
4. Mixed references can save you time when you're repeating the same type of calculation across multiple rows or columns. ☐
5. Arguments must be enclosed in parentheses. ☐
6. The AutoSum command can also be accessed from the function tab. ☐

D Match the followings.

- | | |
|------------------------|--|
| 1. \$A\$2 | a. Relative references. |
| 2. A\$2 | b. Do not change when they are copied. |
| 3. Colons | c. Column and row do not change when copied. |
| 4. Commas | d. The row does not change when copied. |
| 5. Cell references | e. The column does not change when copied. |
| 6. Absolute references | f. Separated by either colons or commas. |
| 7. \$A2 | g. Create a reference to a range of cells. |
| 8. Parentheses | h. Separate individual values. |

E Answer the following questions.

1. What is the use of cell references?
2. What is relative or absolute cell reference?
3. Differentiate between formula and function.
4. What is the use of equal sign in creating a formula?
5. What is parentheses?
6. What is the benefit of AutoSum?

LAB ACTIVITY



Create a spreadsheet using the following information to generate a stationery order for the month of March.

(NEP) Experimental Learning

- i. Enter the data given below, applying as many presentation features (Font, Size, Color, Number Formats, Cell Shading, Text Rotation, etc) to it as you wish.

Description	Quantity (Units)	Cost Per Unit
Note Pad	202	2.85
Highlighter Pen	280	2.95
Ball Point Blue	520	2.50
Ball Point Red	340	2.50
Ball Point Green	250	2.50
Exercise Book	35	4.95
Cello Tape	75	2.95
Manila Folders	100	2.95

- ii. Apply the appropriate number formats to numbers.
- iii. Give your spreadsheet an appropriate title and centre it across your spreadsheet.
- iv. Create formulas to calculate total amount for each stationery item.
- v. Create a formula to calculate total order amount for the month.
- vi. Create a formula to calculate the G.S.T or V.A.T charges of the order as 12.5% of the order amount.

Working with Presentation

5

Lesson at a Glance

Inserting Images
Transitions

Inserting Audio
Slide Show

Inserting Video

A presentation is an act of showing and explaining the contents of a topic to the group of people. It allows you to create presentations by using text, pictures, audio, video, sound, animation, charts and graphics with a number of special effects.

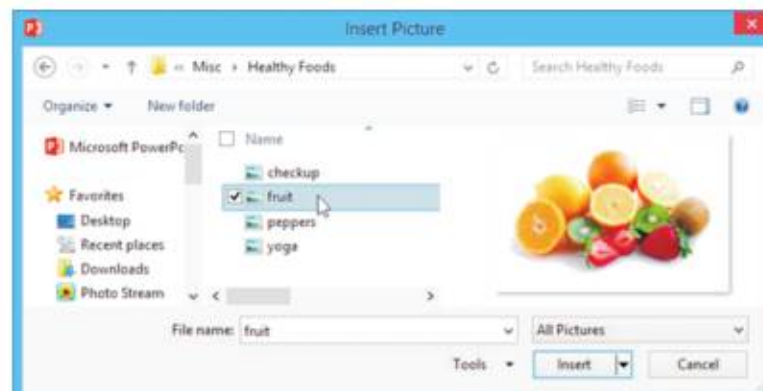
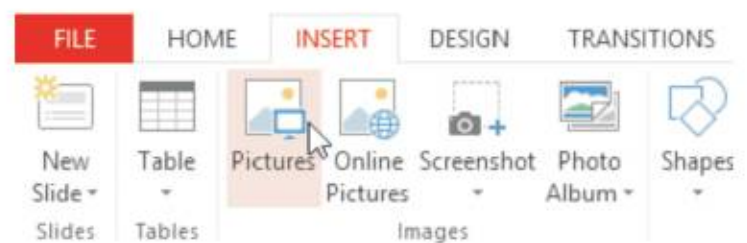
Inserting Images

If you would like to add an additional dimension to a presentation you can add Images to you slides. Images are another way to include graphics into presentation.

Many people like using this feature to incorporate pictures from their personal collections to distinguish their presentation.

To insert a picture from a file:

1. Select the **Insert** tab, then click the **Pictures** command in the **Images** group.



2. A dialog box will appear. Locate and select the desired image file, then click Insert.

3. The picture will appear on the currently selected slide.

New Wellness Program

- ▶ Geb BioFuels will be implementing its new Wellness Program in January.
- ▶ The program will provide resources and opportunities for employees to improve their overall health.



You can also click the Pictures command in a placeholder to insert images.

Must Know!



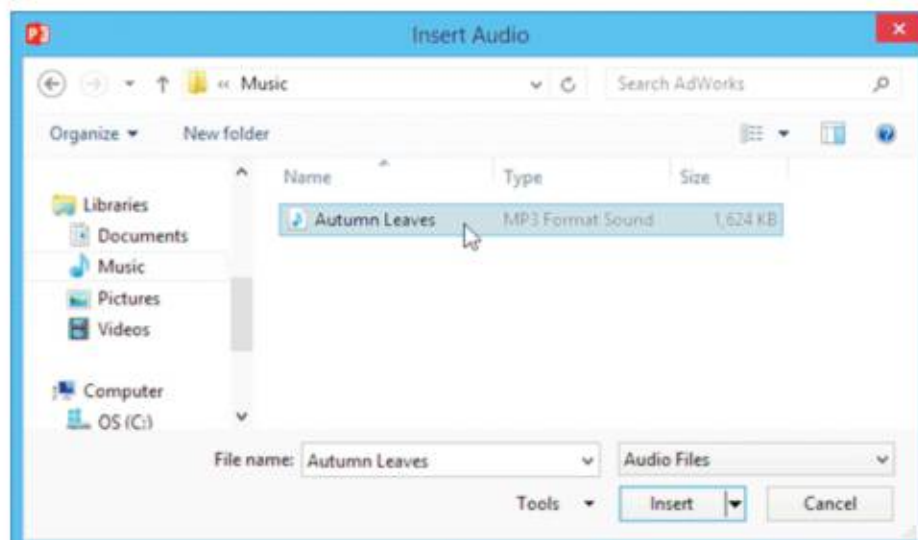
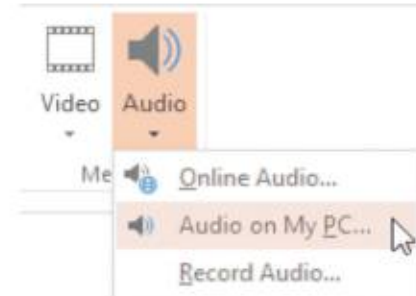
When inserting a graphic using the icon in a placeholder causes the graphic to be automatically sized and positioned.

Inserting Audio

PowerPoint allows you to add sound to your presentation, which opens up many possibilities. You could add background music to one slide, a sound effect to another, and even record your own narration or commentary. You can then edit the audio to customize it for your presentation.

To insert audio from a file :

1. From the Insert tab, click the Audio drop-down arrow, then select Audio on My PC.
2. Locate and select the desired audio file, then click Insert.

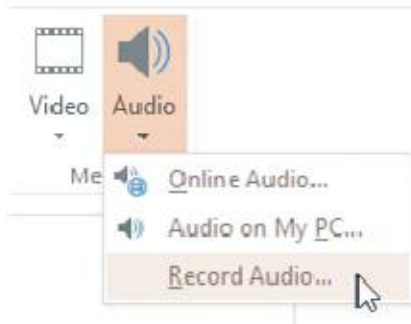


3. The audio file will be added to the slide.

Recording your own audio

Sometimes you may want to record audio directly into a presentation. For example, you might want the presentation to include narration. Before you begin, make sure you have a microphone that is compatible with your computer; many computers have built-in microphones or ones that can be plugged in to the computer.

To record audio :



1. From the Insert tab, click the Audio drop-down arrow, then select Record Audio.

2. Type a name for the audio recording, if desired.

3. Click the Record button to start recording.



4. When you're finished recording, click the Stop button.

5. To preview your recording, click the Play button.



6. When you're done, click OK. The audio file will be inserted into the slide.

To preview an audio file :

1. Click an audio file to select it.
2. Click the Play/Pause button below the audio file. The sound will begin playing, and the timeline next to the Play/Pause button will advance.



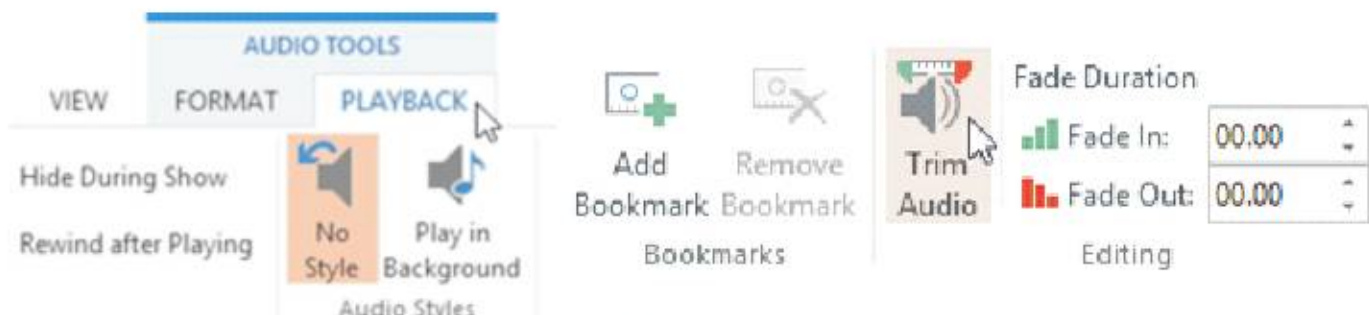
3. To jump to a different part of the file, click anywhere on the timeline.

Editing audio

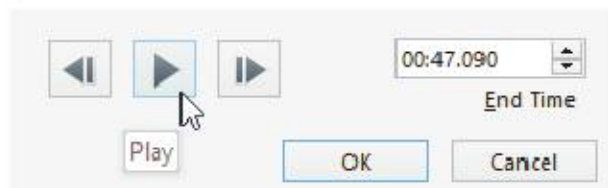
You can modify your audio files using the commands on the Playback tab. For example, if you add a song to a slide you can trim it to play only a brief section. You can also add a fade in and fade out, and add bookmarks that allow you to jump to specific points in the audio file.

To trim an audio file :

1. Select the audio file, then click the **Playback** tab on the Ribbon.
2. Click the **Trim Audio** command.



3. A dialog box will appear. Use the green handle to set the start time and the red handle to set the end time.

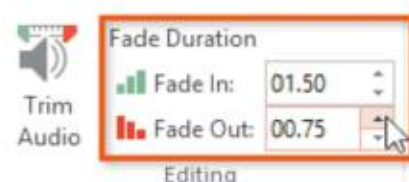


4. To preview the audio file, click the Play button.

5. Adjust the handles again if necessary, then click OK.

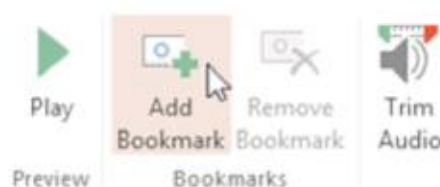
To add a fade in and fade out :

1. On the Playback tab, locate the Fade In: and Fade Out: fields.
2. Type the desired values, or use the up and down arrows to adjust the times.

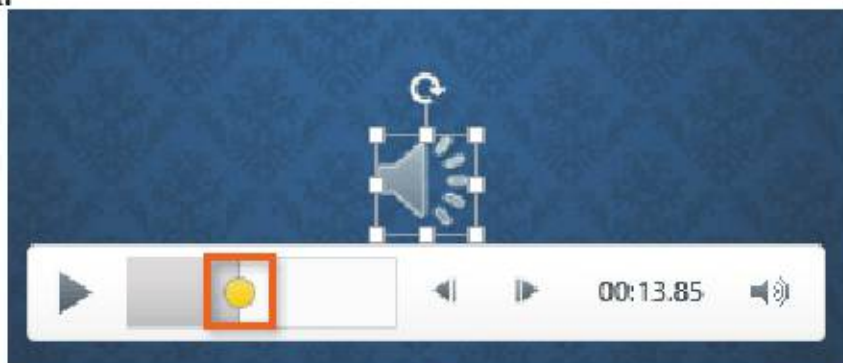


To add a bookmark :

1. Click the timeline to locate the desired part of the audio file.
2. From the Playback tab, click the Add Bookmark command.

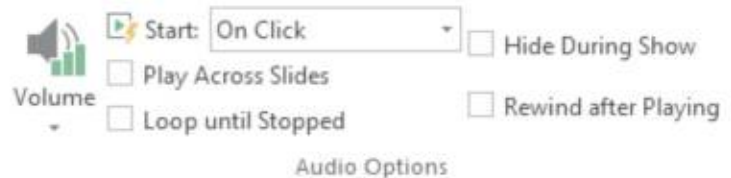


3. The bookmark will appear on the timeline. Click the bookmark to jump to that location.



Audio options

There are other options you can set to control how your audio file will play. These are found in the Audio Options group on the Playback tab.



- **Volume** : Changes the audio volume.
- **Start** : Controls whether the audio file starts automatically or when the mouse is clicked.
- **Hide During Show**: Hides the audio icon while the slide show is playing.
- **Play Across Slides** : Continues playing the audio file across multiple slides instead of just the current slide.
- **Loop until Stopped** : Replays the audio file until stopped.
- **Rewind after Playing** : Returns the audio file to the beginning when it is finished playing.

Must Know!

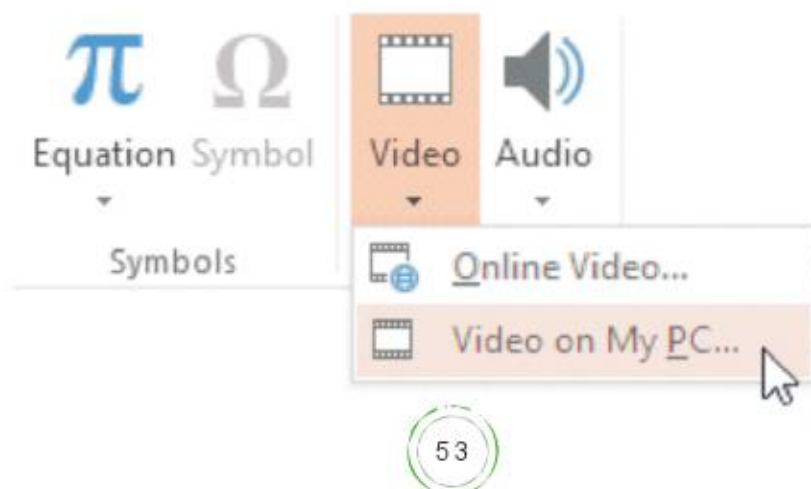


Not all audio file formats are compatible with PowerPoint 2007, but AU, AIFF, MP3, MIDI, WAV, and WMA are acceptable formats.

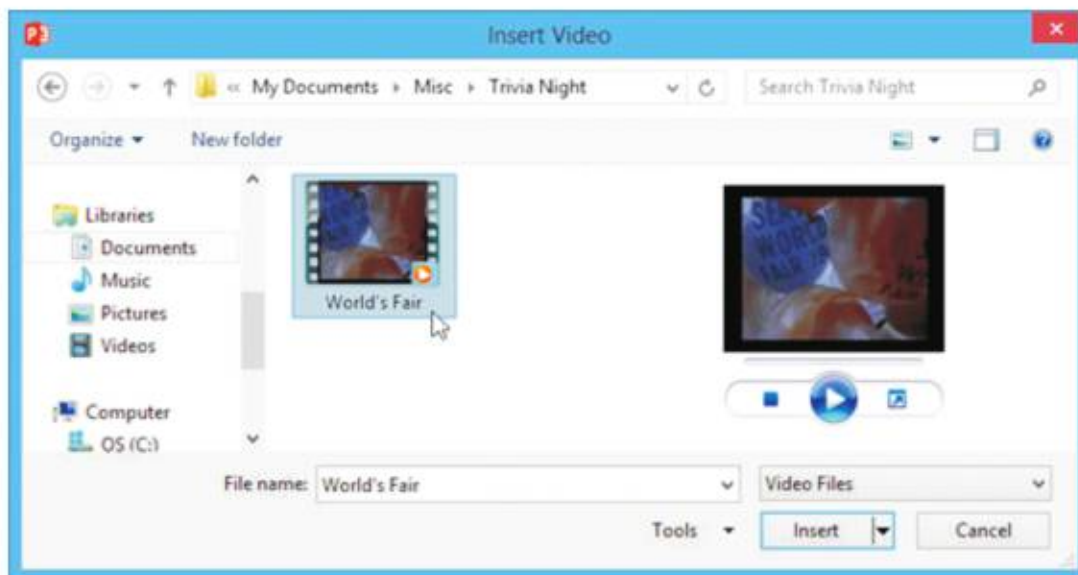
Inserting Video

PowerPoint allows you to insert a video onto a slide and play it during your presentation. This is a great way to make your presentation more engaging for your audience. You can even edit the video within PowerPoint and customize its appearance.

To insert a video from a file:



1. From the Insert tab, click the Video drop-down arrow, then select Video on My PC.
2. Locate and select the desired video file, then click Insert.



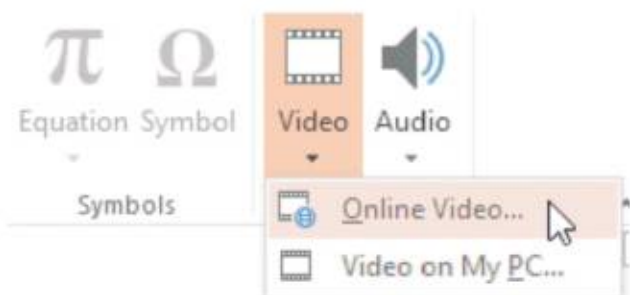
3. The video will be added to the slide.

To insert an online video:

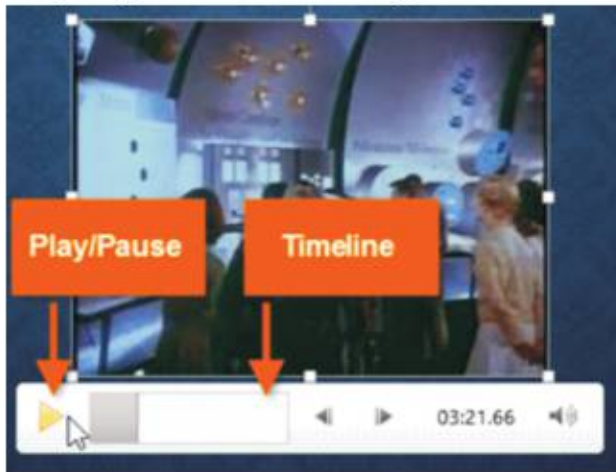
Some websites like YouTube allow you to embed videos into your slides. An embedded video will still be hosted on its original website, meaning the video itself won't be added to your file. Embedding can be a convenient way to reduce the file size of your presentation, but you'll also need to be connected to the Internet for the video to play.

To preview a video:

1. Click a video to select it.
2. Click the Play/Pause button below the video. The video will begin playing, and the timeline next to the Play/Pause button will advance.



3. To jump to a different part of the video, click anywhere on the timeline.



To resize a video :

- Click and drag the corner sizing handles until the video is the desired size.
- The corner sizing handles will resize a video while preserving its original aspect ratio. If you use the side sizing handles, the video will become distorted.



Must Know!



Not all video file formats are compatible with PowerPoint 2007, but AVI, MPEG, and WMV are acceptable formats.

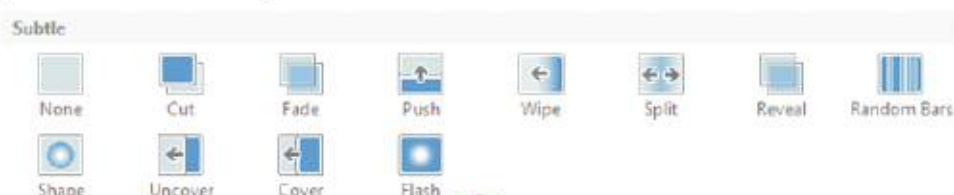
Transitions

Transitions are motion effects that when in Slide Show view add movement to your slides as you advance from one slide to another. There are many transitions to choose from, each one of which allows you to control the speed and even add sound.

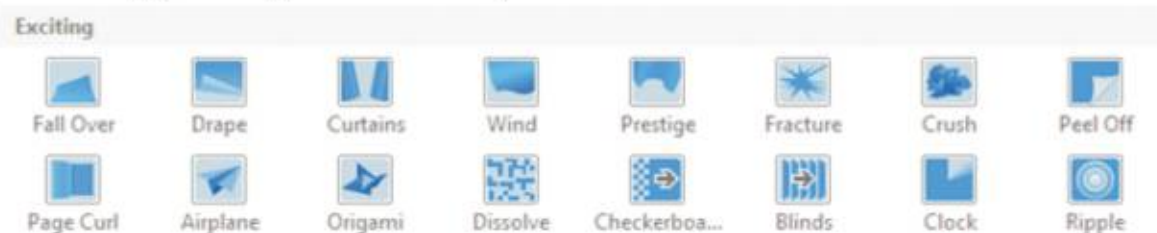
About transitions

There are three categories of unique transitions to choose from, all of which can be found on the Transitions tab:

- Subtle (slight transitions)



- **Exciting (strong transitions)**



- **Dynamic Content (strong transitions that affect only the content, such as text or images)**



To apply a transition:

1. Select the desired slide from the Slide Navigation pane. This is the slide that will appear after the transition.



2. Click the Transitions tab, then locate the Transition to This Slide group. By default, None is applied to each slide.
3. Click the More drop-down arrow to display all transitions.



4. Click a transition to apply it to the selected slide. This will automatically preview the transition.



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βU¿/€[AoH+?•úN



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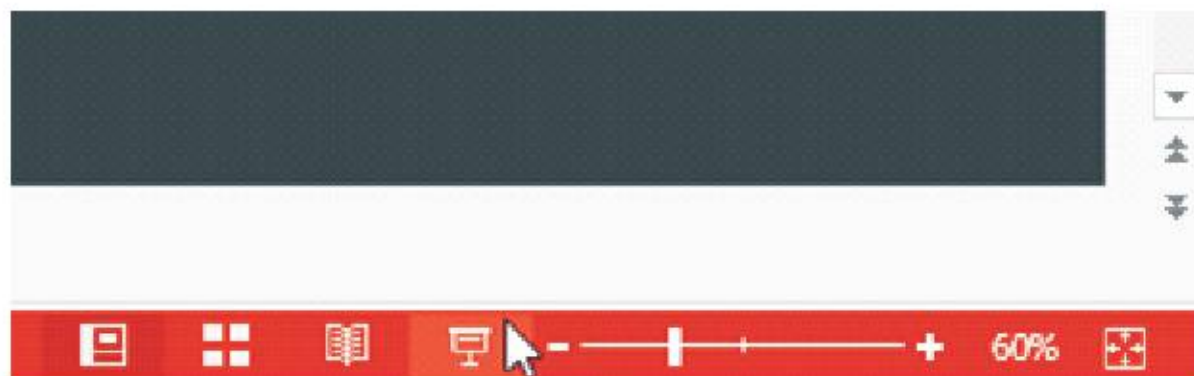


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Click the Slide Show tab on the Ribbon to access even more options. From here, you can start the presentation from the current slide and access advanced presentation options.



A Tick(3) the correct option.

(NEP) Analytical Skills

- To save time, you can create your presentation from an:

(a) Online office.com	☐	(c) Google	☐
(b) Theme	☐	(d) Office.com template	☐
- It is a predefined combination of colors, fonts, and effects that can be applied to your presentation.

(a) Transition	☐	(c) Theme	☐	(b) Slide show	☐	(d) Template	☐
----------------	--------------------------	-----------	--------------------------	----------------	--------------------------	--------------	--------------------------
- Motion effects that when in Slide Show view add movement to your slides as you advance from one slide to another.

(a) Transition	☐	(c) Theme	☐	(b) Slide show	☐	(d) Template	☐
----------------	--------------------------	-----------	--------------------------	----------------	--------------------------	--------------	--------------------------
- Available in the Format tab when you click a shape.

(a) Shape Styles	☐	(c) Slide Styles	☐
(b) Dynamic Content	☐	(d) Wizard	☐
- Strong transitions that affect only the content, such as text or images.

(a) Shape Styles	☐	(c) Slide Styles	☐
(b) Dynamic Content	☐	(d) Wizard	☐

B Fill in the blanks.

- PowerPoint files are called _____.
- PowerPoint includes _____ that allow you to easily create professional-looking presentations.
- Adding _____ and _____ to your presentation can be a great way to illustrate important information.

4. There are three categories of transitions _____, _____ and _____.
5. Animated GIF files are not movies, but _____.

C Write T for True or F for False.

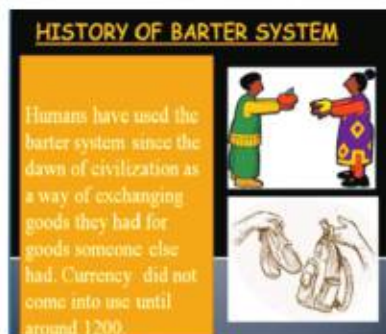
1. Every PowerPoint theme including the default Office slide has its own theme elements. ☐
2. PowerPoint allows you to add sound to your presentation, which opens up many possibilities. ☐
3. To change the audio icon, right click the icon and select Change Picture. ☐
4. PowerPoint allows you to insert a video directly into your presentation. ☐
5. MPG files are categorized as movies in the Clip Organizer. ☐

D Answer the following questions.

1. What is Presentation?
2. Define Themes with its elements.
3. What is the use of Transitions?
4. What is slide show?
5. Define: MP3, MPG

LAB ACTIVITY

Create the following presentation as shown below. Apply animation and slide transition effects to your presentation. Use your creativity. (NEP) Experimental Learning



DIFFERENT ERAS OF METTALIC COINS

- China achieved the ability to cast very refined works of bronze early on as evidence by it's extraordinary Shang and Zhou Dynasty ritual bronzes. Based on its technology, the first metallic coins, which even included "bronze" sea shells, appeared in various forms in different regions of the country during the Western Zhou Dynasty (11th century ~ 771 BC), the Spring and Autumn Period (770-476BC) and the Warring STATES Period (475-221BC).

PROBLEMS OF BARTER SYSTEM

The initial stage of exchange is known as Barter Exchange.

PROBLEMS FACED

- Lack of Double Coincidence of Wants
- Lack of Division /Indivisibility of certain goods
- Lack of Common Measure of Value
- Lack of Store of Value

Working with Flash CS3

Lesson at a Glance

Flash Workplace
Motion Guide Animation

Properties Inspector
Timeline Effects

Panels
Transform Effects

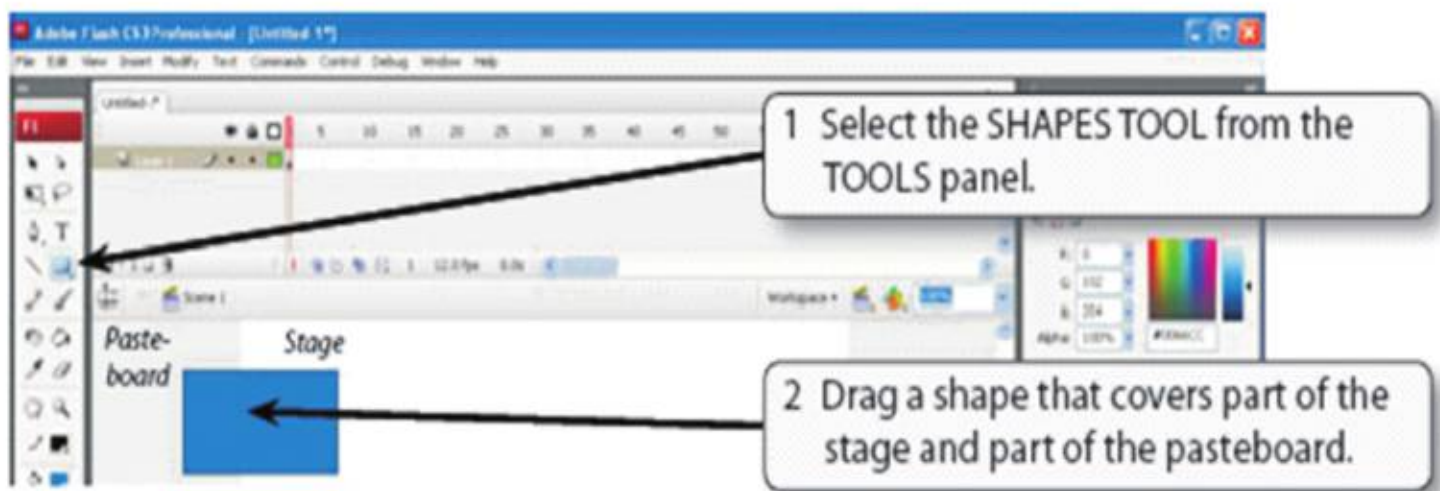
Adobe flash is a popular animation software used for creating animations, games and even full length animation movies, complete with sound and special effects. It was developed by Adobe Systems, Inc. American Software developer Jonathan Gay.

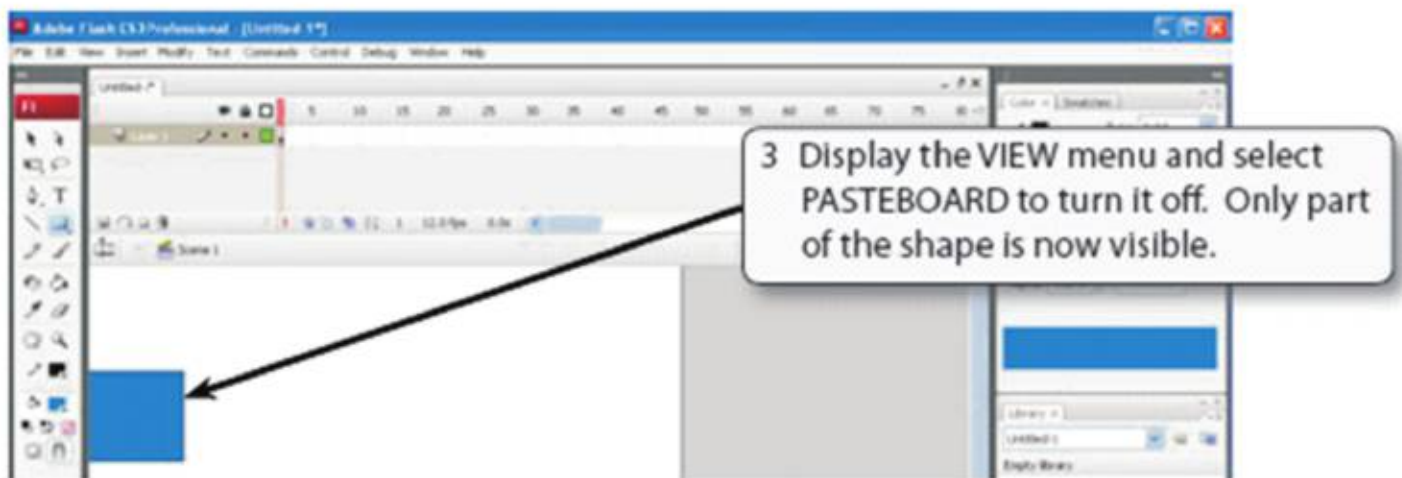
Flash Workspace

There are two parts to the Flash workspace which is at the centre of the screen:
The STAGE AREA which is the white area.

The PASTEBOARD which is the grey area that surrounds the stage.

Only objects within the STAGE AREA are included in a movie. The PASTEBOARD is mainly used to allow objects to move onto the stage then off.

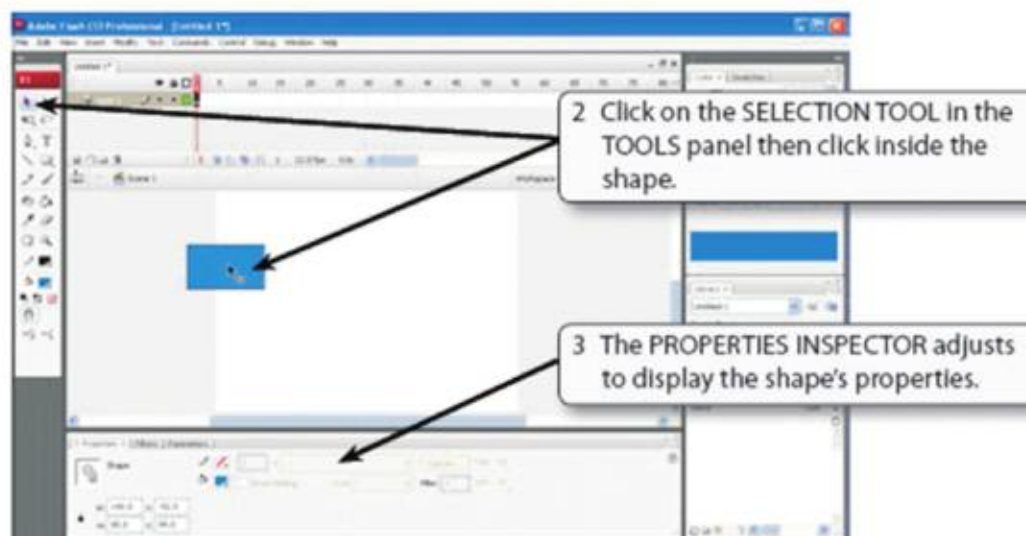
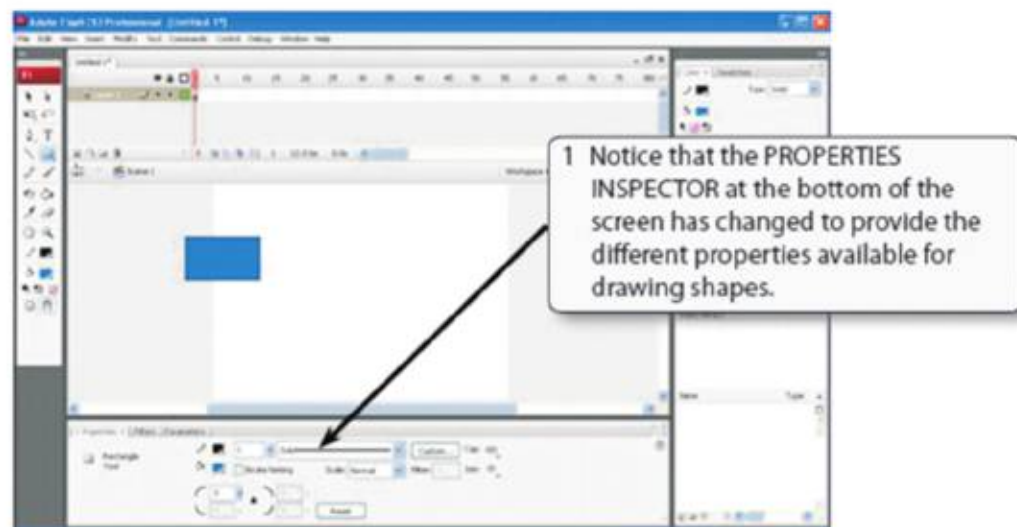




NOTE : This is what the shape would look like if we exported it from Flash.

Properties Inspector

The PROPERTIES INSPECTOR is one of the most used sections of the FLASH screen.



Double click in the shape to select all its parts and press the DELETE key to remove it from the screen.

Panels

Flash provides all the tools needed to create animations in a series of panels that are displayed around the stage and work areas. There are basically five different panel sections.

A. The Tools Panel

The TOOLS panel runs down the left of the screen and it has four sections. The DRAWING tools, the VIEW tools, the COLOUR tools and the tool OPTIONS.

The DRAWING tools are the standard drawing tools that most graphics programs provide. They allow you to create and edit objects.



The VIEW tools comprise the HAND tool which is used to scroll around the screen and the ZOOM tool which is used to magnify sections of the screen.

The COLOUR tools are used to set the FILL or LINE(STROKE) colours of objects.



The TOOL OPTIONS provide extra options for some of the DRAWING tools.



Motion Guide Animations

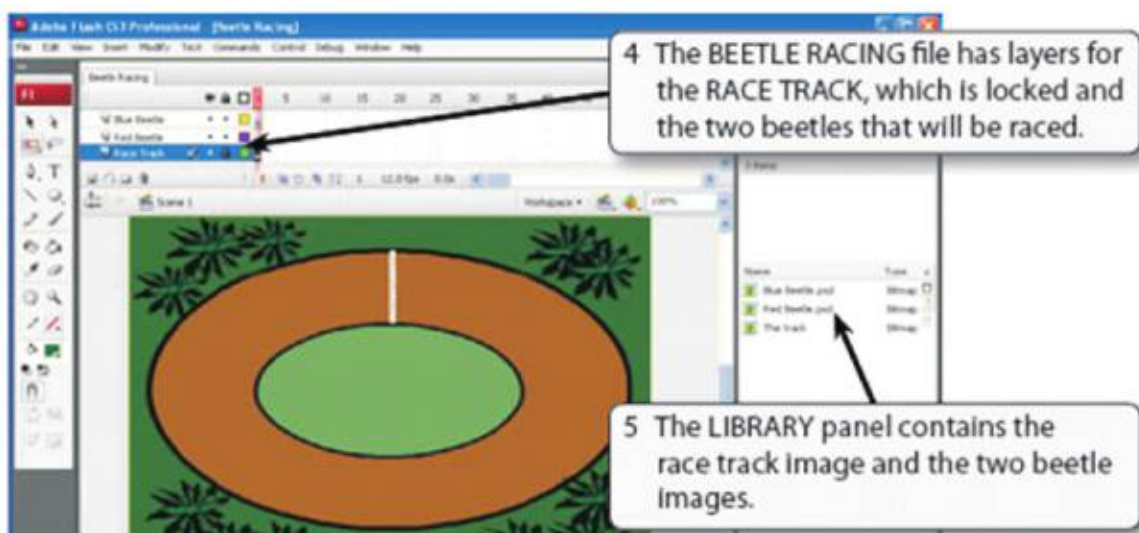
Animations that involve objects orbiting around a point can be done using fixed points, but better results can sometimes be achieved through the use of a circular motion guide.

Motion guides allow you to set the path that an object takes in an animation. To illustrate this a beetle race animation will be created. The race track and the beetles have been prepared for you.

Loading the Prepared File

1. Load Flash and click on the OPEN button, or close the current file and click on the OPEN button.
2. Access the FLASH CS3 SUPPORT FILES, open the CHAPTER 8 folder and load the file: Beetle Racing.

3. If the LIBRARY panel is not open, press CTRL+L to open it.

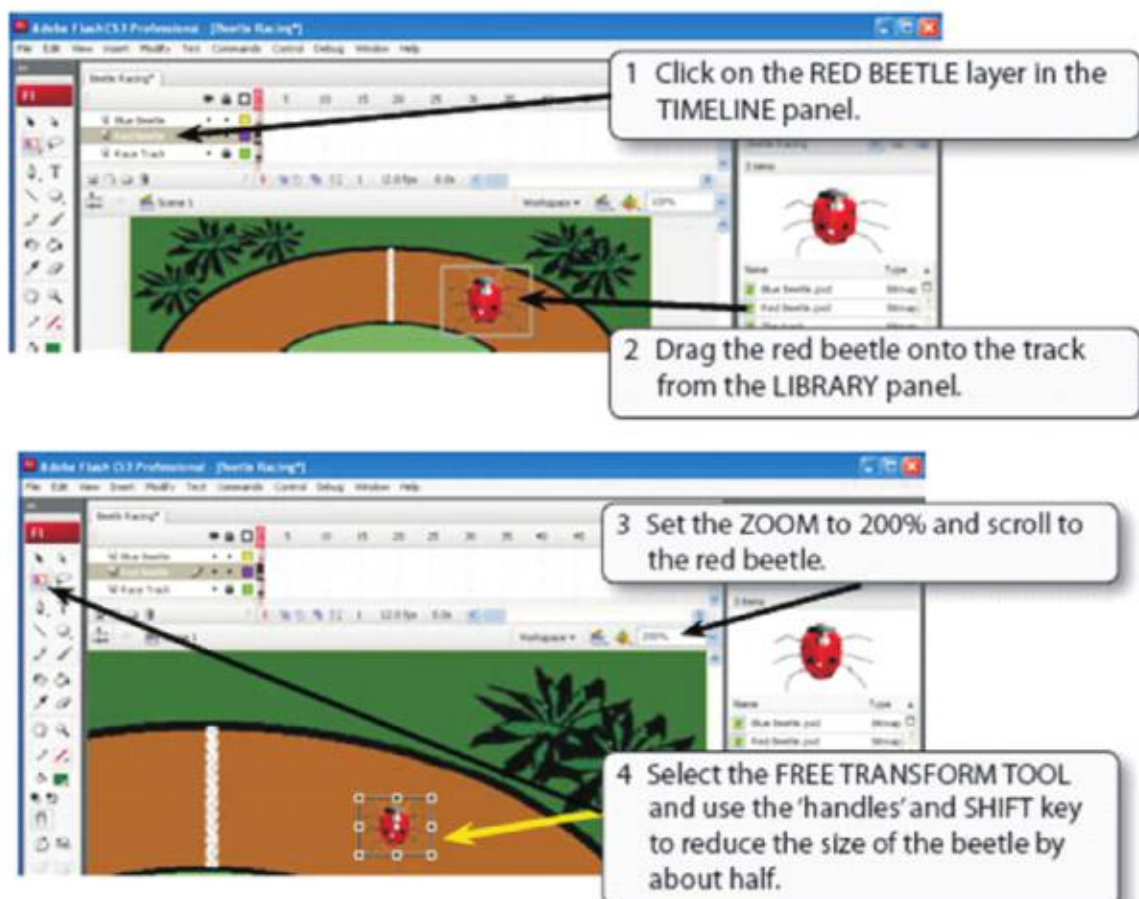


6. Save the file in your STORAGE folder under the name: Beetle Racing

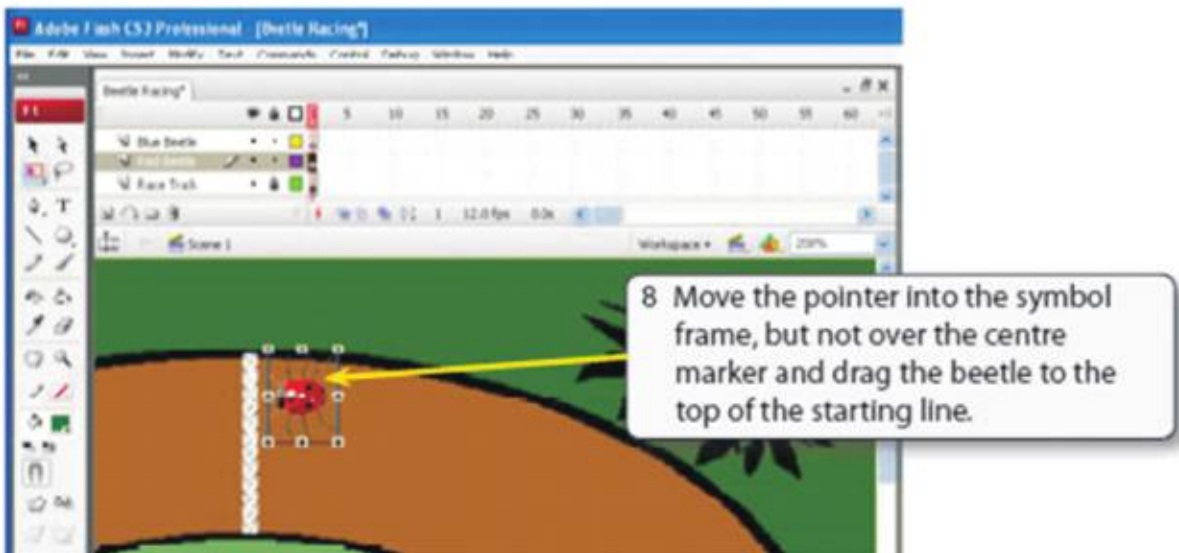
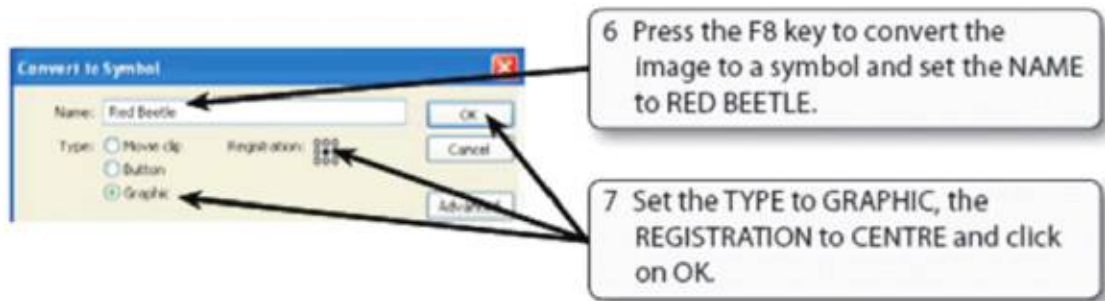
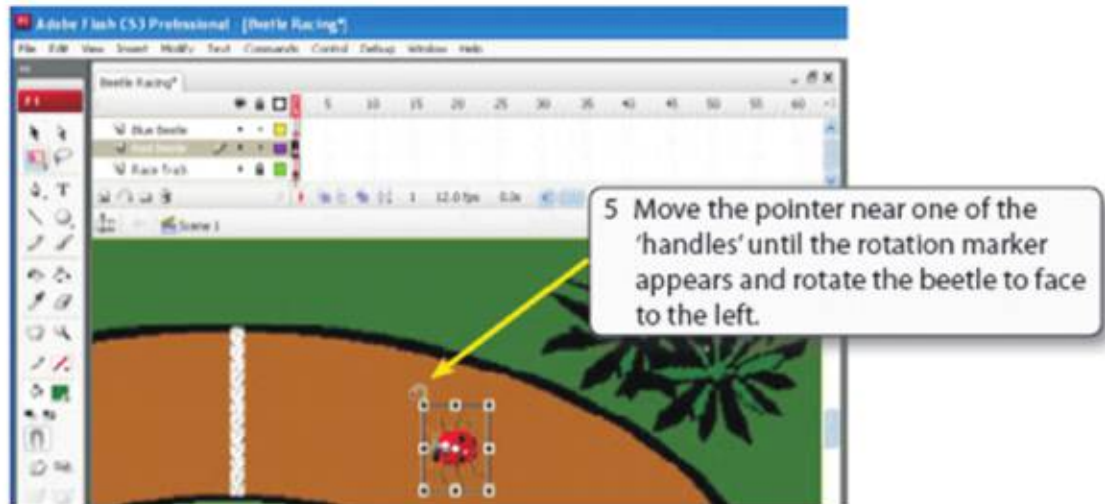
Converting the Beetles to Symbols

The two beetles in the LIBRARY panel are bitmap images. In order to animate them efficiently they need to be converted to symbols.

A. The Red Beetle



NOTE : The SHIFT key keeps the proportions of the beetle intact.

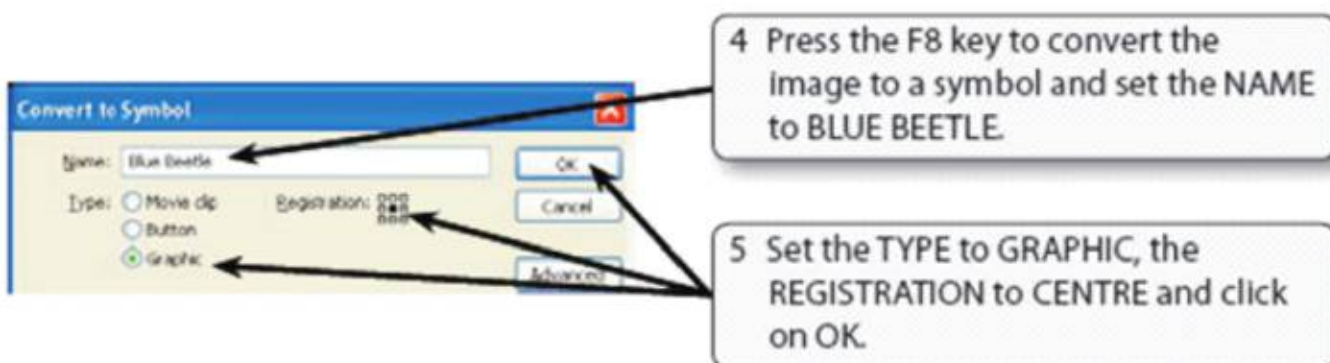
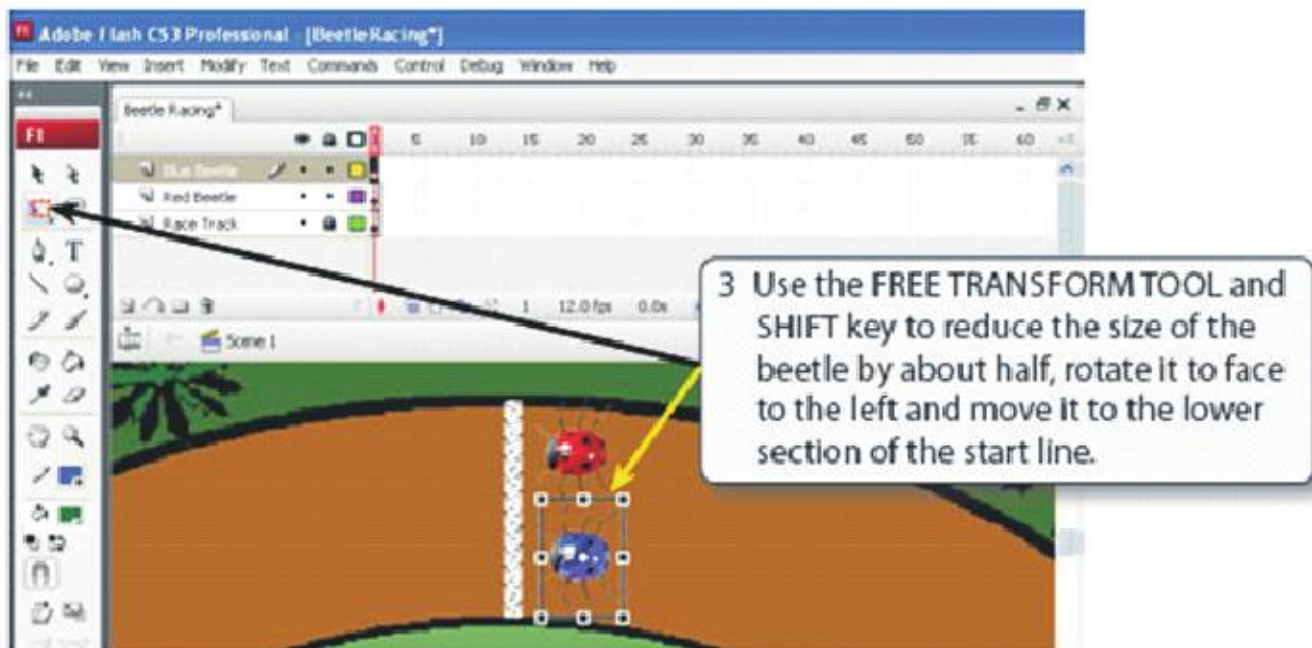
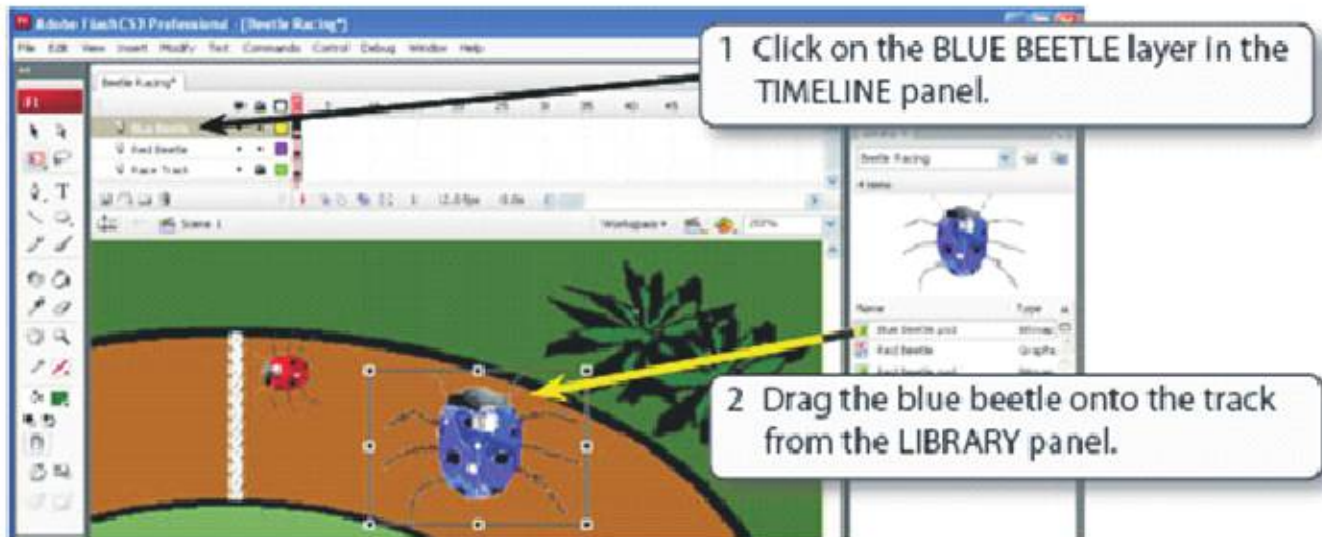


Must Know!



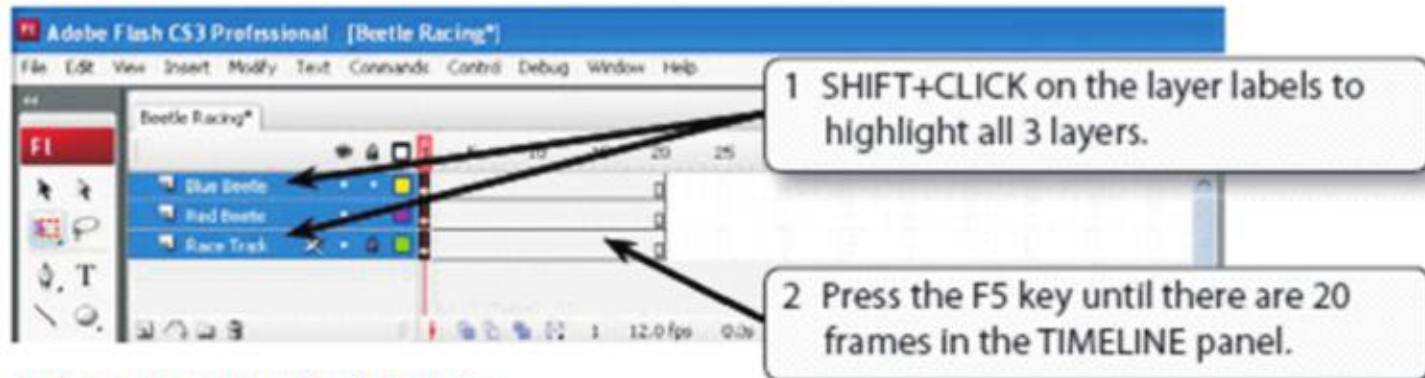
- (i) Be careful not to drag the centre registration marker of the symbol. If you do, press CTRL+Z to undo the move.
- (ii) You have two red beetle items in the LIBRARY panel. The symbol is a copy of the bitmap image. If you delete the bitmap image the symbol will be an empty frame.

A. The Blue Beetle



Setting the Frames

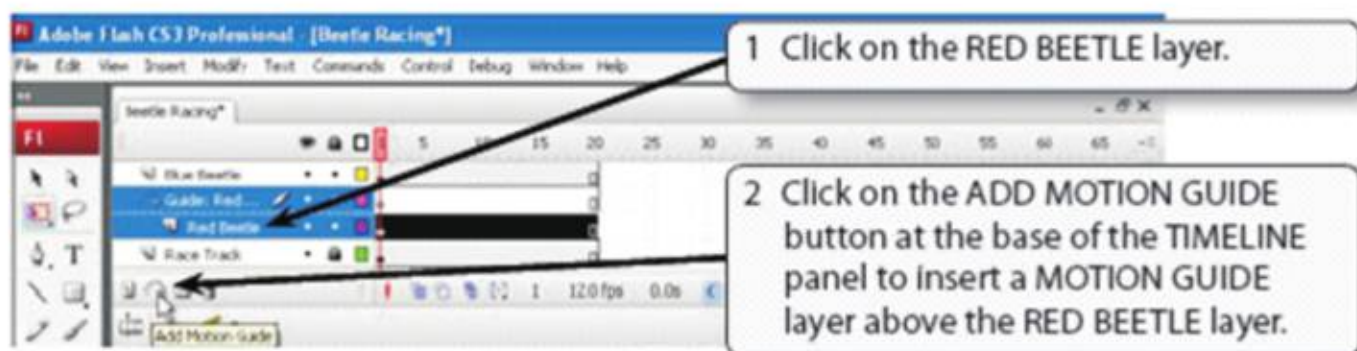
We will add 20 frame to all the layers.



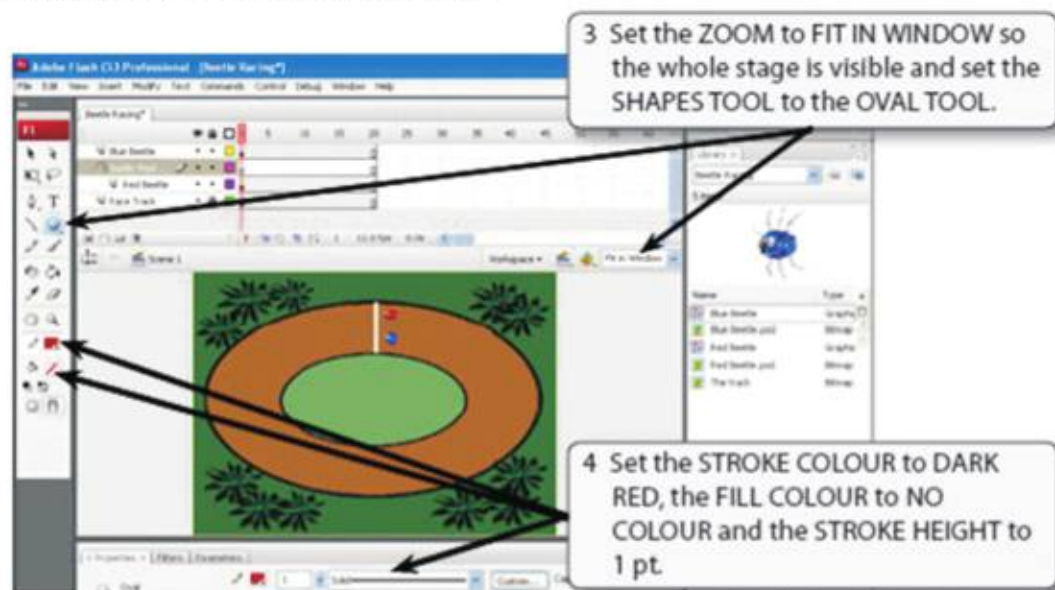
Animating the Red Beetle

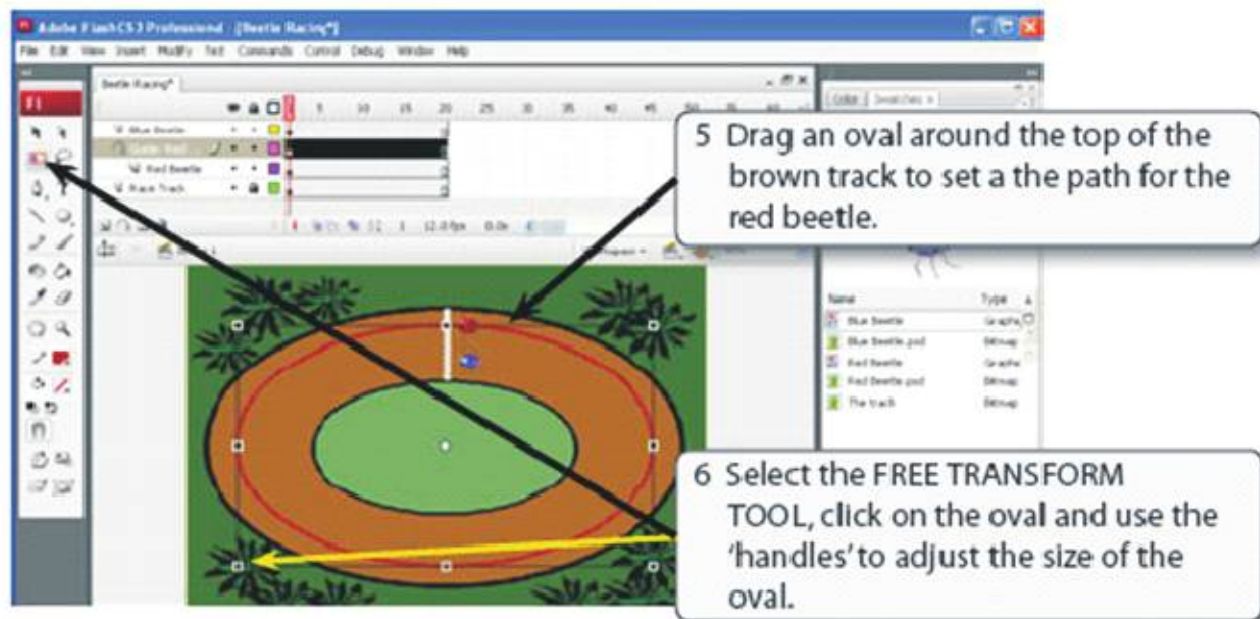
A. Setting the Motion Guide

Ovals will be drawn to set the path of the beetles and a hole cut in each oval so that the start and end points can be set.



NOTE : The motion guide layer is attached to the RED BEETLE layer which is indented to indicate this.





Timeline Effects

Timeline effects allow you to add effects to your animations to make them look more lifelike.

Some examples include added a drop shadow, fade in or out, or a blur effect.

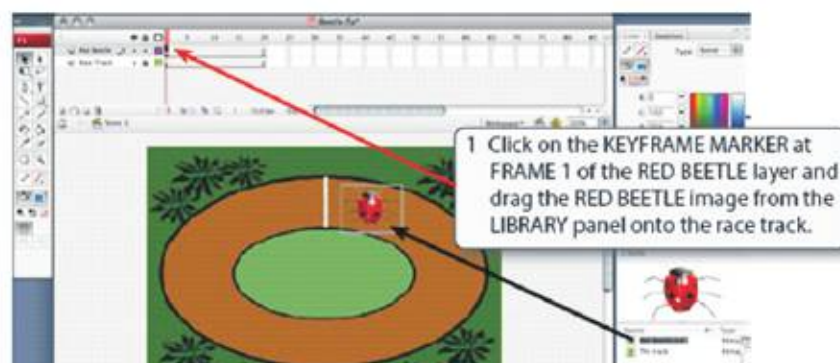
Drop Shadows

All 3-dimensional objects cast a shadow. Timeline effects allow you to easily add a shadow to an object. To illustrate this we will add a shadow to the beetle you raced in an earlier activity.

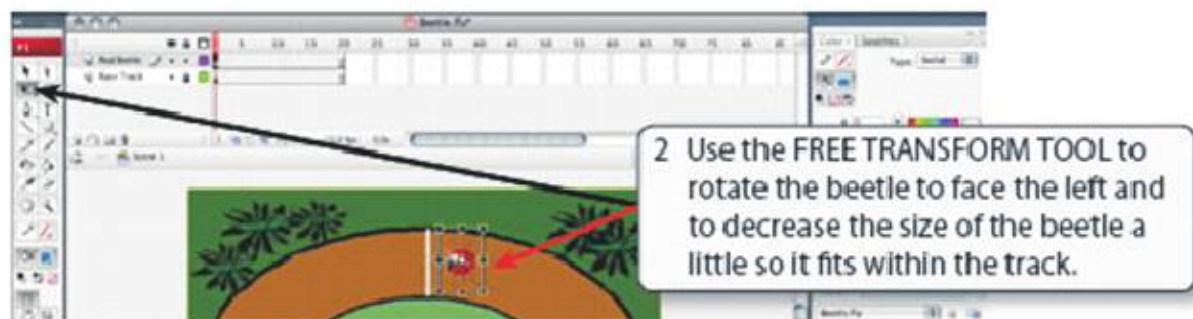
A. Loading the Prepared File

1. Load Flash or close the current file, then click on the OPEN button.
2. Access the FLASHCS3 SUPPORT FILES, open the CHAPTER 15 folder and load the BEETLE file.
3. Save the file in your STORAGE folder as: Beetle

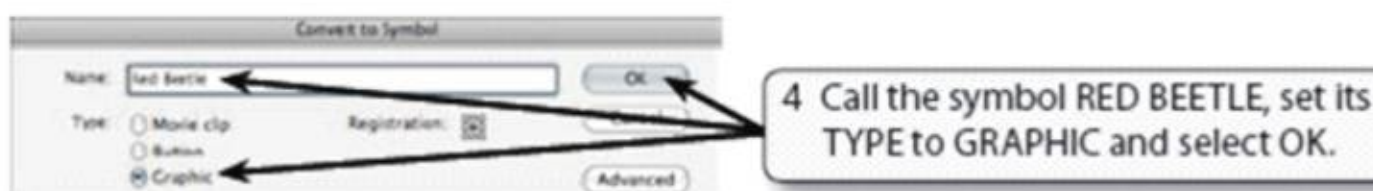
B. Applying the Drop Shadow Effect



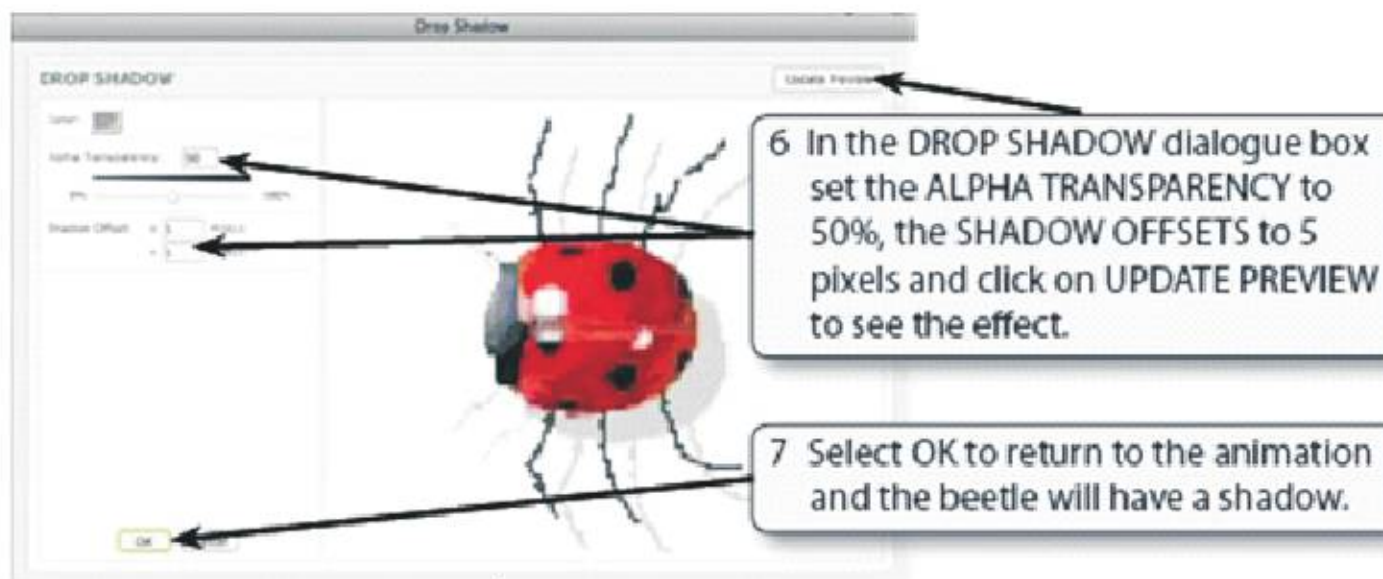
NOTE : If the LIBRARY panel is not open press CTRL+L to open it.



3 Timeline effects only operate on symbols so press the F8 key to convert the beetle to a symbol.



5 Display the INSERT menu, highlight TIMELINE EFFECTS followed by EFFECTS and select DROP SHADOW.



Must Know!

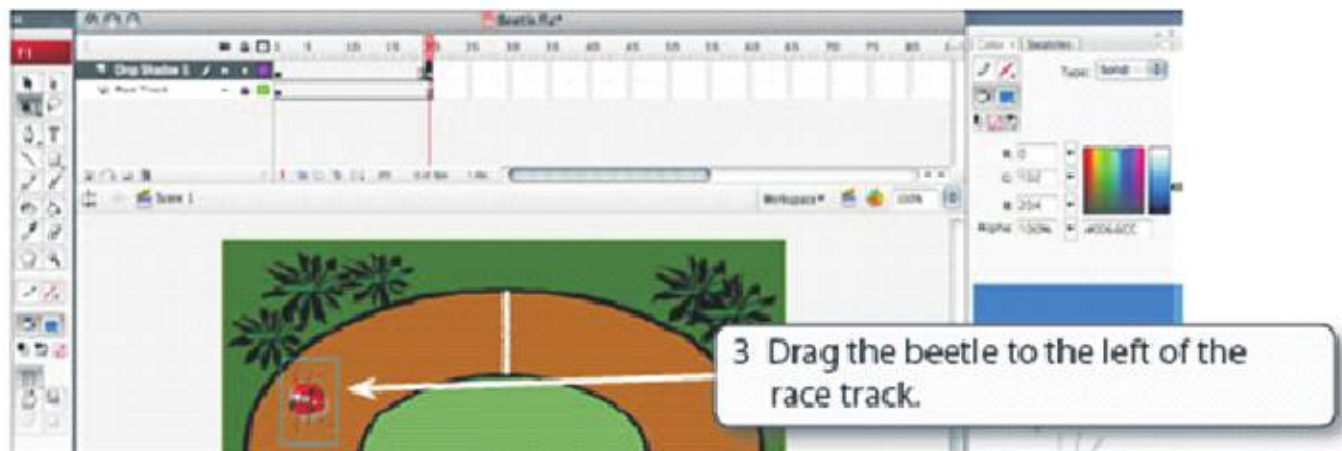


- (i) Alpha transparency is how light the drop shadow is. Shadow off sets are how far across the shadow is from the object.
- (ii) Notice that the drop shadow beetle has been added to the library. The original beetle symbol was not altered.

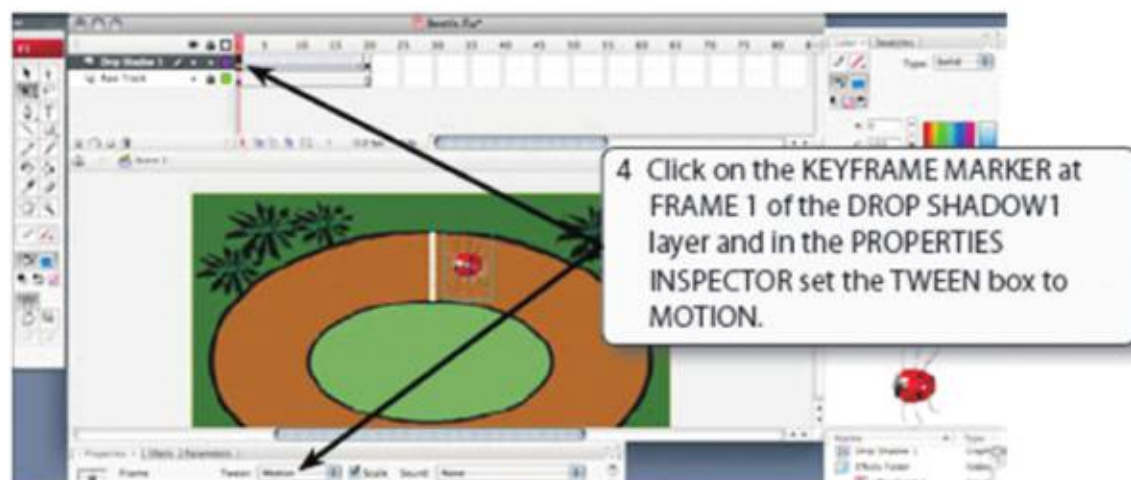
C. Animating the Drop Shadow Beetle

Once a timeline effect has been applied the symbol can be animated in the normal way.

You cannot apply an animation effect to a completed animation. Let's just move the beetle to the left of the race track.



NOTE : The BEETLE layer is renamed DROP SHADOW and the DROP SHADOW symbol is given the name number as the layer that it is inserted into.

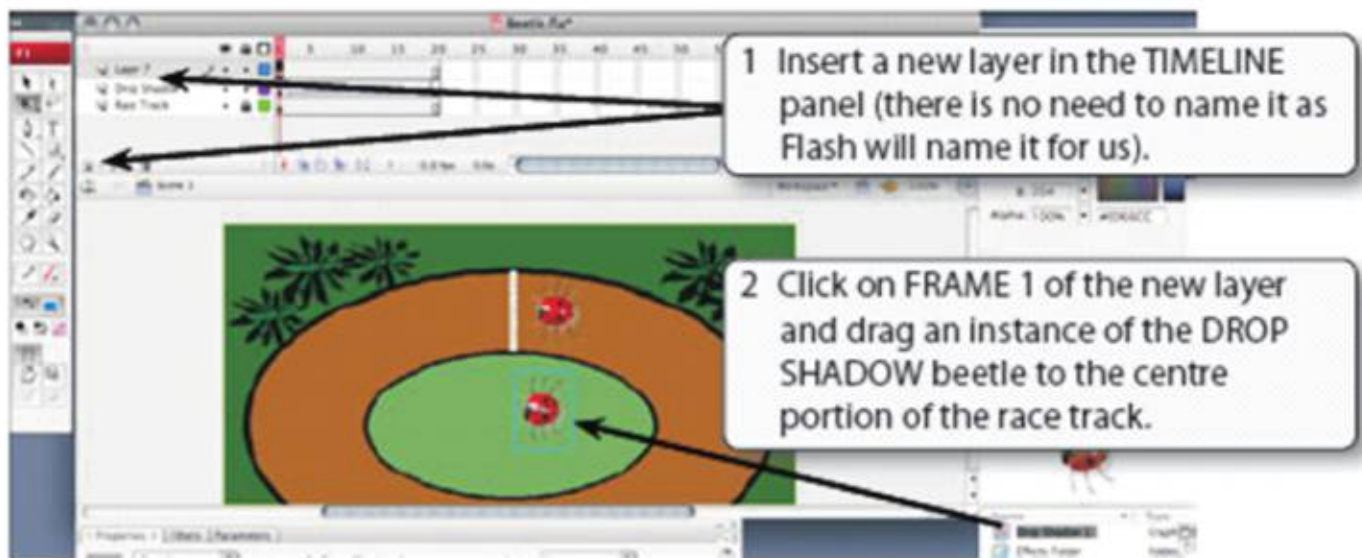


5. Play the animation and the beetle should move with a shadow.

Transform Effects

Transform effects allow you to add some motion as well as applying an effect. Let's move the beetle across the screen and make it fade to give the impression that it is moving away from the race track.

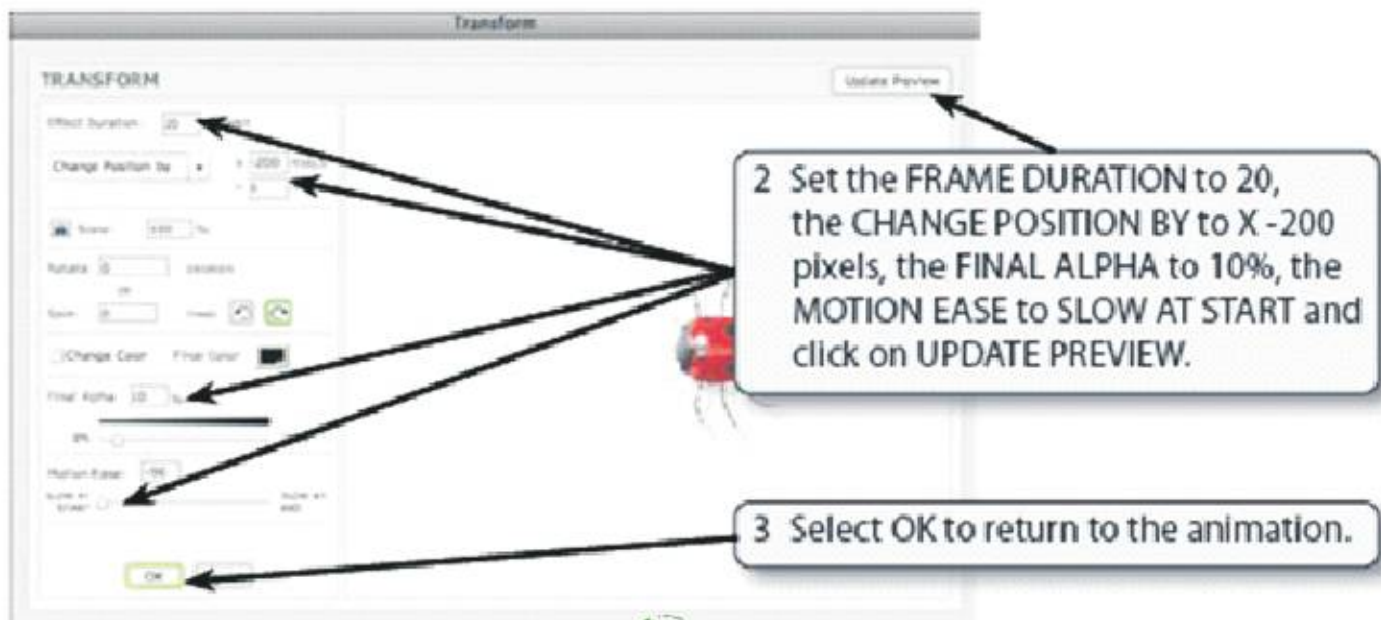
A. Inserting the Symbol



B. Applying the Transform Effect

- 1 Display the INSERT menu, highlight TIMELINE EFFECTS followed by TRANSFORM/TRANSITION and select TRANSFORM.

NOTE : (i) The FRAME DURATION was set to 20 to match the frames in the TIMELINE panel.



- (ii) The CHANGE POSITION BY sets the movement of the symbol in the X or Y direction (in this case 200 pixels to the left).
 - (iii) The FINAL ALPHA is how faded the symbol is at the end of the animation.
 - (iv) The MOTION EASE will start the animation slowly then speed up towards the end.
 - (v) You could rotate, spin or scale the symbol if that is required.
4. Play the animation and the second beetle should move across the stage and gradually fade out.



A Tick(3) the correct option.

(NEP) Analytical Skills

1. This mainly used to allow objects to move onto the stage then off.

(a) Drawing board	☐	(b) Pasteboard	☐
(c) Stage board	☐	(d) File board	☐
2. This is one of the most used sections of the flash screen.

(a) Paste board	☐	(b) Stage board	☐
(c) Panel board	☐	(d) Properties inspector	☐
3. Flash provides all the tools needed to create animations in a series of:

(a) Stage	☐	(b) Motion	☐
(c) Panels	☐	(d) Board	☐
4. This tool comprise the HAND tool which is used to scroll around the screen.

(a) Drawing	☐	(b) Screen	☐
(c) Motion	☐	(d) View	☐
5. This tools are used to set the FILL or LINE colours of objects.

(a) View	☐	(b) Line	☐
(c) Drawing	☐	(d) Colour	☐

6. This provide extra options for some of the DRAWING tools.

(a) Panel options

☐

(b) Stage options

☐

(c) View options

☐

(d) Tool options

☐

B Fill in the blanks.

1. There are two parts to the Flash workspace which is at the centre of the screen _____ and _____.
2. Only objects within the _____ are included in a movie.
3. The _____ tools are the standard drawing tools that most graphics programs provide.
4. _____ that involve objects orbiting around a point can be done using fixed points.
5. _____ effects allow you to add effects to your animations to make them look more lifelike.

C Write T for True or F for False.

1. The Tools panel runs down the right of the screen and it has four sections. ☐
2. Animation effects allow you to easily add a shadow to an object. ☐
3. Once a timeline effect has been applied the symbol can be animated in the normal way. ☐
4. Transform effects allow you to add some motion as well as applying an effect. ☐
5. Motion guides allow you to set the path that an object takes in an animation. ☐

D Answer the following questions.

1. Define Stage area in flash.
2. What is pasteboard in flash?
3. Define Properties Inspector in flash.
4. What are Panels in flash?
5. Write about Motion guide animations in flash.
6. What is Timeline effects in flash?
7. Write the use of Drop Shadows in flash.
8. Define Transform effects in flash.

LAB ACTIVITY

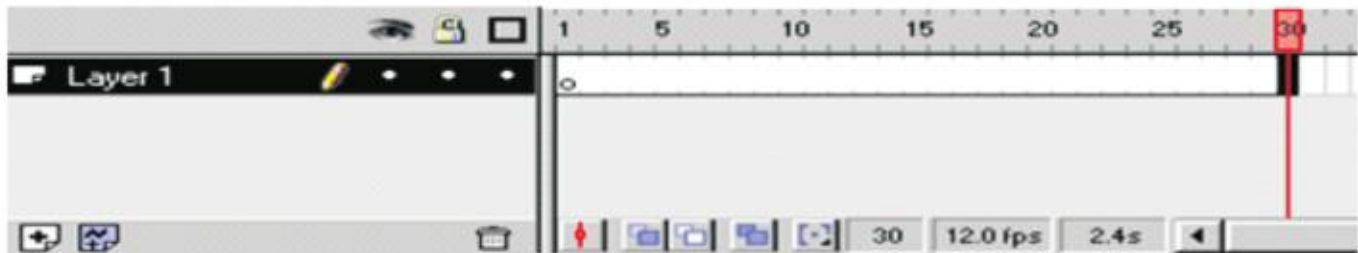


Do the following to make a Motion Guide Tweening.

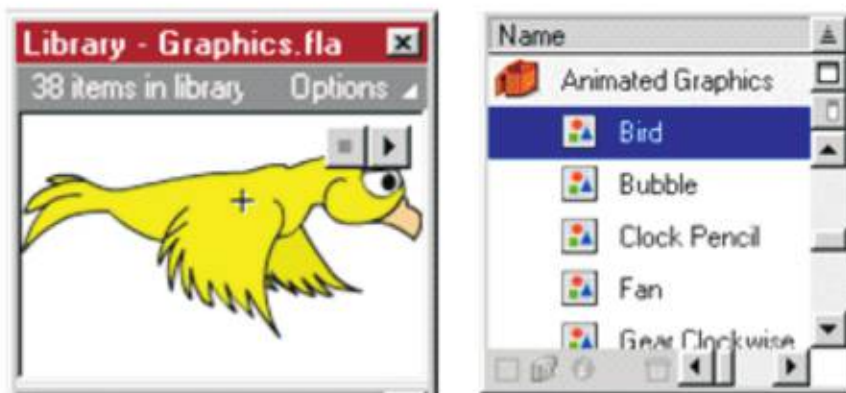
(NEP) Experimental Learning



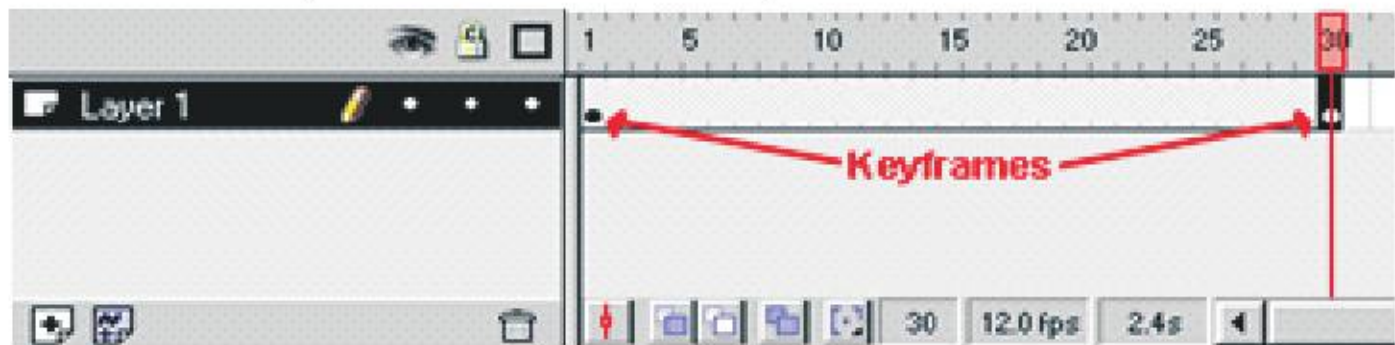
1. To insert 30 frames in your movie Click frame 30 in layer 1 on the timeline - then press F5.



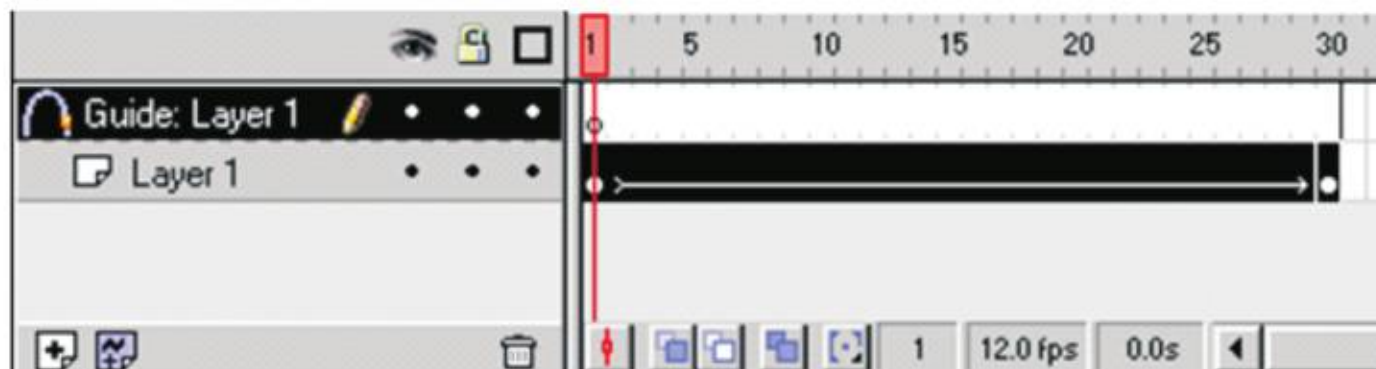
2. Choose Libraries > Graphics in the menu. Find Bird and drag it into the canvas.



3. Click frame 30 in layer one on the timeline. Then press F6.



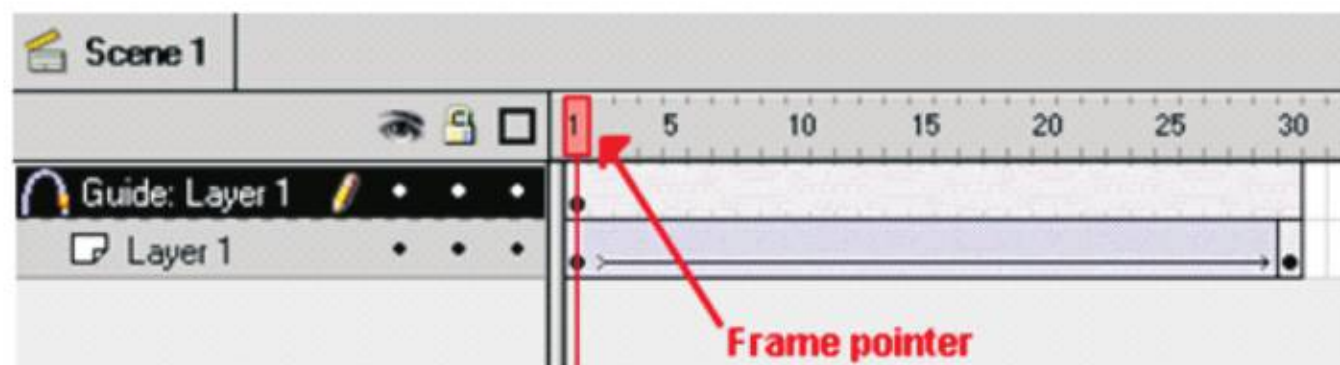
- Next, double click the keyframe in frame 1. A frame Properties Box pops up. Choose "Tweening", and in the Tweening drop-down menu choose "Motion".
- Right click on Layer 1. A pop-up menu appears. Choose Add Motion Guide. Flash now inserts a new layer on top of layer 1, with the Motion Guide icon to the left of the layer's name.



- Click on the Motion Guide Layer to make sure it is the active layer.



- Now adjust the frame pointer so it points to frame 1.



Make sure the Snap button is activated.

- With the mouse, place the bird so that its center is at the beginning of your motion guide. Now adjust the frame pointer so that it points at frame 30. Repeat the process Placing the bird at the end of the motion guide.

Creating Web Pages in HTML

Lesson at a Glance

Introduction	Hyperlinks	HTML Section	Container Tag
Non-container Tag	Base Font Element	BR Element	Paragraph Element
HR and Center Element	LIST in HTML		Inserting Images

Introduction

HTML Stands for Hyper Text Markup Language. It is a markup language for creating webpages. A webpage contains information in the form of text, graphics and animation arranged in attractive manner. Web browsers like Google Chrome and Mozilla Firefox use HTML text to interpret and compose text, images and other content into visual or audible webpages.

HTML is very easy to understand and can be modified easily.

HTML is a platform independent language.

HTML documents are prepared in HTML editors.

HTML document is always saved an extension .htm or .html.

It is the mother language of web.

HTML is not case sensitive, though we should write in small letter.

Hyperlinks

In HTML every single word has its own meaning i.e.

Hypertext : It is a piece of text on a webpage which opens a new webpage when it is clicked. Like the Login button on facebook.com which logs a user into his/her Facebook account after the user has input the email address and password.

Markup Language : It is a language that users markup which are instructions given to the browser on how to display the hypertext, or graphics, or ordinary text or video etc. In HTML, markups are special symbols called 'tags' which are formatting specifications used in this language.



HTML was invented by Tem Berners-Lee. He is known as the inventor of the world wide web.

HTML Section

Basically HTML is divided into two sections i.e. Head section and Body section. Head part contains the title of the page. Body part contains other tags of the web page like title, headings, image, table, etc.

Head section : HTML include pair of tags, and to identify the heading and title of a document. The tag appear in this section.

Body section : The body of HTML contains the text that will show up on web page. The Body section enclosed within and tags. It gives document layout and structure.

Container Tag

Container tag : These tags are in which we define the text or other tag elements.

These actually consist two tags, a start tag and an end tag, which enclose the text.

```
<html>
```

```
</html>
```

In the above Example:

One container tag is defined. Here we can define all the HTML tags inside the given tag.

```
<html>
```

```
<body>
```

```
</body>
```

```
</html>
```

In the above Example:

We putting inside the <html>tag.

Body tag give all description of the document's layout and structure.

Non Container Tag

These are stand alone text do not contain text. Or any other tag element.

`<br/>`

`<hr/>`

`<input type="text"/>`

In the above example:

The two tags are non-container/empty tags. `<br/>` line break tag and `<hr/>` is used to break the line and display line across the screen.

These do not hold any text or any other tags.

Must Know!

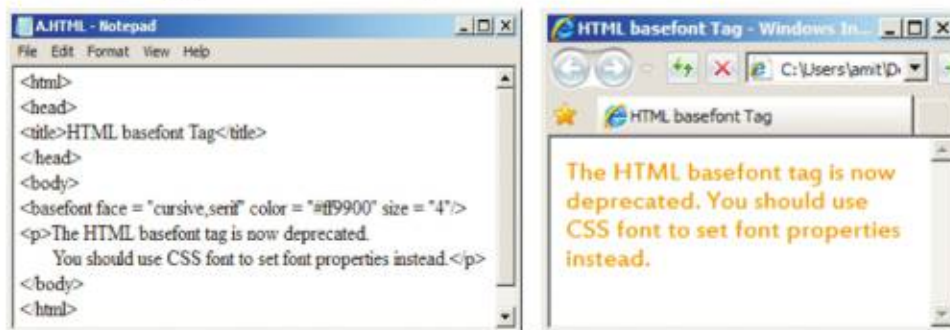


You can view the entire set of codes used in the website simply by pressing CTRL + U Keys on the website active in the browser.

Basefont Element

The HTML `<basefont>` tag is used to specify a base font for the document to use. This base font is applied to complete document.

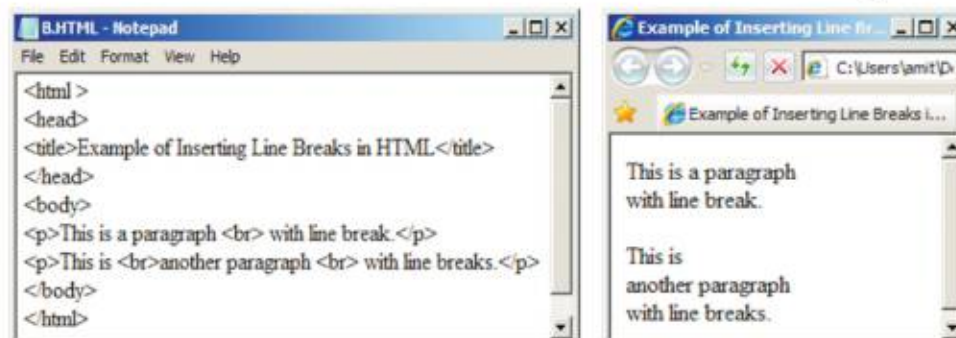
Example :



BR Element

The `<br>` element is used to insert a line break without starting a new paragraph.

Example :



Must Know!



Don't use empty paragraph i.e. `<p></p>` to add extra space in your web pages. The browser may ignore the empty paragraph since it is logical tag.

Paragraph Element

This tag is used to write the paragraph at web page.

It is just like a paragraph written in a book. It is always declared in body section.

Syntax

`<p>Paragraph...</p>`

`<p>another Paragraph...</p>`

`<p>` is the starting of a paragraph `</p>` is end of paragraph.

Must Know!



Content written in middle of `<p>` and `</p>` is paragraph which should be displayed on the web page in a single line.

Align Property(attribute) of paragraph element

Align attribute of tag is used to align the paragraph on the web page.

It have three possible value: left, right, center

`<align="left">Thisisfirst Paragraph...</p>`

`<align="center">Thisis2ndParagraph...</p>`

`<align="right">Thisis3rdParagraph...</p>`

Output

This is first Paragraph...

This is 2nd Paragraph...

This is 3rd Paragraph...

Title property(attribute) of paragraph element

Title attribute of tag is used to show hint box when mouse is hover on paragraph.

`<ptitle="Title will display when mouse is over on the paragraph">Thisisfirst Paragraph...</p>`

Output

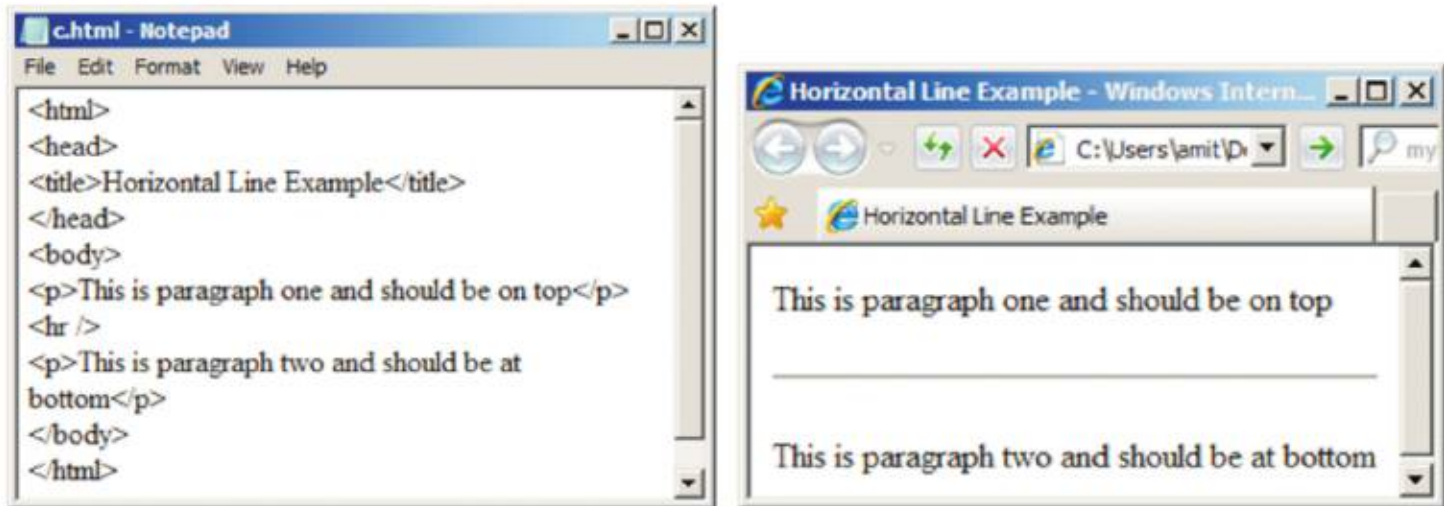
This is first Paragraph...

HR and Center Element

Horizontal lines are used to visually break-up sections of a document. The `<hr>` tag creates a line from the current position in the document to the right margin and breaks the line accordingly.

For example, you may want to give a line between two paragraphs as in the given example below -

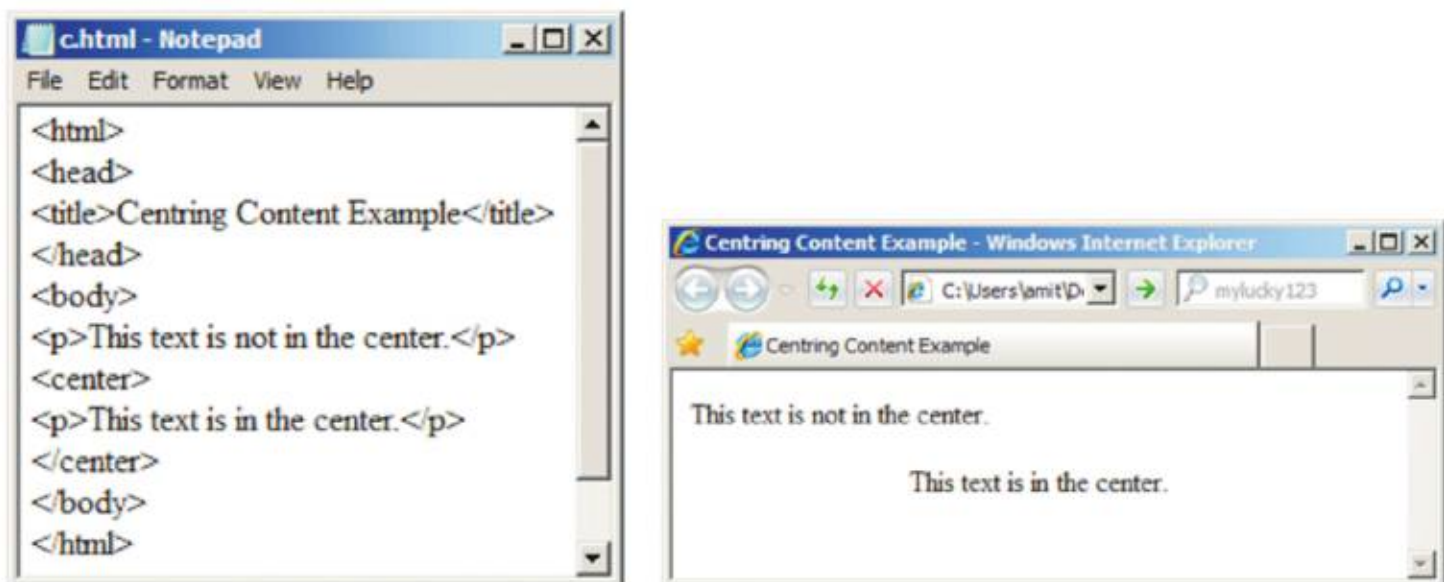
Example :



CENTER Element

You can use `<center>` tag to put any content in the center of the page or any table cell.

Example :



List in HTML

HTML list is used to display some content in ordered manner or unordered manner.

HTML offers three different type of lists. All lists must contain one or more list elements. List may contains :

1. Ordered information
2. Unordered information
3. Definitions

Ordered List

Ordered list is the list which is used to make a list in an ordered manner.

It is also known as the numbered list because it is numbered in a proper sequence. For example,

- | | | | |
|----|-----|----|---|
| 1. | i | a. | A |
| 2. | ii | b. | B |
| 3. | iii | c. | C |

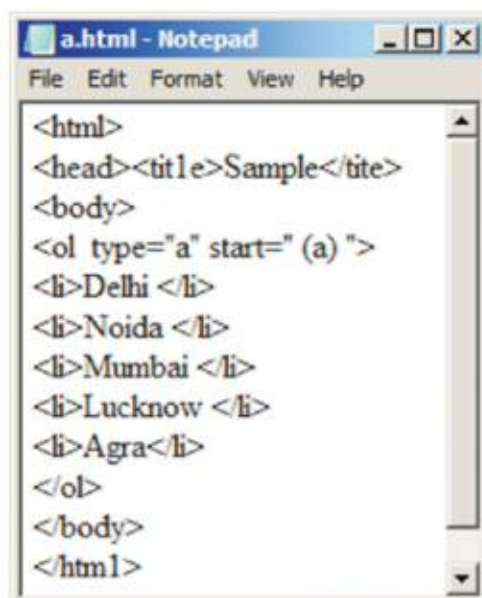
The tag is used to create an ordered list. We can specify the style attribute in the beginning. In HTML, we can use the tag for three kinds of ordered lists:

1. Alphabet
2. Numbers/digits
3. Roman numerals

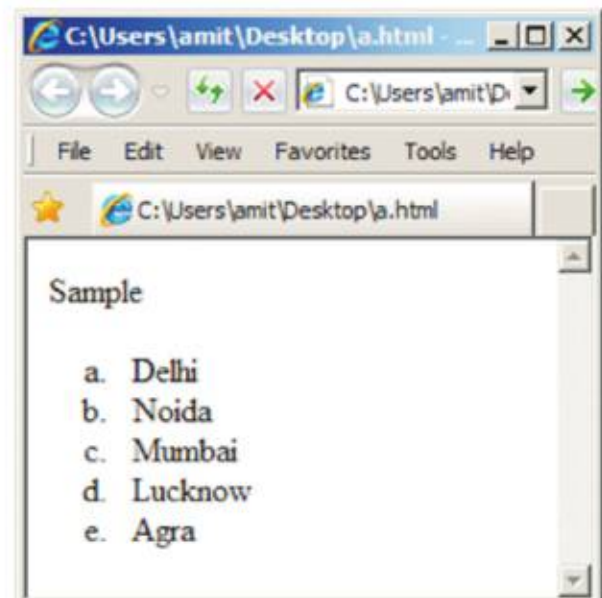
If it is not specified, then by default the numbering begins with 1 and keeps increasing by 1 every time.

Suppose, you want to create an ordered list with alphabets as a, b, ..., then write the HTML documents as:

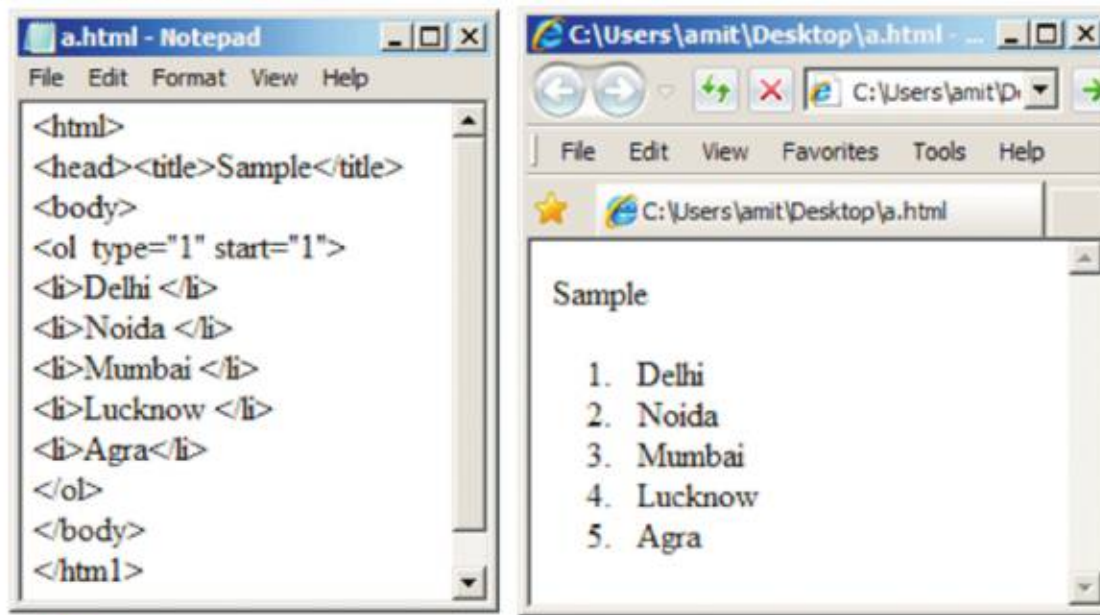
HTML code



```
<html>
<head><title>Sample</title>
<body>
<ol type="a" start="a">
<li>Delhi </li>
<li>Noida </li>
<li>Mumbai </li>
<li>Lucknow </li>
<li>Agra</li>
</ol>
</body>
</html>
```



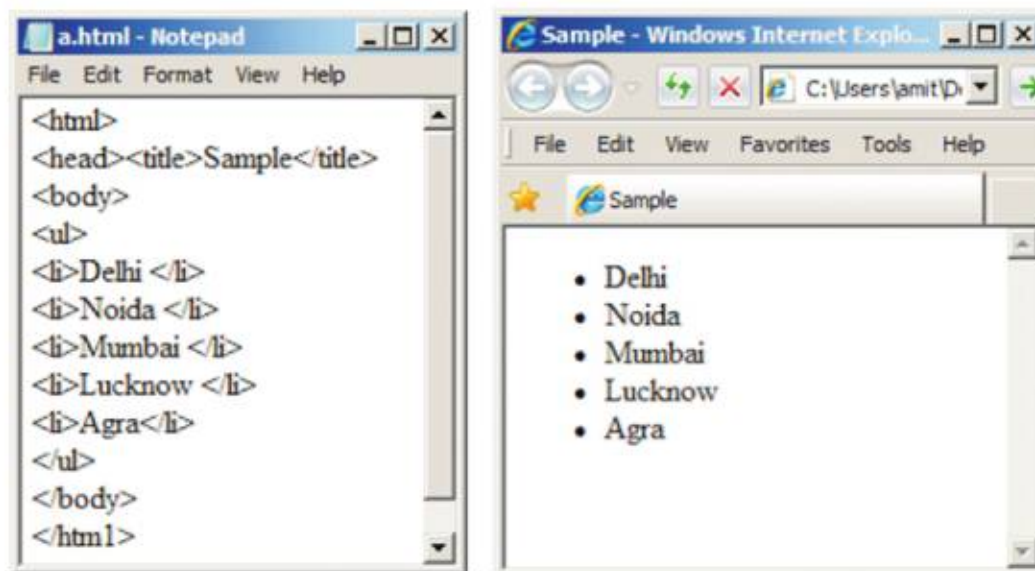
Now, suppose, you want the same list with numbering as 1, 2, , then the HTML document would be:



Unordered List

Unordered list is the list which is used to make a list in an unordered manner. If you want the same list as unordered or unnumbered list, then the tag used would be `<UL>` instead of `<OL>`. The items in this list appear with bullets.

HTML code



Definition List

Definition list containing groups of terms and their definitions. This is a list which is without numbers or bullets and there are specific terms that help to make the list.

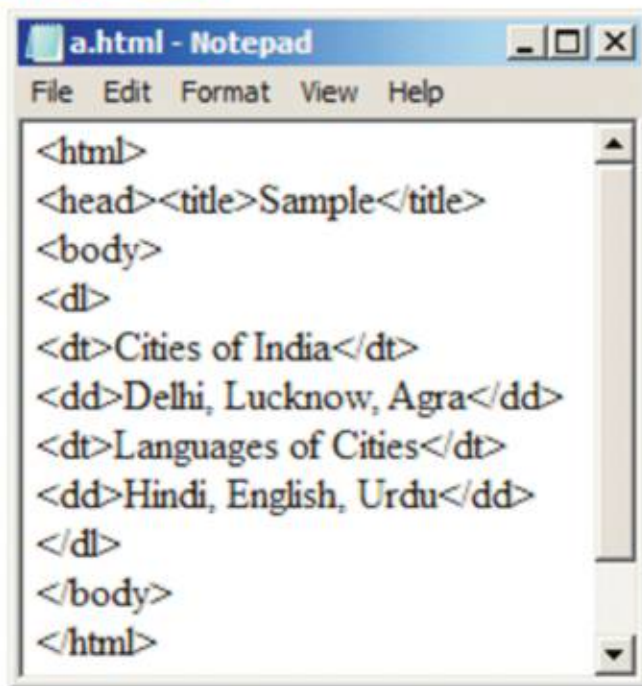
A definition list may contain numerous terms and descriptions, one after the other. Additionally, a definition list may have multiple terms per description as well as multiple descriptions per term. It is useful in work like a dictionary where there are words and their meanings listed without numbers or bullets.

A definition list starts with a <DL> tag (definition list).

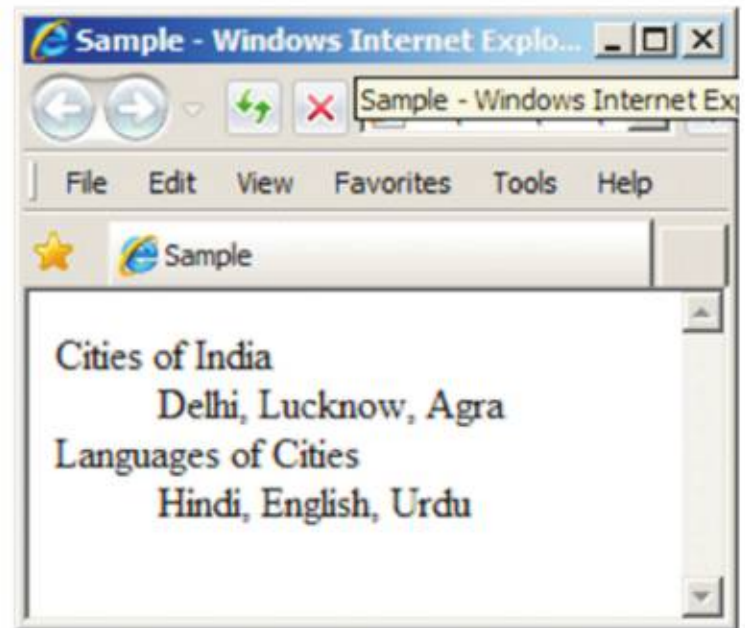
Each term starts with a <DT> tag (definition term).

Each description starts with a <DD> tag (definition description).

HTML code



```
<html>
<head><title>Sample</title>
<body>
<dl>
<dt>Cities of India</dt>
<dd>Delhi, Lucknow, Agra</dd>
<dt>Languages of Cities</dt>
<dd>Hindi, English, Urdu</dd>
</dl>
</body>
</html>
```



Inside the <dd> tag, you can put paragraphs, line breaks, images, links, other lists, etc.

Inserting Images

Images are very important to beautify as well as to depict many complex concepts in simple way on your web page. This tutorial will take you through simple steps to use images in your web pages.

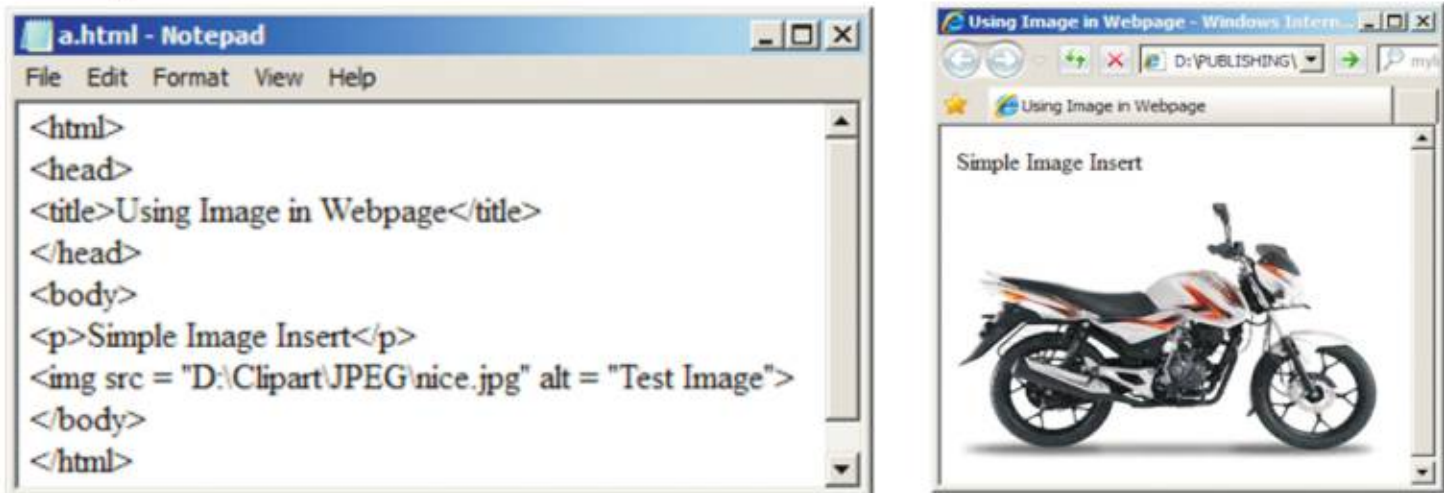
Insert Image

You can insert any image in your web page by using tag. Following is the simple syntax to use this tag.

<imgsrc = "Image URL" ... attributes-list/>

The tag is an empty tag, which means that, it can contain only list of attributes and it has no closing tag.

Example



You can use PNG, JPEG or GIF image file based on your comfort but make sure you specify correct image file name in src attribute. Image name is always case sensitive.

Must Know!

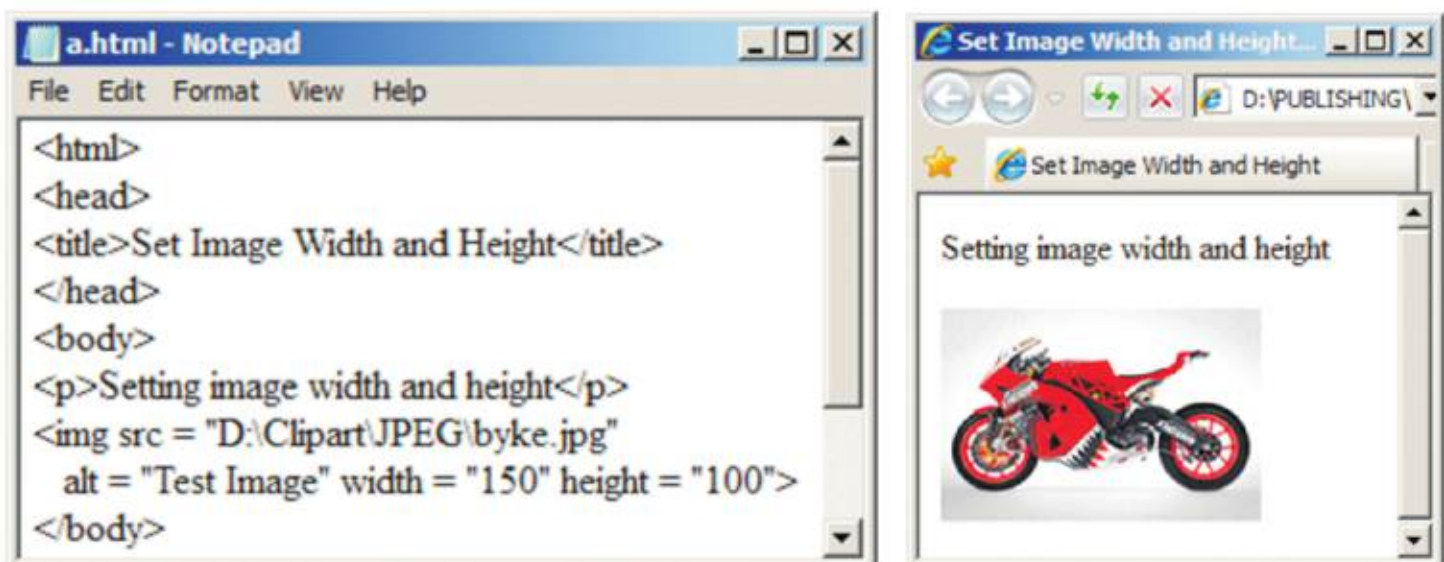


The alt attribute is a mandatory attribute which specifies an alternate text for an image, if the image cannot be displayed.

Set Image Width/Height

You can set image width and height based on your requirement using width and height attributes. You can specify width and height of the image in terms of either pixels or percentage of its actual size.

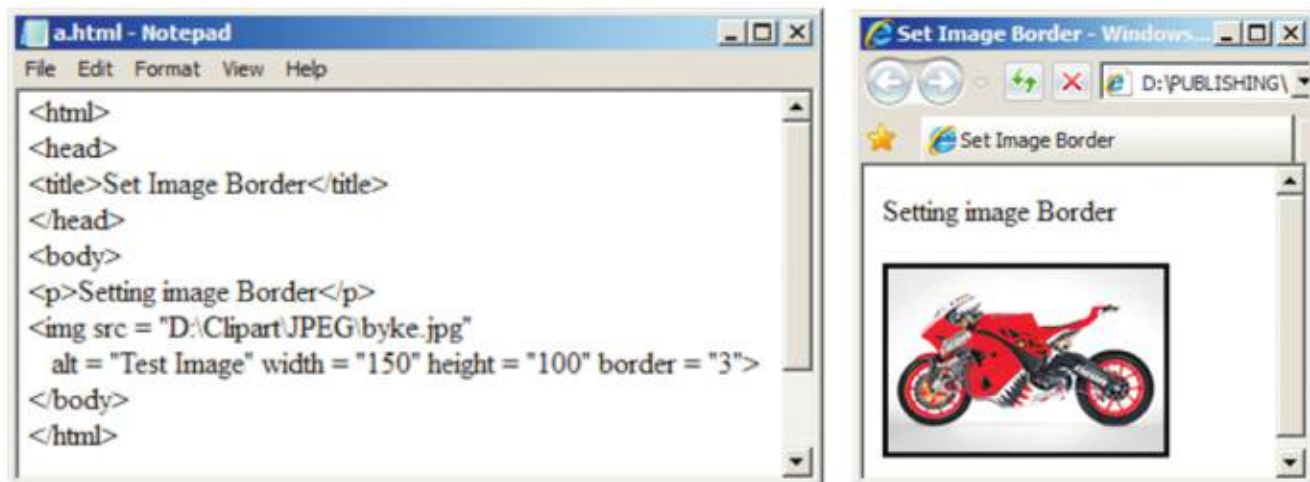
Example



Set Image Border

By default, image will have a border around it, you can specify border thickness in terms of pixels using border attribute. A thickness of 0 means, no border around the picture.

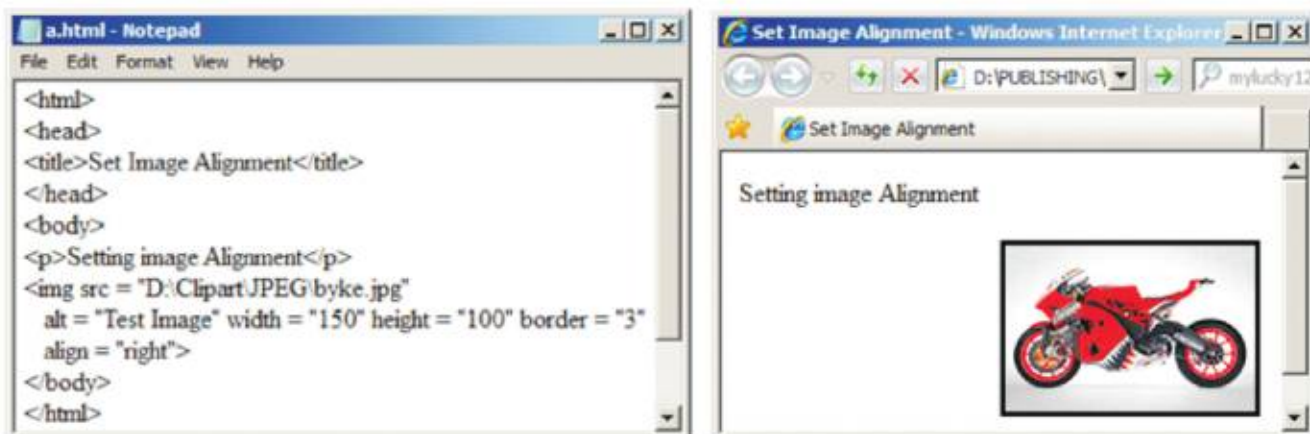
Example



Set Image Alignment

By default, image will align at the left side of the page, but you can use align attribute to set it in the center or right.

Example



A Tick (3) the correct option.

(NEP) Analytical Skills

1. It is a markup language used for creating webpages.

(a) WWW ☐ (b) XML ☐ (c) WML ☐ (d) HTML ☐

2. This HTML tag is used to specify a base font for the document to use.

(a) 	☐ (b) <list>	☐
(c) 	☐ (d) <basefont>	☐
3. This element is used to insert a line break without starting a new paragraph.

(a) <Hr>	☐ (b) 	☐
(c) 	☐ (d) <Td>	☐
4. These tags are in which we define the text or other tag elements.

(a) Element tag	☐ (b) Container tag	☐
(c) Attribute tag	☐ (d) Body tag	☐
5. This tag is used to write the paragraph at web page.

(a) A Element	☐ (b) P Element	☐
(c) B Element	☐ (d) D Element	☐
6. This tag is used to show hint box when mouse is hover on paragraph.

(a) Head attribute	☐ (b) Title attribute	☐
(c) Body attribute	☐ (d) List attribute	☐

B Fill in the blanks.

1. HTML Stands for _____.
2. Every web page designed in HTML has an extension _____.
3. _____ is the list which is used to make a list in an ordered manner.
4. _____ are used to visually break-up sections of a document.
5. _____ containing groups of terms and their definitions.
6. You can insert any image in your web page by using _____ tag.

C Write T for True or F for False.

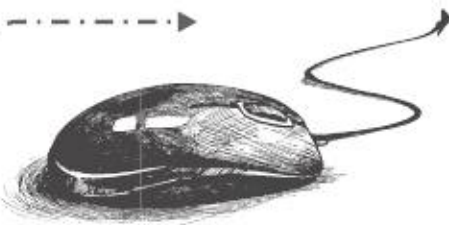
1. HTML is case sensitive, though we should write in small letter. ☐
2. Editor used to design the HTML static web page are notepad. ☐
3. Every HTML script will not be executed in any web browser. ☐
4. HTML is the client side scripting language i.e. used to design static web page. ☐
5. HTML is divided into two sections i.e. Head section and Body section. ☐
6. Ordered list is the list which is used to make a list in an unordered manner. ☐

D Match the followings.

- | | |
|----------|--------------------|
| 1. Hyper | (a) Alternate text |
| 2. <hr> | (b) Insert Image |
| 3. | (c) Unordered List |
| 4. | (d) Reference link |
| 5. | (e) Creates a line |
| 6. Alt | (f) Ordered List |

E Answer the following questions.

1. What is HTML?
2. Explain the two sections of HTML.
3. Differentiate between Container Tag and Non Container Tag.
4. What are Attributes?
5. What tags are used in creating lists?
6. Differentiate between Ordered List and Unordered List.
7. Explain Definition List.



Work sheet



Write down the description of following tags in HTML.

1. <HTML>
2. <HEAD>
3. <TITLE>
4. <Body>
5. <CENTER>
6. <HR>
7.
8. <BASEFONT>
9. <SUB>
10. <SUP>

(NEP) Creative Learning

LAB ACTIVITY



Find the output of the following.

(NEP) Experimental Learning

```
<HTML>
<HEAD><TITLE>UNORDERED LIST</TITLE>
</HEAD>
<BODY>
VARIOUS BULLET STYLES OF UNORDERED LISTS
<UL TYPE=circle>
  <LI> My name
  <LI> My class
  <LI> My school
</UL>

<UL TYPE=disc>
  <LI> My name
  <LI> My class
  <LI> My school
</UL>

<UL TYPE=square>
  <LI> My name
  <LI> My class
  <LI> My school
</UL>
</BODY>
</HTML>
```

```
<HTML>
<HEAD><TITLE>Definition LIST</TITLE>
</HEAD>
<BODY>
<P>Example of Definition Lists are</p>
<DL>
<DT>HTML :
<DD>Hypertext Markup Language<HR>
<DT>Web Browser :
  <DD>It is program that interprets the HTML, arrange & display the
  contents of web-page accordingly.
</DL>
</BODY>
</HTML>
```

Note : <HR> : It is used to display a horizontal line on the screen.

Evolution of AI

Lesson at a Glance

Development in AI

Since ancient times, humans have been trying to make their lives easier by innovating different machines. While creating these machines, he reached a certain stage when these machines were not capable enough to handle certain tasks without a brain. These machines needed to think. This sparked a revolutionary idea for making machines capable of thinking, from there, humans have been making steady progress. Let us look at some of the achievements of the last century.

Developments in AI

The journey to understand if machines could truly think began much before John McCarthy's first academic conference. In Vannevar Bush's seminal work as we may think, he proposed a system which amplifies people's own knowledge and understanding. Five years later, Alan Turing wrote a paper on the notion of machines being able to simulate human beings and the ability to do intelligent things, such as playing chess, Let us take a look at some of the milestones in the previous century.

1950-1960

In this decade, 2 major contributions were observed. Alan Turing submitted a paper about the possibility of creating a machine that could think in 1950. This became a test for machines to check the thinking



capability of a machine. One of the greatest achievements of Alan Turing is the creation of The Bombe, a machine that could decode the encoded messages.

Must Know!



The Bombe was a code breaking machine that was developed to decode encrypted codes from German Enigma machine during the World War II.

In 1956, John McCarthy coined the term Artificial Intelligence in a conference held at Dartmouth College. He created the programming language LISP in 1958 which was extensively used to program AI systems.



1961-1970

In this decade, two new systems were developed which became a great contribution to the field of AI. The first chatbot ELIZA was created in MIT Artificial Intelligence Laboratory by Joseph Weizenbaum in 1966. This is what it looked like.

```
Welcome to
EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LL      II    ZZZ    AAAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LLLLLL IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```


1971-2000

In 1973, WABOT was created by Ichiro Kato and constructed by Unimation. This was the first step towards building a humanoid robot. Given below is picture of WABOT.



In 1984, WABOT-2 was created at Waseda University which was the successor of WABOT-1 and capable of playing a piano.

In this period, Deep blue, created by IBM, defeated the then reigning chess world champion Gary Kasparov in 1997.



2000-2010

In The 2000s, AI technology was successfully established with many successful attempts like: Kismet, a robot developed by Professor Cynthia Breazeal could recognise and simulate emotions with its face.

Honda released ASIMO, an artificial humanoid robot in 2000.



In 2002, i-Robot released Roomba, an autonomous robot vacuum cleaner that could clean and was able to avoid crashing into obstacles.



NASA's Mars Exploration in 2004 was a robotic space mission involving two Mars rovers, Spirit and Opportunity to explore Planet Mars.



Google developed a driverless car in 2009 and by 2014, it passed Nevada's self-driving test.

2010 to Present

This was an era of revolution in the field of artificial intelligence with the following developments:

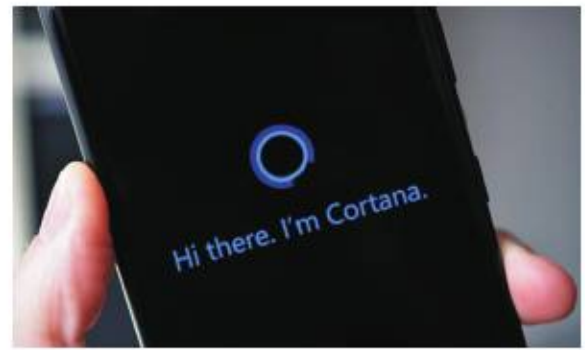


Microsoft launched Kinect for Xbox 360, the first gaming device that tracked human movements using a 3D camera and infrared detection.



Apple released Siri in 2011. It uses a natural language user interface to infer, observe, answer, and give recommendations to the users.

Cortana: Microsoft released Cortana in 2014, a virtual assistant similar to Siri.



Amazon Alexa: In 2014, Amazon created Alexa, a home assistant that is a smart speaker functioning as a personal assistant.

Must Know!



Alexa uses Bing and it is not the only virtual assistant that uses Bing-Apple's Siri and Microsoft's Cortana use it too.

Sophia, a humanoid robot: In 2016, Hanson Robotics created a humanoid named Sophia.



Google released Google Home in 2016. It uses AI to act as a personal assistant helping the user remember tasks, create appointments, and search information by voice.





A Tick(3) the correct option.

(NEP) Analytical Skills

1. Who is known as the father of Artificial Intelligence?

a. John McCarthy	☐ b. Alan Turing	☐
c. Allen Newell	☐ d. All of these	☐
2. Who is created ELIZA?

a. Joseph Weizenbaum	☐ b. Alan Turing	☐
c. John McCarthy	☐ d. None of these	☐
3. Which company secretly developed a driverless car in 2009 which also passed Nevada's self-driving test by 2014?

a. Google	☐ b. Apple	☐
c. Honda	☐ d. None of these	☐
4. When was LISP created?

a. 1956	☐ b. 1957	☐
c. 1958	☐ d. 1959	☐
5. Where was the first AI conference held?

a. MIT	☐ b. DarMouth College	☐
c. IIT	☐ d. None of these	☐
6. Which of the following is a virtual assistant developed by Microsoft?

a. 	b. 	
	☐	☐
c. 	d. 	
	☐	☐

B Fill in the blanks.

1. _____ was built at Waseda University in 1970.
2. John McCarthy created _____ in 1958.

3. The first chatbot _____ was created in 1966.
4. _____ defeated Gary kasparov in 1997.
5. Honda released _____, an artificial humanoid robot in 2000.

C Write T for True or F for False.

1. In 1967, WABOT was created by George Devol and constructed by Unimation. ☐
2. Roomba is a robot used for space exploration. ☐
3. Kismet was developed in year 2000. ☐
4. Google released ASIMO, an artificial humanoid robot. ☐
5. Hanson Robotics created a humanoid named Siri. ☐

D Answer the following questions.

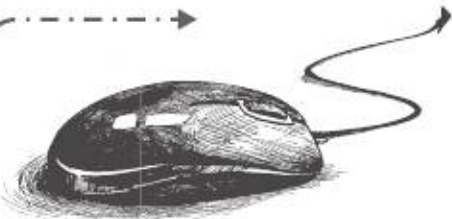
1. Write the names of any 2 voice assistants.

2. What is Google Home?

3. What is AI?

4. Who is Sophia?

5. Who is Siri?



W ork sheet



Answer in one word:

(NEP) Creative Learning

1. Name the two Mars rovers which were a part of NASA's Mars Exploration Mission in 2004.

2. Name the first gaming device that tracked human movement using 3D camera and infrared detection.
3. Name the humanoid created by Hanson Robotics in 2016.
4. Name the artificial humanoid released by Honda in 2000.
5. Name the person who coined the term Artificial Intelligence.

LAB ACTIVITY

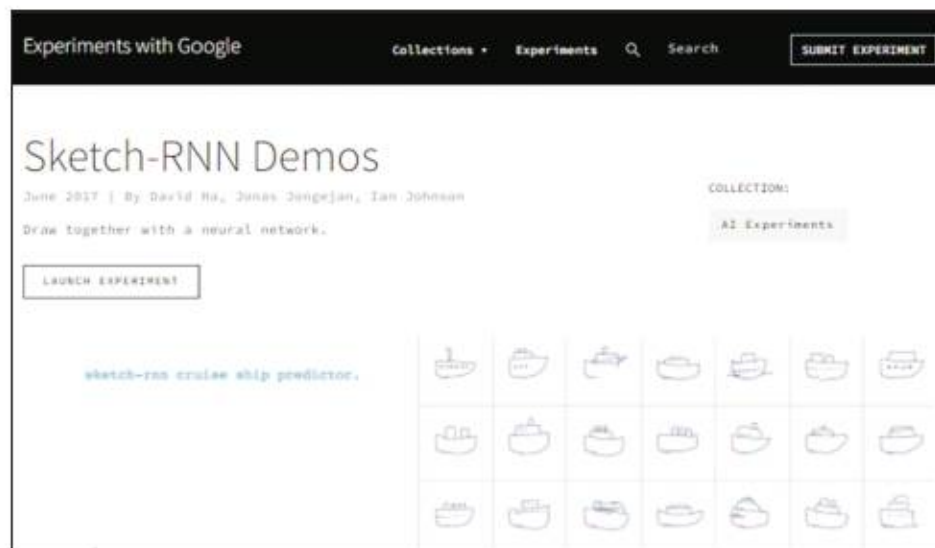
This is an experiment by Google that uses the data from millions of doodles from the Internet and predicts your drawing.

(NEP) Experimental Learning

To play the game, visit the following link:

<https://experiments.withgoogle.com/sketch-rnn-demo>

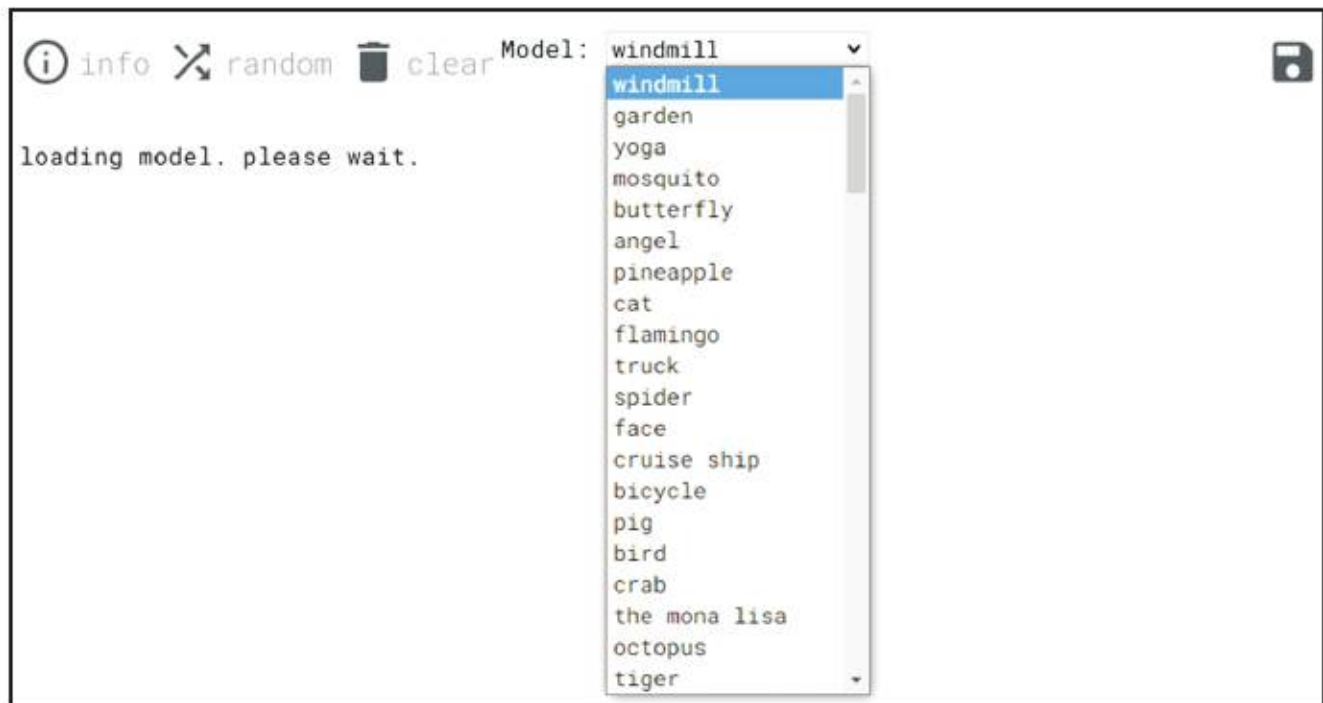
Step 1: Click on the Launch Experiment button.



Step 2: Click on the empty area in the bottom of the window.



Step 3: Click on the drop-down arrow to select what to draw.



Step 4: Now, you can start drawing. While drawing, you will observe that if you draw a stroke and wait for the AI to predict and make the next move.

